LAST BID

Complete project record: v3 upgrade

Every prompt, every step, every command and result, the earlier story-rewrite conversation, and the full source code.

Document date	15 October 2026
Project	Last Bid (HTML5 canvas painting / auction story game)
Current build	Last_Bid_v3.html (about 10.6 MB, music embedded)
Earlier builds	Last_Bid_v2.html (story rewrite), original Last_Bid_v2 (1).html uploaded this session
Contents	1 Overview 2 Earlier conversation 3 Prompts this session 4 Work log with commands and results 5 Design and algorithms 6 Test results 7 Open items Appendices A-G: source code

1. Overview

Last Bid is a single-file browser game. The player is a painter who has ten days to raise $10,000 for their mother's operation. Each day they paint one piece, five buyers rate it and bid, and the player chooses who buys. Who they sell to (collectors, critics, students) shapes the ending.

The work in this record happened in two stages:

- **Stage 1 (earlier conversation, section 2):** the story was rewritten around the mother's operation, the brother Robin, a 15% tax, panic mechanics, replay easter eggs and two endings. Output: Last_Bid_v2.html.
- **Stage 2 (this session, sections 3-6):** a new colour system with mixing, a fill bucket, an automatic art-rating engine, a full music mapping, a credits scene, and a language clean-up. Output: Last_Bid_v3.html.

What v3 adds, at a glance

- Two palette tabs: Additive (RGB light) and Subtractive (pigment RYB plus CMY inks), each with primary, secondary and tertiary colours, plus neutrals and gold.
- A colour mixer with a ratio slider. Additive mixing adds light; subtractive mixing multiplies pigment. Mixes are saved to a "Your mixes" row.
- Fill bucket (scanline flood fill with tolerance and an edge halo) and an Undo button.
- Automatic rating from real pixel data: colour harmony, shapes and lines, gestalt principles, coverage and effort, colour variety. A breakdown is shown in the saleroom.
- RGBA pixel extraction with getImageData, an RGB to HSL converter (hue 0-360), and smart sampling and filtering.
- Nine-slot music map (menu, mother scene, drawing, flashback, day 10, bidding, win, lose, credits), embedded as compressed MP3.
- Credits scene in Papyrus style with two credit cards and a closing card.
- Grammar and wording fixes across story and interface text.

Files delivered

File	What it is
Last_Bid_v3.html	The playable game, music embedded (about 10.6 MB)
This PDF	Record of all work and source code
(earlier) Last_Bid_v2.html	Story-rewrite build from the earlier conversation

2. Earlier conversation (story rewrite), as provided

This is the transcript you attached, reproduced in full. It describes how the game was changed to the current story, characters, tax and endings, and it is the basis on which the v3 work was done.

```
USER (earlier conversation, story rewrite brief, verbatim):

the  basic story has now shifted to
The guys mother need a operation
He got to find money
the player has a brother named robin Williams
the docter says she has around 10 days to live before she starts struggling he leaves the room
and then the mother talks about the father how it was painful and how the father begged the mother to
    kill him talks a little lore about father being a painter and how proud he would be proud to see his
     child following the footsteps and how robin was such a good therapist and how she is porud
she says this is unfair to tell but she says she want to go fast she cant she cant take it the children
    refuse ofcourse saying that if they arrange the money she can easily survive but she says just
    incase after a big pause she then thinks within herself as showed by brackets about how its
    difficult for this to happen and how she is proud of both the children something like that
the bad ending should be how the mother pulls the plug and dies how the brother is so angry he shouts at
     the player they fight but after a while the brother shuts down after some days the brother comes to
     say sorry and says how there parents would want them to be together before gettinf intrupted and
    finds the brother dead then comes the sucide letter about how he blames himself and after the
    expected sadness the brother still mentions well it is your fault but.... something like that
in the good ending the mother is saved but still dies the brother greive but still they get closer and
    cry together for the very first time tears come out make it emotional
now listen the main mechnaix stays the same but the messages in between should be a little longer and
    makes sure it connects the player connect emetionally to the mother and the main player
the brother would be the one to mention about this biding place and how he can cover 8500 and major
    living expensive of the main character tho he has to pay taxes about this art make it difficult to
    achive the good ending nerf everyone a little around 40 percent chance to get the good ending slight
     dioluge changes based on which type of npc you intteracted with both in good ending and bad ending
if the guy looks like he is about to not get good ending around day 7 make it look like he is panicking
    not able to draw straight lines nause time limit etc
good ending-mother survives dies later children survive
bad ending-mother dies brothers fight main character kills him/herself based on gender you know it
    already if you read the chat upward add shaky effect if it looks like he is approching bad ending
    nerf npc
we want to make it look like the main character feels guilty it was his fault
add a place holder where i will add the comic later in the end of bad ending something like it was not
    "your character name" fault it was "your character name" but highlighted in a way that it feels like
     the game is blaming the player then when the player replays add easter eggs like undertale
    deltatrune make it feel like the game knows they have played about how "the previous player name"
    died make it feel like the game knows and trying to point out this ridiculous power of ours then if
    the name is same as before add a difirrent diolougs
DO NOT MAKE IT CORNY
the lore between each day switch should be 4-5 diolugs and emotional we have to make it feel like its
    the player and main character fault for all of this
keep what i told about all of this and FILL IN THE GAPS YOU FIND BY YOURSELF BASED ON WHAT YOU ALREADY
    KNOW

add your elements changed a little if you think its way to good to not change but again make it good
    enough i am trusting you i already gave you a basic good luck

ASSISTANT (earlier conversation, work log shown in that transcript):

Ran 4 commands
Ran 3 commands
Balance settled: $8,500 start, $10,000 goal, a 15% sale tax, nerfed NPC offers, and lower odds, which
    should land near 40%. Now I'm writing the new game in pieces and keeping your original engine.

Ran 9 commands, shared files
I've rewritten the game around the mother's operation and saved it as Last_Bid_v2.html. A stub-DOM test
    ran every ending, in both genders, with and without a previous run, and found no broken text. I
    haven't played it in a real browser, so the timing of the panic effects and the typing speed are
    untested.

I couldn't see the earlier chat you mentioned, so I worked from your message and the HTML file. Anything
     from that chat that isn't in your message above may be missing.

Story

Opening: the doctor gives her ten days and leaves. She then talks about the father begging her to end
    it, his painting, and how proud he'd be of {you}. She says Robin is a good therapist and that she is
     proud of both, and asks to go fast "just in case". Both children refuse. After a long pause, her
    private thoughts appear in brackets. Then Robin, in the corridor, offers $8,500 and covers rent and
    the big bills, and mentions Halden House and the 15% tax.
Between days: each day has 4-5 lines of lore, set in the hospital, the attic or the corridor. A few days
     add a text from Robin that depends on who you've been selling to. The lines are built to make the
    guilt personal, for example "if I were faster", "you count the second time" and Robin's second of
    silence.
Good ending: the mother survives the surgery and lives eleven weeks. She dies at home in October. In the
     attic, the brothers cry together for the first time. You are still a little guilty, but "guilty
```

    together".
Bad ending: she unplugs the machine herself while you and Robin are out getting tea. Robin shouts at you
     and you fight, then he shuts down. Days later he comes to apologise and says their parents would
    want them together. A billing call interrupts him, and you find him dead. His letter blames himself,
     then says "some of it is yours... but you were the one who stayed". Then comes the comic
    placeholder, the highlighted "It was not Robin's fault. It was {you}'s.", and your own death.
Collectors, critics and students: each type changes the clerk scene, what Robin shouts and the letter.
    In the good ending it changes the funeral and hospital lines. Each type has its own ending titles.

Mechanics

Money and tax: you start with $8,500 and need $10,000, with a 15% sale tax. The Community Pass tool is
    now a Tax Waiver that cuts the tax to 8%. Collector, student and local offers are lower, the Silver
    Tongue bonus is smaller and the Patron's Letter pays less.
Odds: I simulated the economy and set the odds so a typical run should land around 40% for the good
    ending. These are my own simulated players, not real ones, so treat 40% as a target to check by
    playing. Falling short on money costs 12 points of odds. The numbers are in the odds() function.
Panic from day 7: if your projected money or ending odds look bad, your lines tremble and stop being
    straight. The canvas sways and blurs, a countdown starts (50 seconds on day 7, 30 by day 10) and
    buyers' offers drop about 18%. When time runs out, the painting is submitted as it is.
Replays: the game remembers your last name, ending, final money and the day your hands started shaking.
    A different name gets one set of "???" lines and the same name gets another, and good and bad past
    runs get different ones. There are small callbacks on days 3, 5, 7 and 9, and extra lines at the
    endings. From the fourth play on, it counts the attempts.

Things you need to do or decide

Comic: set COMIC_SRC in the endings code to your image path. Until then, a dashed box marks the spot.
Suicide handling: both deaths happen off-screen with no method shown. The letter says it's the wrong
    answer and asks the player not to follow. The menu has content notes, and the bad-ending card shows
    a note with Tele-MANAS (14416), 988 and findahelpline.com. I'd keep these in. I did soften the
    game's blame by making it the character's guilt voice, not a claim about the player.
Gap-fills: I made up the hospital, corridor, door and letter scenes, the tools' new effects and the
    Robin text lines. I also gave the brother a billing phone call as the "interruption". Tell me if you
     want any of these changed.
Files are in /mnt/user-data/outputs: Last_Bid_v2.html

ASSISTANT (earlier, after a further reply):
No need to apologise. Changing the story partway through is normal, and the swap was quick to make.
If something about it didn't land, tell me and I'll adjust it. That could be the letter's tone, Robin's
    last scene, or how the blame screen hits. Otherwise the file is ready to play.
USER: kill the main player not his brother robin in the bad endings

ASSISTANT: Ran 2 commands
The EPIPE error comes from my test script's output being cut off by head. The game file built and the
    check passed.
Presented file (Last_Bid_v2.html)
The file already had the main player dying in the bad endings and Robin surviving, from my last update.
    It's possible you opened an older copy, so please download this one again. I also made it clearer,
    because the player's death before was only implied through Robin's reaction.
Robin finds the player's letter and the attic, and now a line states outright that {name} is dead and
    Robin is left standing there. Nothing graphic is shown and no method is given.
If you want something different, such as the player's death shown more directly, or Robin's last scene
    cut, tell me which.
Files are in /mnt/user-data/outputs: Last_Bid_v2.html

3. Prompts given this session (verbatim)

Prompt 1 (colours, bucket, rating). Note the message was cut off at point 6 in the text received.

Instead of providing limited options of color add multiple colors from additive colors and subtractive colors and add a option to combine these colors provide the primary secondary and tertiary color options (for example red blue yellow green etc)
Add a fill bucket tool
Rate each drawing based on
1)Check color harmony
a)compliment colors
b)contrasting colors
c)analog colors
d)etc
2) Check shapes lines
3)Check if any gestalt principles have been used
4) Pixel Data Extraction: A function to extract RGBA pixel data from an HTML5 <canvas> element using getImageData.
5) Color Space Conversion: An efficient RGB-to-HSL conversion utility so we can evaluate colors using the color wheel (0 deg to 360 deg Hue).
6) Smart Sampling & Fil…

Prompt 2 (music)

Menu song
prelude to suite e major

Mother Scene
Megadalene elusive lament

Drawing start
campebala 07

Mid drawing songs
Flashbacks

Day 10
Sonic drowning

Bidding time
Orchestral suit no 2

If they win by day 10
Fire is gone

If they lose by day 10 (fight)
Glacier

Credits
Promise-Reprise

make sure these songs fit perfectly in place (in this game)

Prompt 3 (credits)

add a credit scene in the end

first credit
Rudra Parashar-Ideate music Design
Prince Raj Rawat- Main Ideate Music Finish touches

Rudrapratap singh cauhan- Testing, ideate, Video

MAKE IT LOOK UNIQUE PAPYRUS font and stylish

second credits
Main AI assistance- Claud
Side AI assistance- Gemini

Prompt 4 (open brief)

Fill in the gaps you dont understand if you think you can make something better with guarantee do it change it based on what you think is good we just need the basic expected outcome

Uploaded files

Last_Bid_v2 (1).html (the game, 74 KB) and eight audio files: Prelude to Suite in C Major, Magdalene's Elusive Lament, La Campanella Op. 7, Flashback (Excerpt), Drowning, Orchestral Suite No. 2 in B Minor BWV 1067 VII Badinerie, The Fire Is Gone (piano, saxophone, trumpet), GLACEIR. No file for "Promise-Reprise" was uploaded.

Prompt 5 (language)

there are some gramtically error langaguge issues in few areas, can you view that and make it crisper

Prompt 6 (this document)

can you make a pdf of all the work we did in a pdf file all prompts all work you did all commands run all results came everything for everything we did the last few chats and source code to. Also pick the below comand prompts as well

4. Work log: steps, commands and results

Steps are in the order they happened. Long files written with the file tool (not the shell) are listed by name and printed in full in the appendices.

Step 1. Read the uploaded game

The game file was read in several ranges because the viewer truncates long files. Findings that drove the design:

- Single file, about 490 long lines, with CSS, a Press Start 2P / VT323 / Cormorant font stack, and a dark/light theme.
- Painting happens on a 480x320 canvas. The old quality() function scored stroke count, colour count, coverage, today's colour (+15) and the Gilded Brush tool (+10).
- Nine fixed colours (COLS) and a market colour picked from them each day.
- A stub music system: MUSIC={main:"",good:"",bad:""} plus a "Choose music file" override.
- makeCast() uses quality().q and .cov and .gold to set buyer ratings and offers, so the new rating kept that interface.
- ending() contains the clerk scene, the win branch, the lose branch (fight, letter, comic placeholder) and the final card.

Commands (file viewer calls): view /mnt/user-data/uploads/Last_Bid_v2__1_.html, then view_range [75,250], [127,198], [250,412], [300,363]. Result: full source read.

Step 2. Audio and design inspection

```
$ ls -la /mnt/user-data/uploads/; cat /mnt/skills/public/frontend-design/SKILL.md | head -60; which ffprobe ffmpeg
```

```
total 54828 ... (nine files)
Drowning_spotdown_org_mp3.mpeg                       483073
Flashback__Excerpt__spotdown_org_mp3.mpeg            750117
GLACEIR_spotdown_org_mp3.mpeg                       1282335
La_Campanella__Op_7_spotdown_org_mp3.mpeg          11816296
Last_Bid_v2__1_.html                                  74086
Magdalene_s_Elusive_Lament_spotdown_org_mp3.mpeg    1280605
Orchestral_Suite_No__2 ... Badinerie ...mpeg       18005276
Prelude_to_Suite_in_C_Major_spotdown_org_mp3.mpeg  17644020
The_Fire_Is_Gone ... .mpeg                          4786781
/usr/bin/ffprobe
/usr/bin/ffmpeg
```

```
$ cd /mnt/user-data/uploads; for f in *.mpeg; do echo "$f"; ffprobe -v error -show_entries
   format=duration,bit_rate -of csv=p=0 "$f"; done
```

```
Drowning                14.8 s    260 kbps
Flashback (Excerpt)     32.1 s    187 kbps
GLACEIR                 67.2 s    153 kbps
La Campanella          370.0 s    255 kbps
Magdalene's Lament     123.1 s     83 kbps
Orchestral Suite 2    1201.8 s    120 kbps   (20 minutes)
Prelude to Suite      1167.9 s    121 kbps   (19.5 minutes)
The Fire Is Gone       160.1 s    239 kbps
```

Decision: embedding all files as base64 would give an unusable 75 MB page, so the tracks were compressed to mono 56 kbps MP3 and the two 20-minute files were trimmed.

```
$ python3 - <<'EOF'  # cross-correlation test to see whether the two long files are a short piece repeated
(decode to 4 kHz mono, correlate a 10 s window against the rest, report best repeat lags)
EOF
```

```
Orchestral_Suite  best repeat offsets (s): [70.9, 70.9, 70.9]  correlation [0.14, 0.13, 0.12]
Prelude_to_Suite  best repeat offsets (s): [330.8, 330.8, 330.8] correlation [0.21, 0.20, 0.19]
```

Result: correlations were weak (0.12 to 0.21), so no reliable loop point was found. The files were simply trimmed with fades, not cut at a detected phrase end.

```
$ cd /home/claude/mus; enc(){ ffmpeg -v error -y -i "$U/$1" ${3:+-t $3} -ac 1 -ar 32000 -b:a 56k -af
   "afade=t=in:d=0.4${3:+,afade=t=out:st=$(echo "$3-1.5"|bc):d=1.5}" "$2.mp3"; }
enc Prelude_to_Suite_in_C_Major_spotdown_org_mp3.mpeg menu 150
enc Magdalene_s_Elusive_Lament_spotdown_org_mp3.mpeg mother
enc La_Campanella__Op_7_spotdown_org_mp3.mpeg draw
enc Flashback__Excerpt__spotdown_org_mp3.mpeg flash
enc Drowning_spotdown_org_mp3.mpeg d10
enc Orchestral_Suite_No__2_in_B_Minor__BWV_1067__VII__Badinerie_spotdown_org_mp3.mpeg bid 100
enc The_Fire_Is_Gone__For_Piano__Saxophone_and_Trumpet__spotdown_org_mp3.mpeg win
enc GLACEIR_spotdown_org_mp3.mpeg lose
ls -la; du -ch *.mp3 | tail -1
```

```
bid.mp3 811550   d10.mp3 287243   draw.mp3 2663016   flash.mp3 236466
lose.mp3 482658  menu.mp3 1245718  mother.mp3 993280  win.mp3 1196316
7.6M total
node v22.22.2; playwright (python) available
```

Step 3. Wrote the new code modules (file tool)

Six source files were created. Their full text is in Appendices A-F.

File	Purpose
blockA.js	Palette data, mixing, RGB to HSL, pixel extraction, sampling, harmony, shapes, gestalt, quality(), bucket fill
draw.js	New drawScreen(): tools, palette tabs, mixer, undo, stroke recording, panic timer, flashback trigger
music.js	Music engine: track map, fades, one-shot flashback with resume, autoplay unlock
credits.js	Credits scene (three cards, auto-advance, keyboard support)
style.css	Styles for palette, tools, report bars and the papyrus credits
build.py	Patches the original game with all of the above and embeds the audio

```
$ # in build.py: 26 exact-match string patches against the original file, each asserted to match exactly once
# music block replaced by slicing between '/* ===== MUSIC:' and the '$("#mf").onchange=' line
# old COLS line replaced by blockA.js; old quality() removed; old drawScreen() replaced by draw.js
python3 build.py
```

```
10.656099 MB
```

```
$ python3 - <<'EOF'  # sanity: no duplicate functions, no leftover old music calls
print(s.count('function quality('),s.count('function drawScreen('),s.count("mus('main')"))
EOF
node --check t.js && echo SYNTAX_OK
```

```
1 1 0 1
SYNTAX_OK
chromium ok
```

The trailing "1" is the word "COLS" appearing inside embedded audio text, not code.

Step 4. Browser test of drawing, fill, mixing and rating

A headless Chromium test (test.py, Appendix F2) opened the game, jumped to day 3, drew a circle, filled it, drew lines, switched to the subtractive tab, drew a wave, mixed red and yellow, drew a square, then read the rating and submitted to the saleroom.

```
$ python3 test.py   # first run
```

```
... strokes: 6, SD: 5, FL: 1 ... types: circle, straight, straight, straight, angle
ERRORS ['Failed to load resource: net::ERR_NAME_NOT_RESOLVED']
```

Finding: only 6 of 8 expected actions registered. Cause: Playwright scrolled the page when clicking swatches, so the stored canvas coordinates went stale. This was a test artefact, not a game bug. The test was changed to scroll to top and re-measure the canvas before every stroke, and to use 4-step mouse moves.

```
$ python3 test.py   # after fixing the test
```

```
mix res rgb(241, 90, 7)    (red + yellow, subtractive = orange)
q: 59
 Colour harmony 5.1/25  'Rainbow mix; complementary yellow + blue; flat values...'
 Shapes and lines 16.5/20 '1 circle, 4 straight, 1 angle, 1 rectangle, 1 filled'
 Gestalt 18.3/20  'Uses Proximity, Similarity, Closure, Continuity, Figure and ground'
 Coverage and effort 8.65/15
 Colour variety 10/10  '5 colour families'
strokes 8, SD 7, FL 1, music track now: flash
music after submit: ['bid','bid','bid']
```

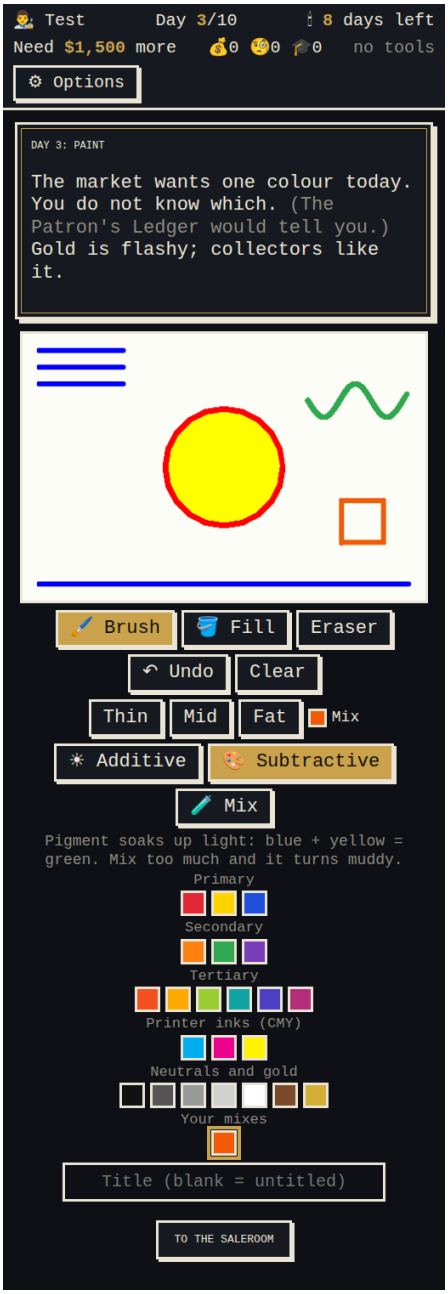
Finding: the green wave was classified as an "angle" because the corner threshold (0.9 rad per 10 px) was too sensitive. The threshold was raised to 1.15 rad.

```
$ sed -i 's/Math.abs(v)>.9;if(hit/Math.abs(v)>1.15;if(hit/' blockA.js && python3 build.py && python3 test.py |
   grep -E 'types|wave|ERRORS'
```

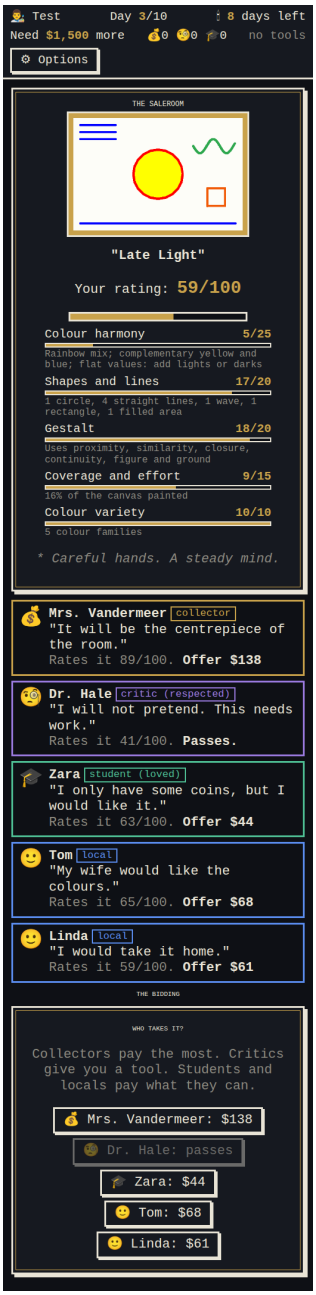
```
10.656101 MB
 "note": "1 circle, 4 straight, 1 wave, 1 rectangle, 1 filled"
 "types": ["wave", ...]
ERRORS ['Failed to load resource: net::ERR_NAME_NOT_RESOLVED']
```

The only console error is Google Fonts failing to load because the sandbox has no network; it does not occur online.

Screenshots from the test



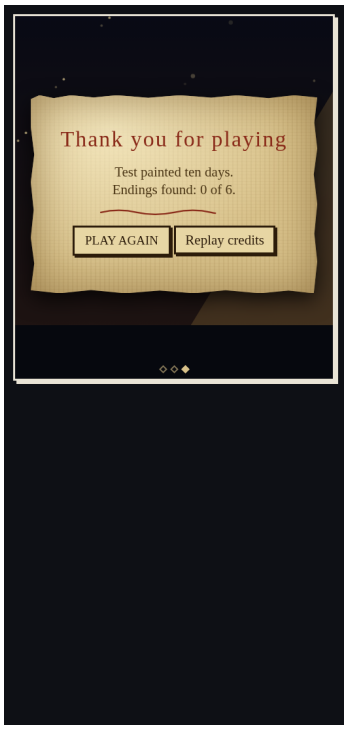
Drawing screen: tools, palette tabs, mixer result swatch in "Your mixes", colour indicator.



Saleroom: rating out of 100 with the five-part breakdown, then buyers and bids.



Credits card 1: papyrus scroll over a night attic (fallback serif shown; Papyrus is not installed in the test machine).



Credits closing card with Play Again and Replay credits.

Step 5. Both endings into credits, music switching

```
$ python3 test2.py   # jump to day 10 via prev('R',true/false), auto-advance dialogue, click through blame screen,
    click CREDITS
```

```
menu mus menu
win  reached credits: True  tracks seen: {'win','menu'}  now: credits  errors: []
menu mus menu
lose reached credits: True  tracks seen: {'lose','menu'} now: credits  errors: []
```

Step 6. Language clean-up (prompt 5)

All player-facing text was read. The story is deliberately written in short fragments, so only clear errors and clumsy phrasing were changed. Fixes were added to build.py so the build is reproducible.

Before	After

Students and locals are the people.	Collectors pay the most. Critics give you a tool. Students and locals pay what they can.
food, bus fare, your phone.	food, bus fare and your phone.
I have been doing numbers since the scan.	I have been running numbers since the scan.
how likely things end well.	how likely it is to end well.
are fiction. (content note)	are fictional.
Tears. The first of your lives, for both of you. They are salt and they are warm, and they are exactly as ordinary as anything.	Tears, the first for both of you. They are salt and warm, and as ordinary as anything.
Speaker "Robin" + text starting "Text from Robin:" (day 2, day 6, and the sales-based texts)	Blank speaker, so it reads as a text message
Rating notes: "4 straight", "complementary yellow + blue", "5 colour families", "Uses Proximity, ..."	"4 straight lines", "complementary yellow and blue", singular/plural handled, lowercase gestalt names, "flat values: add lights or darks"

```
$ python3 build.py && node --check t.js && python3 test.py | grep -E 'note|ERRORS'
```

```
10.656351 MB
SYNTAX_OK
 "note": "Rainbow mix; complementary yellow and blue; flat values: add lights or darks"
 "note": "1 circle, 4 straight lines, 1 wave, 1 rectangle, 1 filled area"
 "note": "Uses proximity, similarity, closure, continuity, figure and ground"
 "note": "16% of the canvas painted"
 "note": "5 colour families"
```

5. Design and algorithms

5.1 Music map

Moment	Track key	Song	Notes
Menu	menu	Prelude to Suite in C Major	Trimmed to 150 s, loops. Your note said "E major"; the file is C Major.
Mother scene (hospital intro; each day's story)	mother	Magdalene's Elusive Lament	Starts at the hospital scene. Also plays during each day's lore and the lose-ending clerk scene.
Drawing starts	draw	La Campanella Op. 7	Full length, loops.
Mid-drawing	flash	Flashback (Excerpt)	Plays once after 6 strokes, then La Campanella resumes at the saved position.
Day 10 (story and drawing)	d10	Drowning	Replaces drawing and flashback tracks on day 10.
Bidding	bid	Orchestral Suite No. 2 (Badinerie)	Trimmed to 100 s, loops. Plays on submit.
Win	win	The Fire Is Gone	From the start of the winning ending.
Lose (fight)	lose	GLACEIR	From the first line of the lose branch ("Then tomorrow...").
Credits	credits	Promise-Reprise	No file supplied. Falls back to the menu song. Options > Load credits song lets you pick the file.

Engine behaviour: tracks crossfade (350 ms out, 800 ms in); a one-shot track (flashback) saves the position of the looping track and resumes it afterwards; browsers block autoplay, so the first tap anywhere starts the current song.

5.2 Palette and mixing

- Additive tab: red/green/blue; yellow/cyan/magenta; orange, chartreuse, spring green, azure, violet, rose. Pure channel values so sums are exact.
- Subtractive tab: red/yellow/blue; orange/green/purple; red-orange, amber, yellow-green, teal, indigo, red-violet; plus CMY printer inks.
- Shared row: black, dark grey, grey, light grey, white, brown, gold. Gold keeps its collector bonus.
- Additive mix: channel = A x 2w + B x 2(1-w), clamped. At 50/50 red + green = yellow.
- Subtractive mix: weighted geometric mean per channel, so red + yellow = orange, blue + yellow = green, red + blue = muted purple.
- The market colour is now a colour family (red, orange, yellow, green, cyan, blue, purple, magenta, black, gold). Any shade of the family counts, if it covers at least 3% of the painted pixels.

5.3 Rating engine (replaces the old quality())

Pipeline: extractPixels (getImageData) -> analyze (sample every 2nd pixel, skip paper, convert to HSL, build 36-bin hue histogram and lightness histogram, cluster hues, drop clusters under 5%) -> harmony / shapes / gestalt scoring -> weighted total clamped to 1-100.

Component	Max	How it is scored
Colour harmony	25	Scheme score x 15 (complementary 1.0, triadic 1.0, split-complementary 0.8-0.95, analogous 0.85, tetradic 0.9, monochrome 0.55, mixed 0.45, busy 0.35, rainbow 0.3) + light/dark contrast x 7 + 3 for a neutral anchor.
Shapes and lines	20	Variety of recognised types (up to 8) + clarity, the share of long strokes that are not scribbles (up to 7) + line-weight variety (2) + fills (3). Types: straight, curve, wave, zigzag, angle, circle, rectangle, triangle, polygon, blob.
Gestalt	20	Six principles, up to 4 points each: proximity, similarity, closure, continuity, figure and ground, symmetry. A principle counts as "used" at strength 0.5 or more.
Coverage and effort	15	Canvas coverage (up to 10) + stroke count (up to 5).

Colour variety	10	2.5 points per colour family above 2% of painted pixels.
Bonuses	+22	Market colour +12, Gilded Brush +10. Total is clamped to 100.

Stroke geometry is recorded as point paths, thinned to 2 px, resampled every 10 px, then classified by straightness (chord/length above 0.93), turning angles, corner count (more than 1.15 rad in one step), inflections, and closure (start near end with about 360 degrees of turning). Closed shapes are told apart by the number of vertices after Ramer-Douglas-Peucker simplification and by circularity. On panic days the pointer jitter makes strokes read as scribbles, so the score drops, which matches the story.

5.4 Fill bucket

Scanline flood fill on the 480x320 canvas using getImageData/putImageData, tolerance 60 (sum of channel differences), followed by a one-pixel halo (tolerance 130) so anti-aliased line edges do not leave a white ring. A fill counts as a stroke, is recorded for analysis, and can be undone.

5.5 Credits

Three cards: "Last Bid was made by" (the three team members and roles), "Shaped with help from" (Claude as main AI, Gemini as side AI), and a closing card with the player's name, endings found, Play Again and Replay credits. Styling: torn papyrus parchment (CSS gradients and a jagged clip-path), font stack Papyrus / Papyrus Condensed / Herculanum / Uncial Antiqua / fantasy, ink-bleed text reveals, brush-stroke dividers that draw themselves, a night-attic background with flickering lamp light. Respects the game's Animations toggle and reduced-motion.

6. Test results summary

Check	Result
JavaScript syntax (node --check)	Pass
Draw, fill, undo, mix, submit in headless Chromium	Pass (after fixing test scroll artefact)
Rating report renders in the saleroom	Pass (screenshot)
Wave and rectangle recognised	Pass after corner threshold change
Win ending reaches credits; music: win	Pass
Lose ending reaches credits; music: lose	Pass
Flashback plays after 6 strokes	Pass (track key switched to flash)
Console errors	Only the offline Google Fonts request
Audio heard by a person	Not tested: no sound device in the sandbox

7. Open items and honest caveats

- Promise-Reprise was not uploaded. The credits use the menu song until you load the file in Options.
- Prompt 1 point 6 was cut off ("Smart Sampling & Fil..."). It was implemented as sampling plus filtering. Send the rest if it meant something else.
- The old odds of about 40% for the good ending were tuned to the old rating formula. The new rating was calibrated by reasoning and by the test paintings (a simple scribble lands near 35-45, a deliberate composition near 55-80), but the 40% target has not been re-simulated. Check by playing, and adjust odds() if needed.
- The two long tracks were trimmed to 150 s and 100 s with fades. Their loop point may not fall on a musical phrase end.
- Papyrus is installed on Windows (with Office) and macOS; other devices use a similar fallback font.
- Spelling corrected from your message: "Rudrapratap Singh Chauhan" (was "cauhan") and "Claude" (was "Claud"). "Megadalene" and "campebala 07" were read as Magdalene's Elusive Lament and La Campanella Op. 7.
- Shape recognition is heuristic. Very unusual drawings may be classified as scribbles or blobs.
- In the earlier story rewrite the comic image is still a placeholder: set COMIC_SRC in the endings code.

Appendix A. blockA.js

Palette, mixing, colour conversion, pixel extraction, sampling, harmony, shapes, gestalt, quality(), bucket fill.

```
/* ===== v3: colour palette (additive + subtractive), mixing, pixel analysis, rating ===== */
const PAL={
add:{tip:'Light adds up: red + green = yellow. Mix all three and you head for white.',groups:[
 ['Primary',[['#ff0000','Red'],['#00ff00','Green'],['#0000ff','Blue']]],
 ['Secondary',[['#ffff00','Yellow'],['#00ffff','Cyan'],['#ff00ff','Magenta']]],
 ['Tertiary',[['#ff8000','Orange'],['#80ff00','Chartreuse'],['#00ff80','Spring
    green'],['#0080ff','Azure'],['#8000ff','Violet'],['#ff0080','Rose']]]]},
sub:{tip:'Pigment soaks up light: blue + yellow = green. Mix too much and it turns muddy.',groups:[
 ['Primary',[['#e32636','Red'],['#ffd400','Yellow'],['#1f4fd8','Blue']]],
 ['Secondary',[['#ff7f11','Orange'],['#2fa84f','Green'],['#7a3db8','Purple']]],
 ['Tertiary',[['#f2501f','Red-orange'],['#ffaa00','Amber'],['#9acd32','Yellow-
    green'],['#0fa3a3','Teal'],['#4b3fc4','Indigo'],['#b32d7a','Red-violet']]],
 ['Printer inks (CMY)',[['#00aeef','Cyan ink'],['#ec008c','Magenta ink'],['#fff200','Yellow ink']]]]}};
const NEU=[['#111111','Black'],['#555555','Dark grey'],['#999999','Grey'],['#d0d0d0','Light
    grey'],['#ffffff','White'],['#7b4a2a','Brown'],['#d4af37','Gold']];
const CN={};[...Object.values(PAL).flatMap(p=>p.groups.flatMap(g=>g[1])),...NEU].forEach(([h,n])=>{CN[h]=n});
let PM='add',MIXED=[],tool='brush',SD=[],FL=[],UN=[],fbDone=false;
const BGHEX='#fdfdf8',GOLD='#d4af37';
const FAMH={red:'#e63946',orange:'#ff8c1a',yellow:'#f5c518',green:'#2fb344',cyan:'#14b8c8',blue:'#2a6fdb',purple:'#8e4
    4ad',magenta:'#e0359e',black:'#111111',gold:'#d4af37'};
const TRENDS=Object.keys(FAMH);
const h2r=h=>[1,3,5].map(i=>parseInt(h.slice(i,i+2),16));
const r2h=(r,g,b)=>'#'+[r,g,b].map(v=>Math.max(0,Math.min(255,Math.round(v))).toString(16).padStart(2,'0')).join('');
/* additive mixing adds light; subtractive mixing multiplies pigment (weighted geometric mean) */
function mixCol(a,b,w,mode){const A=h2r(a),B=h2r(b);return mode==='add'?r2h(...A.map((v,i)=>v*2*w+B[i]*2*(1-
    w))):r2h(...A.map((v,i)=>Math.pow(Math.max(v,1)/255,w)*Math.pow(Math.max(B[i],1)/255,1-w)*255))}

/* ---- colour space: RGB -> HSL (hue 0..360) ---- */
const hd=(a,b)=>{const d=Math.abs(a-b)%360;return d>180?360-d:d};
const
    hn=h=>h<15||h>=345?'red':h<40?'orange':h<70?'yellow':h<165?'green':h<200?'cyan':h<255?'blue':h<290?'purple':'magen
    ta';
function rgbToHsl(r,g,b){r/=255;g/=255;b/=255;const mx=Math.max(r,g,b),mn=Math.min(r,g,b),l=(mx+mn)/2,d=mx-
    mn;if(!d)return{h:0,s:0,l};const s=d/(1-Math.abs(2*l-1));let
    h=mx===r?((g-b)/d)%6:mx===g?(b-r)/d+2:(r-g)/d+4;h*=60;if(h<0)h+=360;return{h,s,l}}
function famOf(r,g,b,c){c=c||rgbToHsl(r,g,b);if(Math.abs(r-212)+Math.abs(g-175)+Math.abs(b-
    55)<60)return'gold';if(c.l<.16)return'black';if(c.s<.14)return c.l>.85?'white':'gray';return hn(c.h)}

/* ---- 1. pixel data extraction ---- */
function extractPixels(cv){return cv.getContext('2d').getImageData(0,0,cv.width,cv.height)}

/* ---- 2. smart sampling + filtering: every 2nd pixel, skip paper, bucket hues, drop noise bins ---- */
function analyze(cv){
 const img=extractPixels(cv),d=img.data,W_=img.width,H_=img.height,ST=2,GX=24,GY=16,cw=W_/GX,chh=H_/GY,NB=36;
 const bc=new Float32Array(NB),bs=new Float32Array(NB),bo=new Float32Array(NB),lh=new Uint32Array(20),fm={},gl=new
    Float32Array(GX*GY),gn=new Uint16Array(GX*GY);
 let tot=0,pt=0,chroma=0,neutral=0,sx=0,sy=0,ls=0;
 for(let y=0;y<H_;y+=ST)for(let x=0;x<W_;x+=ST){const i=(y*W_+x)*4;if(d[i+3]<8)continue;tot++;
  const r=d[i],g=d[i+1],b=d[i+2];
  if(Math.abs(r-253)+Math.abs(g-253)+Math.abs(b-248)<24)continue;
  const c=rgbToHsl(r,g,b);pt++;sx+=x;sy+=y;ls+=c.l;lh[Math.min(19,c.l*20|0)]++;
  const f=famOf(r,g,b,c);fm[f]=(fm[f]||0)+1;
  const ci=(y/chh|0)*GX+(x/cw|0);gl[ci]+=c.l;gn[ci]++;
  if(c.s>=.2&&c.l>.1&&c.l<.93){const
    bi=(c.h/10|0)%NB,rad=c.h*Math.PI/180;bc[bi]++;bs[bi]+=Math.sin(rad);bo[bi]+=Math.cos(rad);chroma++}else neutral++}
 const thr=Math.max(2,chroma*.02),on=Array.from(bc,v=>v>=thr),runs=[];let cur=null;
 if(on.every(Boolean))runs.push([...Array(NB).keys()]);
 else{const s0=on.indexOf(false);for(let k=1;k<=NB;k++){const ix=(s0+k)%NB;if(on[ix])(cur=cur||[]).push(ix);else
    if(cur){runs.push(cur);cur=null}}if(cur)runs.push(cur)}
 let cl=runs.map(r=>{let w=0,sn=0,cs=0;r.forEach(i=>{w+=bc[i];sn+=bs[i];cs+=bo[i]});let h=Math.atan2(sn,cs)*180/Math.P
    I;if(h<0)h+=360;return{h,share:chroma?w/chroma:0}}).filter(c=>c.share>=.05).sort((a,b)=>b.share-a.share);
 const nCl=cl.length;cl=cl.slice(0,4);
 const pc=q=>{if(!pt)return 0;let a=0;const t=pt*q;for(let i=0;i<20;i++){a+=lh[i];if(a>=t)return(i+.5)/20}return 1};
 return{cov:tot?pt/tot:0,pt,tot,fm,clusters:cl,nCl,chroma,neutralShare:pt?neutral/pt:0,p10:pc(.1),p90:pc(.9),cx:pt?sx/
    pt:240,cy:pt?sy/pt:160,ls,gl,gn,GX,GY}}

/* ---- 3. colour harmony: complementary, split, triadic, tetradic, analogous, monochrome + value contrast ---- */
function harmony(A){if(A.pt<30)return{s:0,note:'Nothing to judge yet.'};
 const cl=A.clusters,N=cl.length,Hs=cl.map(c=>c.h),nm=Hs.map(hn),det=[],pairs=[];
 for(let i=0;i<N;i++)for(let j=i+1;j<N;j++)pairs.push([i,j,hd(Hs[i],Hs[j])]);
 for(const[i,j,dd]of pairs){if(dd>=155)det.push(`complementary ${nm[i]} and ${nm[j]}`);else
    if(dd<=45&&nm[i]!==nm[j])det.push(`analogous ${nm[i]} and ${nm[j]}`)}
 let sch='Achromatic',sc=.5;
 if(N===1){sch='Monochromatic ('+nm[0]+')';sc=.55}
 else if(N===2){const dd=pairs[0][2];if(dd>=155){sch='Complementary';sc=1}else if(dd>=125){sch='Split-
    complementary';sc=.8}else if(dd>=90){sch='Contrasting hues';sc=.6}else if(dd>45){sch='Near
    neighbours';sc=.5}else{sch='Analogous';sc=.85}}
 else if(N===3){const ds=pairs.map(p=>p[2]),span=Math.max(...ds);
  const split=[0,1,2].some(i=>{const o=[0,1,2].filter(k=>k!==i);return o.every(k=>{const v=hd(Hs[i],Hs[k]);return
    v>=125&&v<=175})&&hd(Hs[o[0]],Hs[o[1]])<=75});
  if(ds.every(v=>v>=100&&v<=140)){sch='Triadic';sc=1}else if(span<=70){sch='Analogous';sc=.85}else
    if(split){sch='Split-complementary';sc=.95}else if(ds.some(v=>v>=155)){sch='Complementary with an
    accent';sc=.7}else{sch='Mixed palette';sc=.45}}
 else if(N===4){const pr=(a,b,c,d)=>hd(Hs[a],Hs[b])>=150&&hd(Hs[c],Hs[d])>=150;if(pr(0,1,2,3)||pr(0,2,1,3)||pr(0,3,1,2
    )){sch='Tetradic';sc=.9}else{sch='Busy palette';sc=.35}}
 if(A.nCl>4){sch='Rainbow mix';sc=.3}
```

```
 const vc=Math.min(1,(A.p90-A.p10)/.55),anchor=N>=1&&A.neutralShare>=.08&&A.neutralShare<=.6;
 const s=Math.min(25,sc*15+vc*7+(anchor?3:0));
 const note=[sch,det.slice(0,2).join(', '),vc>.6?'strong light and dark contrast':vc<.25?'flat values: add lights or
    darks':'',anchor?'neutrals anchor the colours':''].filter(Boolean).join('; ');
 return{s,note}}

/* ---- 4. shapes and lines ---- */
const pd=(p,a,b)=>{const dx=b.x-a.x,dy=b.y-a.y,L=Math.hypot(dx,dy);if(!L)return Math.hypot(p.x-a.x,p.y-a.y);return
    Math.abs(dy*p.x-dx*p.y+b.x*a.y-b.y*a.x)/L};
function rdp(p,e){if(p.length<3)return p;let mx=0,ix=0;const a=p[0],b=p[p.length-1];for(let
    i=1;i<p.length-1;i++){const v=pd(p[i],a,b);if(v>mx){mx=v;ix=i}}return
    mx>e?rdp(p.slice(0,ix+1),e).slice(0,-1).concat(rdp(p.slice(ix),e)):[a,b]}
function strokeInfo(s){const p=s.p,n=p.length;let len=0,x0=1e9,x1=-1e9,y0=1e9,y1=-1e9;
 for(let i=0;i<n;i++){const q=p[i];if(q.x<x0)x0=q.x;if(q.x>x1)x1=q.x;if(q.y<y0)y0=q.y;if(q.y>y1)y1=q.y;if(i)len+=Math.
    hypot(q.x-p[i-1].x,q.y-p[i-1].y)}
 const a=p[0],b=p[n-1],chord=Math.hypot(b.x-a.x,b.y-a.y),dg=Math.hypot(x1-x0,y1-y0);
 const o={len,chord,cx:(x0+x1)/2,cy:(y0+y1)/2,dg,c:s.c,z:s.z,f:famOf(...h2r(s.c)),t:'dot',p};
 if(len<12)return o;
 const rs=[a];for(let i=1;i<n;i++){const l=rs[rs.length-1];if(Math.hypot(p[i].x-l.x,p[i].y-l.y)>=10)rs.push(p[i])}
 const hs=[];for(let i=1;i<rs.length;i++)hs.push(Math.atan2(rs[i].y-rs[i-1].y,rs[i].x-rs[i-1].x));
 const dth=[];for(let i=1;i<hs.length;i++){let
    v=hs[i]-hs[i-1];while(v>Math.PI)v-=2*Math.PI;while(v<-Math.PI)v+=2*Math.PI;dth.push(v)}
 const sm=dth.map((v,i)=>((dth[i-1]??v)+v+(dth[i+1]??v))/3),turn=dth.reduce((a,b)=>a+b,0);
 let corners=0,csg=[],pv=false;for(const v of dth){const
    hit=Math.abs(v)>1.15;if(hit&&!pv){corners++;csg.push(Math.sign(v))}pv=hit}
 let infl=0,lsg=0;for(const v of sm){if(Math.abs(v)>.12){const g=Math.sign(v);if(lsg&&g!==lsg)infl++;lsg=g}}
 const st=chord/len;
 if(len>60&&(chord<.14*len&&Math.abs(turn)>4.5||chord<.1*len&&corners>=3)){
  let ar=0;for(let i=0;i<n;i++){const q=p[i],r=p[(i+1)%n];ar+=q.x*r.y-r.x*q.y}ar=Math.abs(ar)/2;
  const circ=4*Math.PI*ar/(len*len),v=Math.max(2,rdp(p,Math.max(5,.06*dg)).length-1);
  o.closed=true;o.t=v===3?'triangle':v===4?'rect':circ>.8?'circle':circ>.5?'polygon':'blob';return o}
 if(len>60&&chord>=.14*len&&chord<=.38*len&&Math.abs(turn)>3.6){o.implied=true;o.t='curve';return o}
 if(st>.93){o.t='straight';const ang=Math.abs(Math.atan2(b.y-a.y,b.x-
    a.x)*180/Math.PI);o.dir=ang<15||ang>165?'horizontal':ang>75&&ang<105?'vertical':'diagonal';return o}
 if(corners===0)o.t=infl>=2?'wave':'curve';
 else if(corners<=2)o.t='angle';
 else{let alt=0;for(let
    i=1;i<csg.length;i++)if(csg[i]!==csg[i-1])alt++;o.t=alt>=.7*(csg.length-1)?'zigzag':'scribble'}
 return o}
const SHL={straight:['straight line','straight lines'],curve:['curve','curves'],wave:['wave','waves'],zigzag:['zigzag'
    ,'zigzags'],angle:['angle','angles'],circle:['circle','circles'],rect:['rectangle','rectangles'],triangle:['triang
    le','triangles'],polygon:['polygon','polygons'],blob:['blob','blobs'],scribble:['scribble','scribbles']};
function shapesEval(){const inf=SD.filter(s=>s.p.length).map(strokeInfo),longs=inf.filter(o=>o.len>=30),cnt={};
 inf.forEach(o=>{cnt[o.t]=(cnt[o.t]||0)+1});const dots=cnt.dot||0;
 const kinds=Object.keys(cnt).filter(k=>k!=='scribble'&&k!=='dot');if(dots>=6)kinds.push('dots');
 const good=longs.filter(o=>o.t!=='scribble').length,clar=longs.length?good/longs.length:(dots>=6?.6:0),zs=new
    Set(SD.map(s=>s.z)).size;
 if(!inf.length&&!FL.length)return{s:0,inf,note:'No lines yet.'};
 const s=Math.min(20,Math.min(1,kinds.length/4)*8+clar*7+(zs>=2?1:0)+(zs>=3?1:0)+Math.min(3,FL.length*1.5));
 const parts=Object.entries(cnt).filter(([k])=>k!=='dot').map(([k,v])=>v+'
    '+SHL[k][v>1?1:0]);if(dots)parts.push(dots+' dots');if(FL.length)parts.push(FL.length+(FL.length>1?' filled
    areas':' filled area'));
 const note=(parts.join(', ')||'just marks')+(longs.length>=3&&clar<.4?'; shaky hands turn lines into scribbles':'');
 return{s,inf,note}}

/* ---- 5. gestalt principles ---- */
function gestalt(A,inf){const cl=(v,a=0,b=1)=>Math.max(a,Math.min(b,v)),dist=(a,b)=>Math.hypot(a.x-b.x,a.y-b.y);
 const marks=[...inf.map(o=>({x:o.cx,y:o.cy})),...FL.map(f=>({x:f.x,y:f.y}))];
 let prox=0;if(marks.length>=4){const par=marks.map((_,i)=>i),fd=i=>par[i]===i?i:(par[i]=fd(par[i]));
  for(let i=0;i<marks.length;i++)for(let j=i+1;j<marks.length;j++)if(dist(marks[i],marks[j])<60)par[fd(i)]=fd(j);
  const g={};marks.forEach((_,i)=>(g[fd(i)]=g[fd(i)]||[]).push(i));const gs=Object.values(g);
  if(gs.length>=2&&gs.length<=6&&gs.some(a=>a.length>1)){let gap=1e9;for(let a=0;a<gs.length;a++)for(let
    b=a+1;b<gs.length;b++)gs[a].forEach(i=>gs[b].forEach(j=>{gap=Math.min(gap,dist(marks[i],marks[j]))}));
   let ws=0,wn=0;gs.forEach(a=>{if(a.length>1)a.forEach(i=>{let
    m=1e9;a.forEach(j=>{if(j!==i)m=Math.min(m,dist(marks[i],marks[j]))});ws+=m;wn++})});
   prox=cl((gap/Math.max(10,ws/wn)-1)/3)}}
 const key={};inf.forEach(o=>{const k=o.f+'|'+o.t+'|'+o.z;key[k]=(key[k]||0)+1});const
    vs=Object.values(key).sort((a,b)=>b-a);
 const sim=vs[0]>=3?cl((vs[0]-1)/4+(vs[1]>=2?.15:0)):vs[0]===2?.25:0;
 const imp=inf.filter(o=>o.implied).length,clo=inf.filter(o=>o.closed).length,closure=cl(imp*.6+clo*.25+(FL.length&&cl
    o?.2:0));
 const
    sl=inf.filter(o=>['straight','curve','wave'].includes(o.t)&&o.len>=30),lc=sl.filter(o=>o.len>=100).length,ends=[];
 sl.forEach((o,k)=>{const p=o.p,n=p.length,m=Math.min(5,n-1);ends.push({k,x:p[0].x,y:p[0].y,a:Math.atan2(p[0].y-
    p[m].y,p[0].x-p[m].x)*180/Math.PI},{k,x:p[n-1].x,y:p[n-1].y,a:Math.atan2(p[n-1].y-p[n-1-m].y,p[n-1].x-p[n-1-
    m].x)*180/Math.PI})});
 let cp=0;for(let i=0;i<ends.length;i++)for(let
    j=i+1;j<ends.length;j++)if(ends[i].k!==ends[j].k&&dist(ends[i],ends[j])<18&&hd(ends[i].a,ends[j].a+180)<35)cp++;
 const cont=cl(lc*.35+cp*.5);
 const cenS=cl(1-Math.hypot(A.cx-240,A.cy-160)/150),meanL=A.pt?A.ls/A.pt:1,con=cl(Math.abs(.99-
    meanL)/.4),covF=A.cov<.05?A.cov/.05:A.cov>.65?cl((.95-A.cov)/.3):1;
 const fg=A.pt<30?0:cl(cenS*1.2)*con*covF;
 let best=0;for(const mode of[0,1]){let cons=0,mt=0;for(let cy=0;cy<A.GY;cy++)for(let
    cx=0;cx<A.GX;cx++){if(mode?cy>=A.GY/2:cx>=A.GX/2)continue;
  const
    i=cy*A.GX+cx,j=mode?(A.GY-1-cy)*A.GX+cx:cy*A.GX+(A.GX-1-
    cx),pa=A.gn[i]>=8,pb=A.gn[j]>=8;if(!pa&&!pb)continue;cons++;
  if(pa&&pb&&Math.abs(A.gl[i]/A.gn[i]-A.gl[j]/A.gn[j])<.2)mt++}if(cons>=10)best=Math.max(best,mt/cons)}
 const sym=cl((best-.55)/.35);
 const list=[['Proximity',prox],['Similarity',sim],['Closure',closure],['Continuity',cont],['Figure and
    ground',fg],['Symmetry',sym]];
```

```
 const det=list.filter(l=>l[1]>=.5).map(l=>l[0]);
 return{s:Math.min(20,list.reduce((a,l)=>a+l[1]*4,0)),note:det.length?'Uses '+det.join(', ').toLowerCase():'No clear
    principle yet. Try grouping, repeating or closing shapes.'}}

/* ---- overall rating (the "quality" the rest of the game already uses) ---- */
function quality(cv){const A=analyze(cv),hm=harmony(A),sh=shapesEval(),ge=gestalt(A,sh.inf);
 const cov=A.pt?Math.min(10,A.cov*30)+Math.min(5,strokes*.5):0;
 const fams=Object.keys(A.fm).filter(f=>f!=='gray'&&A.fm[f]/A.pt>=.02).length,vr=Math.min(10,fams*2.5);
 const tr=A.pt&&(A.fm[S.trend]||0)/A.pt>=.03?12:0,br=has('brush')?10:0,raw=hm.s+sh.s+ge.s+cov+vr+tr+br;
 const rep=[{k:'Colour harmony',s:hm.s,max:25,note:hm.note},{k:'Shapes and
    lines',s:sh.s,max:20,note:sh.note},{k:'Gestalt',s:ge.s,max:20,note:ge.note},{k:'Coverage and
    effort',s:cov,max:15,note:Math.round(A.cov*100)+'% of the canvas painted'},{k:'Colour
    variety',s:vr,max:10,note:fams+(fams===1?' colour family':' colour families')}];
 const bonus=[tr?`Market colour (${S.trend}) +12.`:'',br?'Gilded Brush +10.':''].filter(Boolean);
 return{q:(devRating!=null?devRating:Math.max(1,Math.min(100,Math.round(raw)))),cov:A.cov,gold:(A.fm.gold||0)/Math.max
    (1,A.pt)>=.01,rep,bonus}}
const repHtml=m=>`<div class="rep">${m.rep.map(r=>`<div
    class="rr"><span>${r.k}</span><b>${Math.round(r.s)}/${r.max}</b><div class="rb"><div style="width:${Math.round(r.s
    /r.max*100)}%"></div></div><small>${esc(r.note)}</small></div>`).join('')}${m.bonus.length?`<p class="dim"
    style="margin:6px 0 0;font-size:18px">${m.bonus.map(esc).join(' ')}</p>`:''}</div>`;

/* ---- fill bucket: scanline flood fill with tolerance + 1px halo so anti-aliased edges are covered ---- */
function bucket(ctx,sx,sy,hex){const W_=480,H_=320;sx|=0;sy|=0;if(sx<0||sy<0||sx>=W_||sy>=H_)return 0;
 const im=ctx.getImageData(0,0,W_,H_),d=im.data,si=(sy*W_+sx)*4,tr=d[si],tg=d[si+1],tb=d[si+2],[fr,fg,fb]=h2r(hex);
 if(Math.abs(tr-fr)+Math.abs(tg-fg)+Math.abs(tb-fb)<6)return 0;
 const df=i=>Math.abs(d[i]-tr)+Math.abs(d[i+1]-tg)+Math.abs(d[i+2]-tb),match=i=>df(i)<=60,seen=new
    Uint8Array(W_*H_),stack=[sx,sy];let cnt=0;
 const put=p=>{const i=p*4;d[i]=fr;d[i+1]=fg;d[i+2]=fb;d[i+3]=255};
 while(stack.length){const y=stack.pop();let x=stack.pop();while(x>=0&&!seen[y*W_+x]&&match((y*W_+x)*4))x--;x++;let
    up=false,dn=false;
  for(;x<W_&&!seen[y*W_+x]&&match((y*W_+x)*4);x++){seen[y*W_+x]=1;put(y*W_+x);cnt++;
   if(y>0){const m=!seen[(y-1)*W_+x]&&match(((y-1)*W_+x)*4);if(m&&!up){stack.push(x,y-1);up=true}else if(!m)up=false}
   if(y<H_-1){const m=!seen[(y+1)*W_+x]&&match(((y+1)*W_+x)*4);if(m&&!dn){stack.push(x,y+1);dn=true}else
    if(!m)dn=false}}}
 const grow=[];for(let y=0;y<H_;y++)for(let x=0;x<W_;x++){const p=y*W_+x;if(seen[p])continue;
  if((x>0&&seen[p-1]||x<W_-1&&seen[p+1]||y>0&&seen[p-W_]||y<H_-1&&seen[p+W_])&&df(p*4)<=130)grow.push(p)}
 grow.forEach(put);ctx.putImageData(im,0,0);return cnt}
```

Appendix B. draw.js

The new drawScreen() function.

```
function drawScreen(){clearInterval(TMR);strokes=0;used=new
    Set();SD=[];FL=[];UN=[];tool='brush';fbDone=false;col='#111111';sz=6;const P=S.panic,J=P?2.2+(S.day-6)*1.1:0;
mus(S.day===DAYS?'d10':'draw');
const hint=has('radar')?`Today the market wants <b style="border-bottom:6px solid ${FAMH[S.trend]}">${S.trend}</b>.
    Any shade of it counts. Use it for a big bonus.`:`The market wants one colour today. You do not know which. <span
    class="dim">(The Patron's Ledger would tell you.)</span>`;
W.className=P?'pw':'';
W.innerHTML=`<div class="box"><div class="px" style="font-size:12px">DAY ${S.day}: PAINT</div><p>${hint} Gold is
    flashy; collectors like it.</p>${has('eye')?'<p>Live rating: <b id="lv">0</b>/100</p>':''}${P?'<div class="bar"
    style="max-width:100%"><div id="tmb" style="width:100%;background:var(--bad)"></div></div>':''}<div id="msg"
    class="${P?'pnc':''}" style="${P?'':'color:var(--bad)'}">${P?pick(PNC):''}</div></div><canvas id="cv"
    class="${P?'nau':''}" width="480" height="320"></canvas>
<div class="row"><button data-t="brush"><U+1F58C><U+FE0F> Brush</button><button data-t="bucket"><U+1FAA3>
    Fill</button><button data-t="erase">Eraser</button><button id="un">↶ Undo</button><button
    id="clr">Clear</button></div>
<div class="row"><button data-s="3">Thin</button><button data-s="7">Mid</button><button data-s="16">Fat</button><span
    class="curw"><span class="cur" id="cur"></span><span id="curn"></span></span></div>
<div class="row"><button class="ptab" data-pm="add">✻ Additive</button><button class="ptab" data-pm="sub"><U+1F3A8>
    Subtractive</button><button id="mixb"><U+1F9EA> Mix</button></div>
<p class="c dim" id="ptip" style="margin:2px 0;font-size:18px"></p><div id="pal"></div>
<div class="box mixp" id="mixp" style="display:none"><div class="row"><button class="slot" id="sa" aria-label="Colour
    A"></button><b>+</b><button class="slot" id="sb" aria-label="Colour B"></button><b>=</b><span class="slot res"
    id="sr"></span></div><div class="row dim" style="font-size:18px">more A <input type="range" id="mr" min="10"
    max="90" value="50"> more B</div><div class="row"><button id="madd">Use this mix</button></div><p class="dim c"
    style="font-size:17px;margin:0">Tap colours above to fill slot A, then slot B. The mix follows the tab you are
    on.</p></div>
<div class="row"><input id="ti" maxlength="30" placeholder="Title (blank = untitled)" style="max-width:380px;font-
    size:20px"></div><p class="c"><button class="big" id="sub">TO THE SALEROOM</button></p>`;
const cv=$('#cv'),x=cv.getContext('2d',{willReadFrequently:true});x.fillStyle=BGHEX;x.fillRect(0,0,480,320);x.lineCap=
    x.lineJoin='round';
let mxA='#e32636',mxB='#ffd400',mxN='A',mxR=50,mxRes='#ff7f11',mxOpen=false,cs=null;
const sync=()=>{document.querySelectorAll('[data-
    t]').forEach(b=>b.classList.toggle('sel',b.dataset.t===tool));document.querySelectorAll('[data-
    pm]').forEach(b=>b.classList.toggle('sel',b.dataset.pm===PM));
 $('#cur').style.background=tool==='erase'?BGHEX:col;$('#curn').textContent=tool==='erase'?'Eraser':(CN[col]||'Mix');c
    v.style.cursor=tool==='bucket'?'cell':'crosshair'};
const mxUpd=()=>{mxRes=mixCol(mxA,mxB,mxR/100,PM);$('#sa').style.background=mxA;$('#sb').style.background=mxB;$('#sr')
    .style.background=mxRes;$('#sa').classList.toggle('nx',mxN==='A');$('#sb').classList.toggle('nx',mxN==='B')};
const sel=c=>{col=c;if(tool==='erase')tool='brush';renderPal();sync()};
const swClick=h=>{if(mxOpen){if(mxN==='A'){mxA=h;mxN='B'}else{mxB=h;mxN='A'}mxUpd()}else sel(h)};
function renderPal(){const sw=([h,n])=>`<button class="sw${h===col&&tool!=='erase'?' on':''}" data-c="${h}"
    style="background:${h}" title="${n}" aria-label="${n}"></button>`;
 $('#ptip').textContent=PAL[PM].tip;
 $('#pal').innerHTML=PAL[PM].groups.map(([g,a])=>`<div class="pgl"><span class="dim">${g}</span><div
    class="pal">${a.map(sw).join('')}</div></div>`).join('')+`<div class="pgl"><span class="dim">Neutrals and
    gold</span><div class="pal">${NEU.map(sw).join('')}</div></div>`+(MIXED.length?`<div class="pgl"><span
    class="dim">Your mixes</span><div class="pal">${MIXED.map(h=>sw([h,'Mix'])).join('')}</div></div>`:'');
 document.querySelectorAll('.sw').forEach(b=>b.onclick=()=>swClick(b.dataset.c));
 document.querySelectorAll('[data-pm]').forEach(b=>b.classList.toggle('sel',b.dataset.pm===PM))}
const live=()=>{if(has('eye')&&strokes)$('#lv').textContent=quality(cv).q};
const wipe=()=>{x.fillStyle=BGHEX;x.fillRect(0,0,480,320);strokes=0;used=new
    Set();SD=[];FL=[];UN=[];if(has('eye'))$('#lv').textContent=0};
const
    snap=()=>{UN.push({im:x.getImageData(0,0,480,320),sd:SD.length,fl:FL.length,st:strokes});if(UN.length>12)UN.shift(
    )};
document.querySelectorAll('[data-s]').forEach(b=>b.onclick=()=>sz=+b.dataset.s);
document.querySelectorAll('[data-t]').forEach(b=>b.onclick=()=>{tool=b.dataset.t;renderPal();sync()});
document.querySelectorAll('[data-pm]').forEach(b=>b.onclick=()=>{PM=b.dataset.pm;renderPal();mxUpd()});
$('#clr').onclick=wipe;
$('#un').onclick=()=>{const
    u=UN.pop();if(!u)return;x.putImageData(u.im,0,0);SD.length=u.sd;FL.length=u.fl;strokes=u.st;used=new
    Set([...SD.map(s=>s.c),...FL.map(f=>f.c)]);live();blip(200)};
$('#mixb').onclick=()=>{mxOpen=!mxOpen;$('#mixp').style.display=mxOpen?'block':'none';$('#mixb').classList.toggle('sel
    ',mxOpen);mxUpd()};
$('#sa').onclick=()=>{mxN='A';mxUpd()};$('#sb').onclick=()=>{mxN='B';mxUpd()};
$('#mr').oninput=e=>{mxR=+e.target.value;mxUpd()};
$('#madd').onclick=()=>{if(!MIXED.includes(mxRes))MIXED.push(mxRes);if(MIXED.length>10)MIXED.shift();mxOpen=false;$('#
    mixp').style.display='none';$('#mixb').classList.remove('sel');sel(mxRes);blip(420)};
renderPal();mxUpd();sync();
/* panic: the pointer drifts and trembles, so no line is straight */
const pos=e=>{const r=cv.getBoundingClientRect(),tt=performance.now();let a=(e.clientX-
    r.left)*480/r.width,b=(e.clientY-r.top)*320/r.height;if(J){a+=Math.sin(tt/260)*J*.8+(Math.random()-
    .5)*J;b+=Math.cos(tt/310)*J*.8+(Math.random()-.5)*J}return[a,b]};
cv.onpointerdown=e=>{cv.setPointerCapture(e.pointerId);const[a,b]=pos(e);
 if(tool==='bucket'){snap();const
    n=bucket(x,a,b,col);if(n){strokes++;used.add(col);FL.push({c:col,x:a,y:b,n});blip(300);live()}else
    UN.pop();return}
 drawing=true;snap();x.strokeStyle=x.fillStyle=tool==='erase'?BGHEX:col;x.lineWidth=sz;x.beginPath();x.arc(a,b,sz/2,0,
    7);x.fill();x.beginPath();x.moveTo(a,b);strokes++;
 if(tool==='erase')cs=null;else{used.add(col);cs={p:[{x:a,y:b}],c:col,z:sz};SD.push(cs)}};
cv.onpointermove=e=>{if(!drawing)return;const[a,b]=pos(e);x.lineTo(a,b);x.stroke();if(cs){const
    l=cs.p[cs.p.length-1];if(Math.hypot(a-l.x,b-l.y)>=2)cs.p.push({x:a,y:b})}};
cv.onpointerup=cv.onpointercancel=()=>{if(!drawing)return;drawing=false;cs=null;live();
 /* mid-drawing flashback (not on Day 10, which has its own track) */
 if(!fbDone&&S.day!==DAYS&&strokes>=6){fbDone=true;mus('flash',true)}};
const send=()=>{clearInterval(TMR);S.lf=0;submit(cv,$('#ti').value.trim())};
```

```
$('#sub').onclick=()=>{if(strokes<2||!used.size){$('#msg').textContent='The terminal needs a painting. Draw something
    first.';return}
/* registration glitch: ~15% of the time, never twice in a row, never while panicking */
if(!S.lf&&!P&&Math.random()<.15){S.lf=1;wipe();blip(110);$('#msg').textContent='Registration error: the terminal could
     not read your piece. Nothing was saved. Paint it again.';scrollTo(0,0);return}
send()};
if(P){const T=Math.max(30,50-(S.day-7)*5)*1000,t0=Date.now();let ln=0;
TMR=setInterval(()=>{if(!$('#tmb')){clearInterval(TMR);return}const
    f=1-(Date.now()-t0)/T;$('#tmb').style.width=Math.max(0,f*100)+'%';
if(Date.now()-ln>5000){ln=Date.now();$('#msg').textContent=pick(PNC)}
if(f<=0){clearInterval(TMR);if(strokes<2||!used.size){x.strokeStyle='#111';x.lineWidth=6;for(let i=0;i<3;i++){const
    pts=[{x:rnd(40,440),y:rnd(40,280)}];x.beginPath();x.moveTo(pts[0].x,pts[0].y);for(let j=0;j<5;j++){const q={x:rnd(
    40,440),y:rnd(40,280)};pts.push(q);x.lineTo(q.x,q.y)}x.stroke();SD.push({p:pts,c:'#111111',z:6})}strokes=Math.max(
    strokes,3);used.add('#111111')}
$('#msg').textContent='Time. Your hands stop.';setTimeout(send,700)}},100)}}
```

Appendix C. music.js

Music engine. The MUSIC object holds the embedded MP3 data (omitted here).

```
/* ===== MUSIC: tracks are embedded in this file. Credits song (Promise-Reprise): use Options > Load credits song
    ===== */
const MUSIC={/*MUSIC_DATA*/};
const VOL=.4,bg=$('#bgm');bg.volume=VOL;
let lastK='',curK='',rest='',restAt=0,want=null,fT=0;
const track=k=>MUSIC[k]||(k==='credits'?MUSIC.menu:'');
function ramp(to,ms,done){clearInterval(fT);const from=bg.volume,t0=performance.now();fT=setInterval(()=>{const
    f=Math.min(1,(performance.now()-t0)/ms);bg.volume=Math.max(0,Math.min(1,from+(to-
    from)*f));if(f>=1){clearInterval(fT);done&&done()}},40)}
function loadTrack(k,at){bg.dataset.k=k;bg.src=track(k);if(at)bg.addEventListener('loadedmetadata',()=>{try{bg.current
    Time=at}catch(e){}},{once:true});bg.play().catch(()=>{})}
function commit(){const w=want;want=null;if(!w)return;const{k,once}=w;
 if(once){if(curK===rest)restAt=bg.currentTime}else{rest=k;restAt=0}
 curK=k;bg.loop=!once;
 bg.onended=once?()=>{if(rest){curK=rest;bg.loop=true;bg.onended=null;loadTrack(rest,restAt);bg.volume=VOL}}:null;
 loadTrack(k,0);bg.volume=0;ramp(VOL,800)}
function mus(k,once){if(!once)lastK=k;if(!O.music){bg.pause();curK='';return}
 if(!track(k))return;
 if(!once&&k===curK&&!bg.paused&&!want)return;
 if(once&&k===curK)return;
 want={k,once};
 if(!bg.paused&&curK)ramp(0,350,commit);else commit()}
/* browsers block autoplay until the first tap: start the current song then */
['pointerdown','keydown'].forEach(ev=>document.addEventListener(ev,()=>{if(O.music&&lastK&&bg.paused&&!want)mus(lastK)
    },{passive:true}));
$('#mf').onchange=e=>{const f=e.target.files[0];if(f){MUSIC.credits=URL.createObjectURL(f)}};
```

Appendix D. credits.js

Credits scene.

```
/* ===== credits ===== */
let CRT=0;
function credits(){clearTimeout(CRT);H.style.display='none';W.className='';mus('credits');
 const dv=`<svg class="dv" viewBox="0 0 300 14" width="230" height="14" aria-hidden="true"><path d="M4 8Q40 0 76 8T150
     8T224 8T296 7" fill="none" stroke="#8a2b1a" stroke-width="3" stroke-linecap="round" stroke-
    dasharray="320"/></svg>`;
 const hum=[['Rudra Parashar','Ideation, Music Design'],['Prince Raj Rawat','Main Ideation, Music Finishing
    Touches'],['Rudrapratap Singh Chauhan','Testing, Ideation, Video']];
 const ai=[['Main AI assistance','Claude'],['Side AI assistance','Gemini']];
 const sky=`<svg class="bgs" viewBox="0 0 400 230" preserveAspectRatio="xMidYMid slice" aria-
    hidden="true"><defs><linearGradient id="cg" x1="0" y1="0" x2="0" y2="1"><stop offset="0" stop-
    color="#0a0c18"/><stop offset="1" stop-color="#2a1a14"/></linearGradient></defs><rect width="400" height="230"
    fill="url(#cg)"/>${Array.from({length:26},(_,i)=>`<circle class="tw" cx="${(i*53)%400}" cy="${(i*37)%110}"
    r="${i%4?.8:1.4}" fill="#ffe7a8" style="animation-delay:${(i%7)*.3}s"/>`).join('')}<polygon class="flick"
    points="330,0 400,0 400,230 190,230" fill="#f0c060"/><rect y="196" width="400" height="34" fill="#07090f"/><g
    transform="translate(24 120)" stroke="#3a2a1c" stroke-width="3" fill="none"><path d="M10 76L40 0M70 76L40 0M40
    0V76"/><rect x="8" y="24" width="64" height="38" fill="#120d0a"/></g></svg>`;
 const ST=[
  `<p class="ct">Last Bid</p><p class="cs">was made by</p>${hum.map(([n,r],i)=>`<div class="ce" style="animation-
    delay:${.8+i*1.2}s;--d:${.8+i*1.2}s"><div class="cn">${esc(n)}</div><div
    class="cro">${esc(r)}</div>${i<2?dv:''}</div>`).join('')}`,
  `<p class="ct">Shaped with help from</p>${ai.map(([r,n],i)=>`<div class="ce" style="animation-
    delay:${.8+i*1.4}s;--d:${.8+i*1.4}s"><div class="cro">${esc(r)}</div><div class="cn
    big">${esc(n)}</div>${i<1?dv:''}</div>`).join('')}`,
  `<p class="ct">Thank you for playing</p><div class="ce" style="animation-delay:.9s;--d:.9s"><p
    class="cs">${esc(S.name)} painted ten days.<br>Endings found: ${new Set(O.seen).size} of 6.</p>${dv}<p class="cs"
    style="margin-top:14px"><button class="big" id="cpa">PLAY AGAIN</button> <button id="crp">Replay
    credits</button></p></div>`];
 let i=-1;const DUR=[11000,8500];
 const key=e=>{if(e.target&&e.target.tagName==='BUTTON')return;if(e.code==='Space'||e.code==='Enter'){e.preventDefault
    ();go()}};
 const leave=()=>{clearTimeout(CRT);document.removeEventListener('keydown',key)};
 function go(n){clearTimeout(CRT);i=n!==undefined?n:Math.min(i+1,ST.length-1);
  W.innerHTML=`<div class="cr fadein" id="cr">${sky}<div class="scrw"><div class="scr">${ST[i]}</div></div><div
    class="cdots">${ST.map((_,k)=>`<span class="${k===i?'on':''}"></span>`).join('')}</div></div>${i<ST.length-1?'<p
    class="c"><button id="cnx">next ▸</button> <button id="csk">skip ▸▸</button></p>':''}`;
  $('#cr').onclick=e=>{if(e.target.tagName!=='BUTTON'&&i<ST.length-1)go()};
  if($('#cnx')){$('#cnx').onclick=()=>go();$('#csk').onclick=()=>go(ST.length-1)}
  if($('#cpa')){$('#cpa').onclick=()=>{leave();menu()};$('#crp').onclick=()=>go(0)}
  if(i<ST.length-1)CRT=setTimeout(()=>go(),DUR[i]);scrollTo(0,0)}
 document.addEventListener('keydown',key);go(0)}
```

Appendix E. style.css

Styles added in v3.

```
/* ===== v3 additions: palette, tools, rating report, credits ===== */
.pgl{text-align:center;font-size:17px;margin:3px 0}.pgl .pal{margin:2px 0 4px}.pgl .sw{width:30px;height:30px}
.ptab.sel,#mixb.sel{background:var(--ac);color:#111}
.curw{display:inline-flex;align-items:center;gap:6px;font-size:18px}.cur{width:22px;height:22px;border:3px solid
    var(--line);display:inline-block}
.mixp{max-width:420px;margin:8px auto;padding:10px}
.slot{width:44px;height:44px;border:3px solid var(--line);padding:0;box-shadow:none;display:inline-
    block}.slot.nx{outline:4px solid var(--ac);outline-offset:1px}.slot.res{box-shadow:3px 3px 0 var(--line)}
input[type=range]{width:150px;max-width:150px;padding:0;border:0;background:none;accent-color:var(--ac)}
.rep{text-align:left;margin:10px auto;max-width:380px}
.rr{display:grid;grid-template-columns:1fr auto;gap:0 8px;margin:7px 0;font-size:20px}.rr b{color:var(--ac)}
.rr .rb{grid-column:1/-1;height:8px;border:2px solid var(--line);background:var(--bg);margin:2px 0}.rb
    div{height:100%;background:var(--ac)}
.rr small{grid-column:1/-1;color:var(--dim);font-size:16px;line-height:1.15}
.cr{position:relative;min-height:560px;background:#07080d;border:3px solid var(--line);box-shadow:5px 5px 0
    var(--line);overflow:hidden;display:grid;place-items:center;padding:24px 10px 34px;cursor:pointer;margin-
    bottom:14px;font-family:Papyrus,'Papyrus Condensed',Herculanum,'Uncial
    Antiqua',fantasy;--ink:#2b1b0a;--rust:#8a2b1a}
.cr .bgs{position:absolute;inset:0;width:100%;height:100%;opacity:.8}
.scrw{position:relative;width:min(94%,500px);filter:drop-shadow(0 10px 16px rgba(0,0,0,.75))}
.scr{padding:46px 30px 42px;color:var(--ink);text-align:center;line-height:1.25;background:radial-gradient(ellipse at
    28% 18%,rgba(255,246,210,.7),transparent 62%),repeating-linear-gradient(90deg,rgba(120,85,30,.08) 0
    2px,transparent 2px 7px),repeating-linear-gradient(0deg,rgba(120,85,30,.05) 0 1px,transparent 1px
    5px),#d8c18a;box-shadow:inset 0 0 70px rgba(90,55,10,.5);clip-path:polygon(0 2%,3% 0,8% 1.5%,14% 0,22% 1.2%,30%
    0,41% 1.4%,52% 0,63% 1.3%,72% 0,83% 1.5%,92% 0,100% 1.2%,99% 14%,100% 27%,98.8% 40%,100% 55%,99% 70%,100%
    84%,98.8% 100%,90% 98.6%,80% 100%,68% 98.5%,57% 100%,45% 98.6%,33% 100%,21% 98.4%,10% 100%,0 98.5%,1.2% 85%,0
    72%,1% 58%,0 44%,1.2% 30%,0 16%)}
.ct{font-size:34px;margin:0 0 4px;color:var(--rust);letter-spacing:3px;animation:rv 2.2s both}
.cs{font-size:21px;margin:4px 0 14px;color:#4a3515}
.ce{margin:16px 0;animation:ink 1.7s both}.ce .dv{display:block;margin:12px auto 0}.dv path{animation:drw 1.6s
    calc(var(--d,0s) + .9s) both}
.cn{font-size:31px;color:var(--ink);letter-spacing:1px}.cn.big{font-size:44px;letter-spacing:4px}
.cro{font-size:20px;color:var(--rust);margin-top:2px}
.cr button{font-family:inherit;color:var(--ink);background:#e7d6a4;border-color:var(--ink);box-shadow:3px 3px 0
    var(--ink)}.cr button:hover{background:var(--rust);color:#f3e6bd}
.cdots{position:absolute;bottom:10px;left:0;right:0;display:flex;gap:8px;justify-content:center}.cdots
    span{width:9px;height:9px;transform:rotate(45deg);border:2px solid #d8c18a;opacity:.6}.cdots
    .on{background:#d8c18a;opacity:1}
@keyframes ink{from{opacity:0;filter:blur(7px);transform:translateY(8px)}}
@keyframes drw{from{stroke-dashoffset:320}}
@media(max-width:480px){.cn{font-size:25px}.cn.big{font-size:36px}.ct{font-size:27px}.scr{padding:38px 18px 36px}}
.cr button.big{font-size:19px;padding:8px 16px}
```

Appendix F1. build.py

Build script that patches the original game and embeds the audio (includes the language fixes).

```
import base64,re
src=open('/mnt/user-data/uploads/Last_Bid_v2__1_.html',encoding='utf-8').read()
def rep(old,new,cnt=1):
    global src
    assert src.count(old)==cnt,(src.count(old),old[:70])
    src=src.replace(old,new)
rd=lambda f:open(f,encoding='utf-8').read()

rep('family=Cormorant+Garamond:ital,wght@1,500&display=swap','family=Cormorant+Garamond:ital,wght@1,500&family=Uncial+
    Antiqua&display=swap')
rep('</style></head><body>',rd('style.css')+'</style></head><body>')

# music block
a=src.index('/* ===== MUSIC:');b=src.index("$('#mf').onchange=")
b=src.index('\n',b)
data=','.join("%s:'data:audio/mpeg; base64,%s'"%(k,base64.b64encode(open('mus/%s.mp3'%k,'rb').read()).decode()) for k
    in ['menu','mother','draw','flash','d10','bid','win','lose'])
src=src[:a]+rd('music.js').replace('/*MUSIC_DATA*/',data)+src[b:]

rep("if(k==='music')mus('main')","if(k==='music')mus(lastK||'menu')")
rep('''<button onclick="$('#mf').click()">Choose music file</button>''','''<button onclick="$('#mf').click()">Load
    credits song</button>''')
rep('<p>Food, bus fare and phone cost money every night.</p>','<p>Food, bus fare and phone cost money every
    night.</p><p>Your painting is rated on colour harmony, shapes and lines, and gestalt design. Mix colours, fill
    shapes with the bucket, and read the notes in the saleroom.</p>')
rep('<p><i>The market favours one colour a day. Try several until you find it.</i></p>','<p><i>The market favours one
    colour family a day (any shade counts). Try several until you find it. Buyers also judge harmony, shapes and
    design.</i></p>')
rep("trend:'#111111'","trend:'black'")
rep("$('#nm').onkeydown=e=>{if(e.key==='Enter')$('#go').click()}","$('#nm').onkeydown=e=>{if(e.key==='Enter')$('#go')
    .click()};mus('menu')}")
rep("O.g=g;save();mus('main');intro()}","O.g=g;save();intro()}")
rep("if(O.prev)await dialog(replayLines());","if(O.prev)await dialog(replayLines());mus('mother');")
rep("S.trend=pick(COLS.filter(c=>c!=='#ffffff'));","mus(S.day===DAYS?'d10':'mother');S.trend=pick(TRENDS);")

# palette line -> block A
a=src.index('const COLS=');b=src.index('\n',a)
src=src[:a]+rd('blockA.js')+src[b:]
# old quality
a=src.index('function quality(cv){\nconst d=') if 'function quality(cv){\nconst d=' in src else src.index("function
    quality(cv){const d=cv.getContext")
b=src.index('\nconst PNC=',a)
src=src[:a]+src[b+1:]
# drawScreen
a=src.index('function drawScreen(){');b=src.index('\nconst ROAST=',a)
src=src[:a]+rd('draw.js')+src[b+1:]

rep("async function submit(cv,title){const m=quality(cv),img=cv.toDataURL();","async function
    submit(cv,title){mus('bid');const m=quality(cv),img=cv.toDataURL();")
rep('<div class="bar"><div style="width:${m.q}%"></div></div><p class="dim"><i>* ${rateLine(m.q)}</i></p></div><div
    id="rm"></div>','<div class="bar"><div style="width:${m.q}%"></div></div>${repHtml(m)}<p class="dim"><i>*
    ${rateLine(m.q)}</i></p></div><div id="rm"></div>')
rep("mus(win?'good':'bad');","mus(win?'win':'mother');")
rep("}else{\nawait dialog([\n['Robin','Then tomorrow.","}else{\nmus('lose');\nawait dialog([\n['Robin','Then
    tomorrow.")
rep('''<p><button class="big" onclick="menu();mus('main')">PLAY AGAIN</button></p>''','''<p><button class="big"
    onclick="credits()">CREDITS ▸</button></p>''')
rep('/* ===== replay: dev quick-jump ===== */',rd('credits.js')+'\n/* ===== replay: dev quick-jump ===== */')

rep('Collectors pay most. Critics give you a tool. Students and locals are the people.','Collectors pay the most.
    Critics give you a tool. Students and locals pay what they can.')
rep('so all you carry is food, bus fare, your phone.','so all you carry is food, bus fare and your phone.')
rep('I have been doing numbers since the scan.','I have been running numbers since the scan.')
rep("'* Tears. The first of your lives, for both of you. They are salt and they are warm, and they are exactly as
    ordinary as anything.'","'* Tears, the first for both of you. They are salt and warm, and as ordinary as
    anything.'")
rep('Robin and the guilt in this story are fiction.','Robin and the guilt in this story are fictional.')
rep("['Robin','Text from Robin: \"She ate half","['','Text from Robin: \"She ate half")
rep("['Robin','Text from Robin: \"Mum asked where","['','Text from Robin: \"Mum asked where")
rep("return[['Robin',DYN[topType()][i]]]","return[['',DYN[topType()][i]]]")
rep('and how likely things end well.','and how likely it is to end well.')
open('/mnt/user-data/outputs/Last_Bid_v3.html','w',encoding='utf-8').write(src)
print(len(src)/1e6,'MB')
```

Appendix F2. test.py

Headless browser test for drawing, fill, mix, rating and credits.

```
import math,json
from playwright.sync_api import sync_playwright
with sync_playwright() as p:
    b=p.chromium.launch(args=['--autoplay-policy=no-user-gesture-required'])
    pg=b.new_page(viewport={'width':520,'height':1100})
    errs=[];pg.on('console',lambda m:errs.append(m.text) if m.type=='error' else None);pg.on('pageerror',lambda
    e:errs.append(str(e)))
    pg.goto('file:///mnt/user-data/outputs/Last_Bid_v3.html');pg.wait_for_timeout(800)
    pg.evaluate("self.FAST=1;S.name='Test';S.day=3;S.lf=1;hud();drawScreen()")
    pg.wait_for_timeout(300)
    box=pg.locator('#cv').bounding_box();sx=box['width']/480;sy=box['height']/320
    def P(x,y):return (box['x']+x*sx,box['y']+y*sy)
    def stroke(pts):
        global box,sx,sy
        pg.evaluate('scrollTo(0,0)');box=pg.locator('#cv').bounding_box();sx=box['width']/480;sy=box['height']/320
        pg.mouse.move(*P(*pts[0]));pg.mouse.down()
        for q in pts[1:]:pg.mouse.move(*P(*q),steps=4)
        pg.mouse.up()
    pg.click('[data-c="#ff0000"]')
    stroke([(240+70*math.cos(t/20*6.283),160+70*math.sin(t/20*6.283)) for t in range(21)])  # circle

    pg.click('[data-t="bucket"]');pg.click('[data-
    c="#ffff00"]');pg.evaluate('scrollTo(0,0)');pg.mouse.click(*P(240,160))
    pg.click('[data-t="brush"]');pg.click('[data-c="#0000ff"]')
    stroke([(20,300),(460,300)])
    stroke([(20,20),(120,20)]);stroke([(20,40),(120,40)]);stroke([(20,60),(120,60)])
    pg.click('[data-pm="sub"]');pg.click('[data-c="#2fa84f"]')
    stroke([(350,60+i*3+20*math.sin(i/6)) for i in range(0,60)] and [(340+i*2,80+20*math.sin(i/6)) for i in
    range(0,60)])
    # mixer
    pg.click('#mixb');pg.click('[data-c="#e32636"]');pg.click('[data-c="#ffd400"]')
    print('mix res',pg.evaluate("mxRes") if False else
    pg.locator('#sr').evaluate("e=>getComputedStyle(e).backgroundColor"))
    pg.click('#madd')
    stroke([(380,250),(430,250),(430,200),(380,200),(380,251)])
    r=pg.evaluate("(()=>{const m=quality($('#cv'));return
    {q:m.q,rep:m.rep,bonus:m.bonus,strokes,SD:SD.length,FL:FL.length,types:SD.map(s=>strokeInfo(s).t),mus:curK}})()")
    print(json.dumps(r,indent=1))
    pg.screenshot(path='shot_draw.png',full_page=True)
    pg.click('#sub');pg.wait_for_timeout(500)
    pg.evaluate("self.FAST=1");pg.wait_for_timeout(6000)
    pg.screenshot(path='shot_sale.png',full_page=True)
    print('mus after submit',pg.evaluate("[curK,bg.dataset.k,lastK]"))
    pg.evaluate("ET='R';EW=true;credits()");pg.wait_for_timeout(3500)
    pg.screenshot(path='shot_cr1.png')
    pg.click('#csk');pg.wait_for_timeout(2500);pg.screenshot(path='shot_cr3.png')
    pg.evaluate("go=null");
    print('ERRORS',errs)
    b.close()
```

Appendix F3. test2.py

Headless test of both endings into credits.

```
from playwright.sync_api import sync_playwright
def run(win):
    with sync_playwright() as p:
        b=p.chromium.launch(args=['--autoplay-policy=no-user-gesture-required'])
        pg=b.new_page(viewport={'width':520,'height':900})
        errs=[];pg.on('pageerror',lambda e:errs.append(str(e)))
        pg.goto('file:///mnt/user-data/outputs/Last_Bid_v3.html');pg.wait_for_timeout(600)
        print('menu mus',pg.evaluate("curK"))

    pg.evaluate("self.FAST=1;S.name='Test';O.ts=0;S.day=10;S.money=9000;S.rep={R:2,S:0,P:0};hud();prev('R',%s)"%('true
    ' if win else 'false'))
        seen=set();done=False
        for i in range(900):
            seen.add(pg.evaluate("curK"))
            if pg.locator('#bn').count() and pg.locator('#bn').is_visible():
                pg.wait_for_timeout(120)
                try: pg.click('#bn',timeout=500)
                except: pass
            elif pg.locator('button:has-text("CREDITS")').count():
                pg.click('button:has-text("CREDITS")');done=True;break
            else: pg.keyboard.press('Space')
            pg.wait_for_timeout(15)
        pg.wait_for_timeout(500)
        print('win' if win else 'lose','reached credits:',done,'tracks
    seen:',seen,'now:',pg.evaluate("curK"),'errors:',errs)
        b.close()
run(True);run(False)
```

Appendix G. Complete game source (Last_Bid_v3.html)

The full final file. The eight embedded MP3 data URIs are replaced with a placeholder, because they are about 10 MB of text. Lines are wrapped for print; continuation lines are indented four spaces.

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Last Bid</title>
<link rel="stylesheet" href="https://fonts.googleapis.com/css2?family=Press+Start+2P&family=VT323&family=Cormorant+Gar
    amond:ital,wght@1,500&family=Uncial+Antiqua&display=swap">
<style>
:root{--bg:#0e1015;--fg:#e9e4d6;--card:#161920;--line:#e9e4d6;--ac:#c9a24b;--dim:#8d8a80;--good:#6fcf97;--bad:#d1566a}
:root[data-theme=light]{--bg:#e8e2d3;--fg:#1a1a1f;--card:#f5f0e4;--line:#1a1a1f;--ac:#8f6a1a;--dim:#6b665a;--
    good:#1f8a4c;--bad:#b02a40}
*{box-sizing:border-box}html,body{margin:0;min-height:100%}
body{background:var(--bg);color:var(--fg);font:22px/1.2 'VT323',monospace;background-image:radial-gradient(var(--dim)
    .5px,transparent .5px);background-size:16px 16px;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-
    area-inset-bottom,0px)}
h1,h2,.px{font-family:'Press Start 2P',monospace;line-height:1.5;font-weight:400}
.serif{font-family:'Cormorant Garamond',serif;font-style:italic;font-size:24px}
#hud{display:none;position:sticky;top:0;z-index:5;gap:8px 14px;flex-wrap:wrap;align-items:center;justify-
    content:space-between;padding:8px 12px;background:var(--card);border-bottom:3px solid var(--line);font-size:20px}
#hud b{color:var(--ac)}
#wrap{max-width:720px;margin:0 auto;padding:14px}
.box,.dbox,.tc{background:var(--card);border:3px solid var(--line);box-shadow:5px 5px 0 var(--line),inset 0 0 0 4px
    var(--card),inset 0 0 0 5px var(--ac)}
.box{padding:16px;margin-bottom:14px}
button,input{font:inherit;color:var(--fg)}
button{background:var(--card);border:3px solid var(--line);padding:6px 14px;cursor:pointer;box-shadow:3px 3px 0
    var(--line)}
button:hover,button:focus-visible{background:var(--ac);color:#111;outline:none}
button:active{transform:translate(3px,3px);box-shadow:none}
button.big{font-family:'Press Start 2P',monospace;font-size:13px;padding:12px 18px}
button:disabled{opacity:.4;cursor:not-allowed}
input{background:var(--bg);border:3px solid var(--line);padding:8px;width:100%;max-width:300px;text-align:center;font-
    size:26px}
.c{text-align:center}.dim{color:var(--dim)}
.scene{height:230px;border:3px solid var(--line);background:var(--bg);position:relative;overflow:hidden;margin-
    bottom:12px}
.scene svg{position:absolute;bottom:0;width:100%;height:100%;animation:fin .8s}@keyframes
    fin{from{opacity:0}to{opacity:1}}
.scene.cine::before,.scene.cine::after{content:"";position:absolute;left:0;right:0;height:16px;background:#000;z-
    index:2}.scene.cine::before{top:0}.scene.cine::after{bottom:0}
.dbox{padding:14px;min-height:130px;cursor:pointer;position:relative}.dbox.hs{padding-bottom:42px}
.dbox .sp{font-family:'Press Start 2P',monospace;font-size:12px;margin-bottom:8px;color:var(--ac)}
.dbox p{margin:0;font-size:26px}.dbox i{display:block;text-align:right;color:var(--dim);font-size:18px;margin-top:6px}
.sk{position:absolute;left:10px;bottom:8px;font-size:16px;padding:1px 10px;box-shadow:2px 2px 0 var(--line)}
.pam{background:#d9cfae;color:#241a04;padding:28px 22px;transform:rotate(-.6deg);box-shadow:0 0 0 3px #241a04,0 0 0
    7px #d9cfae,0 0 0 8px #241a04,8px 8px 0 #000}
.pam h2{font-size:14px;margin:0 0 4px;color:#5e0f1a}.pam .serif{margin:0 0 12px;color:#3a2c10}.pam li{margin:9px
    0}.pam hr{border:0;border-top:2px dashed #241a04;opacity:.5}
.stamp{display:inline-block;border:3px solid #5e0f1a;color:#5e0f1a;padding:2px 10px;transform:rotate(-5deg);font-
    family:'Press Start 2P',monospace;font-size:11px;margin-top:6px}
.tag{display:inline-block;padding:0 7px;border:2px solid;font-size:17px;margin-left:6px;vertical-align:middle}
canvas#cv{width:100%;max-width:480px;aspect-ratio:3/2;background:#fdfdf8;border:3px solid var(--line);touch-
    action:none;cursor:crosshair;display:block;margin:0 auto}
.pal{display:flex;gap:6px;flex-wrap:wrap;justify-content:center;margin:10px 0}
.sw{width:34px;height:34px;padding:0;box-shadow:none}.sw.on{outline:4px solid var(--ac);outline-offset:1px}
.row{display:flex;gap:8px;flex-wrap:wrap;justify-content:center;align-items:center;margin:8px 0}
.npc{display:flex;gap:10px;align-items:flex-start;border:3px solid;padding:8px 10px;margin:8px
    0;background:var(--bg);animation:fin .5s}
.npc em{font-style:normal;font-size:32px}
.bar{height:16px;border:3px solid var(--line);background:var(--bg);margin:6px auto;max-width:300px}.bar
    div{height:100%;background:var(--ac)}
.art{max-width:300px;width:100%;border:8px solid var(--ac);outline:3px solid var(--line);display:block;margin:6px
    auto}
.flash{animation:fl .6s}@keyframes fl{0%{background:var(--ac);color:#111}}
button.sel{background:var(--ac);color:#111}
.ovl{position:fixed;inset:0;z-index:30;background:rgba(0,0,0,.7);display:none;place-items:center;padding:14px}
.orow{display:flex;justify-content:space-between;align-items:center;gap:8px;flex-wrap:wrap;margin:10px
    0}.orow>span{min-width:110px}.orow .row{margin:0}.orow button{font-size:18px;padding:3px 10px}
/* animation */
.rain{stroke-dasharray:7 9;animation:rain .5s linear infinite}@keyframes rain{to{stroke-dashoffset:-16}}
.flick{animation:flick 3.1s steps(1) infinite}@keyframes
    flick{0%,100%{opacity:.22}8%{opacity:.08}10%{opacity:.22}46%{opacity:.14}72%{opacity:.1}74%{opacity:.22}}
.flick2{animation:f2 3.3s steps(1) infinite}@keyframes
    f2{0%,100%{opacity:1}12%{opacity:.6}14%{opacity:1}60%{opacity:.75}62%{opacity:1}}
.steam{animation:steam 2.2s ease-in-out infinite}@keyframes
    steam{0%,100%{transform:translateY(0);opacity:.9}50%{transform:translateY(-4px);opacity:.4}}
.bob{transform-box:fill-box;animation:bob 2.6s ease-in-out infinite}@keyframes bob{50%{transform:translateY(-3px)}}
.tw{animation:tw 2s ease-in-out infinite}@keyframes tw{50%{opacity:.15}}
.mote{animation:mote 4.5s linear infinite}@keyframes
    mote{0%{transform:translateY(0);opacity:0}20%{opacity:.9}100%{transform:translateY(-110px);opacity:0}}
.sun{transform-box:fill-box;animation:sun 5s ease-out both}@keyframes sun{from{transform:translateY(40px)}}
.tc{min-height:340px;background:#05060a;color:var(--ac);display:grid;align-content:center;justify-items:center;text-
    align:center;padding:24px;cursor:pointer;margin-bottom:14px;position:relative;overflow:hidden}
```

```
.tcs{position:absolute;inset:0;width:100%;height:100%;opacity:.55}.tx{position:relative}
.t1{font-size:30px;letter-spacing:6px}.t1 span{display:inline-block;animation:blur 1.4s both}@keyframes
    blur{from{opacity:0;filter:blur(10px);transform:translateY(10px)}}
.t2{color:#e9e4d6;font-size:26px;animation:tin 1.2s 2.4s both}.t3{color:#8d8a80;animation:tin 1.2s 3.4s
    both}@keyframes tin{from{opacity:0;transform:translateY(8px)}}
.rv{animation:rv 2.4s both}@keyframes rv{from{opacity:0;letter-spacing:18px;filter:blur(6px)}to{letter-spacing:2px}}
.fadein{animation:fin 2s}
body.noanim *,body.noanim *::before,body.noanim *::after{animation:none!important}
@media(prefers-reduced-motion:reduce){*{animation:none!important}}
@media(max-width:480px){body{font-size:20px}.dbox p{font-size:23px}h1{font-size:18px}.t1{font-size:22px}}
/* ===== v2 additions ===== */
.dbox.meta{border-color:var(--bad)}.dbox.meta .sp{color:var(--bad);animation:gl 1.3s steps(2) infinite}.dbox.meta
    p{color:var(--bad)}
@keyframes gl{50%{transform:translateX(2px);opacity:.65}}
.pw .box,.pw .pal,.pw .row{animation:shk .2s linear infinite}
@keyframes shk{0%,100%{transform:translate(0,0)}25%{transform:translate(1.6px,-1px)}50%{transform:translate(-
    1px,1.6px)}75%{transform:translate(1px,1px)}}
.shk2{animation:shk2 .5s linear 1}@keyframes shk2{0%,100%{transform:translate(0)}20%{transform:translate(-
    7px,3px)}40%{transform:translate(6px,-4px)}60%{transform:translate(-5px,-2px)}80%{transform:translate(4px,3px)}}
.nau{animation:nau 3.4s ease-in-out infinite}@keyframes
    nau{0%,100%{transform:rotate(0);filter:blur(0)}30%{transform:rotate(.7deg) scale(1.01);filter:blur(1.3px) hue-
    rotate(12deg)}60%{transform:rotate(-.8deg);filter:blur(.4px)}}
.pnc{color:var(--bad);font-style:italic}
.ecg{stroke-dasharray:60 30;animation:ecg 1.3s linear infinite}@keyframes ecg{to{stroke-dashoffset:-90}}
.blame{font-family:'Press Start 2P',monospace;font-size:13px;line-height:2.2;text-align:center;transition:opacity
    1.6s}
.blame mark{background:var(--bad);color:#fff;padding:2px 8px;animation:bl 2.2s steps(1) infinite}@keyframes
    bl{50%{background:#7a1020}}
.comic{border:3px dashed var(--dim);min-height:220px;display:grid;place-items:center;color:var(--dim);margin:10px 0
    18px;padding:10px}.comic img{max-width:100%;display:block}
.note{font-size:17px;color:var(--dim);line-height:1.25;max-width:520px;margin:10px auto 0}
/* ===== v3 additions: palette, tools, rating report, credits ===== */
.pgl{text-align:center;font-size:17px;margin:3px 0}.pgl .pal{margin:2px 0 4px}.pgl .sw{width:30px;height:30px}
.ptab.sel,#mixb.sel{background:var(--ac);color:#111}
.curw{display:inline-flex;align-items:center;gap:6px;font-size:18px}.cur{width:22px;height:22px;border:3px solid
    var(--line);display:inline-block}
.mixp{max-width:420px;margin:8px auto;padding:10px}
.slot{width:44px;height:44px;border:3px solid var(--line);padding:0;box-shadow:none;display:inline-
    block}.slot.nx{outline:4px solid var(--ac);outline-offset:1px}.slot.res{box-shadow:3px 3px 0 var(--line)}
input[type=range]{width:150px;max-width:150px;padding:0;border:0;background:none;accent-color:var(--ac)}
.rep{text-align:left;margin:10px auto;max-width:380px}
.rr{display:grid;grid-template-columns:1fr auto;gap:0 8px;margin:7px 0;font-size:20px}.rr b{color:var(--ac)}
.rr .rb{grid-column:1/-1;height:8px;border:2px solid var(--line);background:var(--bg);margin:2px 0}.rb
    div{height:100%;background:var(--ac)}
.rr small{grid-column:1/-1;color:var(--dim);font-size:16px;line-height:1.15}
.cr{position:relative;min-height:560px;background:#07080d;border:3px solid var(--line);box-shadow:5px 5px 0
    var(--line);overflow:hidden;display:grid;place-items:center;padding:24px 10px 34px;cursor:pointer;margin-
    bottom:14px;font-family:Papyrus,'Papyrus Condensed',Herculanum,'Uncial
    Antiqua',fantasy;--ink:#2b1b0a;--rust:#8a2b1a}
.cr .bgs{position:absolute;inset:0;width:100%;height:100%;opacity:.8}
.scrw{position:relative;width:min(94%,500px);filter:drop-shadow(0 10px 16px rgba(0,0,0,.75))}
.scr{padding:46px 30px 42px;color:var(--ink);text-align:center;line-height:1.25;background:radial-gradient(ellipse at
    28% 18%,rgba(255,246,210,.7),transparent 62%),repeating-linear-gradient(90deg,rgba(120,85,30,.08) 0
    2px,transparent 2px 7px),repeating-linear-gradient(0deg,rgba(120,85,30,.05) 0 1px,transparent 1px
    5px),#d8c18a;box-shadow:inset 0 0 70px rgba(90,55,10,.5);clip-path:polygon(0 2%,3% 0,8% 1.5%,14% 0,22% 1.2%,30%
    0,41% 1.4%,52% 0,63% 1.3%,72% 0,83% 1.5%,92% 0,100% 1.2%,99% 14%,100% 27%,98.8% 40%,100% 55%,99% 70%,100%
    84%,98.8% 100%,90% 98.6%,80% 100%,68% 98.5%,57% 100%,45% 98.6%,33% 100%,21% 98.4%,10% 100%,0 98.5%,1.2% 85%,0
    72%,1% 58%,0 44%,1.2% 30%,0 16%)}
.ct{font-size:34px;margin:0 0 4px;color:var(--rust);letter-spacing:3px;animation:rv 2.2s both}
.cs{font-size:21px;margin:4px 0 14px;color:#4a3515}
.ce{margin:16px 0;animation:ink 1.7s both}.ce .dv{display:block;margin:12px auto 0}.dv path{animation:drw 1.6s
    calc(var(--d,0s) + .9s) both}
.cn{font-size:31px;color:var(--ink);letter-spacing:1px}.cn.big{font-size:44px;letter-spacing:4px}
.cro{font-size:20px;color:var(--rust);margin-top:2px}
.cr button{font-family:inherit;color:var(--ink);background:#e7d6a4;border-color:var(--ink);box-shadow:3px 3px 0
    var(--ink)}.cr button:hover{background:var(--rust);color:#f3e6bd}
.cdots{position:absolute;bottom:10px;left:0;right:0;display:flex;gap:8px;justify-content:center}.cdots
    span{width:9px;height:9px;transform:rotate(45deg);border:2px solid #d8c18a;opacity:.6}.cdots
    .on{background:#d8c18a;opacity:1}
@keyframes ink{from{opacity:0;filter:blur(7px);transform:translateY(8px)}}
@keyframes drw{from{stroke-dashoffset:320}}
@media(max-width:480px){.cn{font-size:25px}.cn.big{font-size:36px}.ct{font-size:27px}.scr{padding:38px 18px 36px}}
.cr button.big{font-size:19px;padding:8px 16px}
</style></head><body>
<div id="hud"></div><div id="wrap"></div><div id="opt" class="ovl"></div>
<audio id="bgm" loop></audio><input type="file" id="mf" accept="audio/*" hidden>
<button id="dvb" onclick="dvToggle()"
    style="display:none;position:fixed;left:8px;bottom:8px;z-index:20">DEV</button><div id="dvp" class="box"
    style="display:none;position:fixed;left:8px;bottom:52px;z-index:20;max-height:62vh;overflow:auto;max-
    width:340px;font-size:18px"></div>
<script>
const $=s=>document.querySelector(s),W=$('#wrap'),H=$('#hud'),pick=a=>a[Math.random()*a.length|0],sleep=ms=>new Promis
    e(r=>setTimeout(r,self.FAST?30:ms)),rnd=(a,b)=>a+Math.random()*(b-
    a),esc=t=>String(t).replace(/[&<>"]/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;'}[c]));

/* ===== MUSIC: tracks are embedded in this file. Credits song (Promise-Reprise): use Options > Load credits song
    ===== */
const MUSIC={menu:'data:audio/mpeg; base64,<<MP3 DATA OMITTED>>',mother:'data:audio/mpeg; base64,<<MP3 DATA
    OMITTED>>',draw:'data:audio/mpeg; base64,<<MP3 DATA OMITTED>>',flash:'data:audio/mpeg; base64,<<MP3 DATA
    OMITTED>>',d10:'data:audio/mpeg; base64,<<MP3 DATA OMITTED>>',bid:'data:audio/mpeg; base64,<<MP3 DATA
    OMITTED>>',win:'data:audio/mpeg; base64,<<MP3 DATA OMITTED>>',lose:'data:audio/mpeg; base64,<<MP3 DATA OMITTED>>'};
```

```
const VOL=.4,bg=$('#bgm');bg.volume=VOL;
let lastK='',curK='',rest='',restAt=0,want=null,fT=0;
const track=k=>MUSIC[k]||(k==='credits'?MUSIC.menu:'');
function ramp(to,ms,done){clearInterval(fT);const from=bg.volume,t0=performance.now();fT=setInterval(()=>{const
    f=Math.min(1,(performance.now()-t0)/ms);bg.volume=Math.max(0,Math.min(1,from+(to-
    from)*f));if(f>=1){clearInterval(fT);done&&done()}},40)}
function loadTrack(k,at){bg.dataset.k=k;bg.src=track(k);if(at)bg.addEventListener('loadedmetadata',()=>{try{bg.current
    Time=at}catch(e){}},{once:true});bg.play().catch(()=>{})}
function commit(){const w=want;want=null;if(!w)return;const{k,once}=w;
 if(once){if(curK===rest)restAt=bg.currentTime}else{rest=k;restAt=0}
 curK=k;bg.loop=!once;
 bg.onended=once?()=>{if(rest){curK=rest;bg.loop=true;bg.onended=null;loadTrack(rest,restAt);bg.volume=VOL}}:null;
 loadTrack(k,0);bg.volume=0;ramp(VOL,800)}
function mus(k,once){if(!once)lastK=k;if(!O.music){bg.pause();curK='';return}
 if(!track(k))return;
 if(!once&&k===curK&&!bg.paused&&!want)return;
 if(once&&k===curK)return;
 want={k,once};
 if(!bg.paused&&curK)ramp(0,350,commit);else commit()}
/* browsers block autoplay until the first tap: start the current song then */
['pointerdown','keydown'].forEach(ev=>document.addEventListener(ev,()=>{if(O.music&&lastK&&bg.paused&&!want)mus(lastK)
    },{passive:true}));
$('#mf').onchange=e=>{const f=e.target.files[0];if(f){MUSIC.credits=URL.createObjectURL(f)}};


/* ===== options ===== */
const O={sound:1,music:1,ts:28,anim:1,theme:'dark',name:'',g:'m',seen:[],prev:null,plays:0};
try{Object.assign(O,JSON.parse(localStorage.getItem('lastbid2')||'{}'))}catch(e){}
const save=()=>{try{localStorage.setItem('lastbid2',JSON.stringify(O))}catch(e){}};
const
    apply=()=>{document.documentElement.dataset.theme=O.theme;document.body.classList.toggle('noanim',!O.anim)};apply(
    );
const fx=t=>{const f=S.g==='f',p=S.prev||{},pf=p.g==='f';return t.replaceAll('{n}',S.name).replaceAll('{He}',f?'She':'
    He').replaceAll('{he}',f?'she':'he').replaceAll('{him}',f?'her':'him').replaceAll('{prev}',p.n||'').replaceAll('{p
    Him}',pf?'her':'him').replaceAll('{pHis}',pf?'her':'his').replaceAll('{pHe}',pf?'she':'he')};
const optShown=()=>$('#opt').style.display==='grid';
const seg=(l,k,ch)=>`<div class="orow"><span>${l}</span><div class="row">${ch.map(([n,v])=>`<button
    class="${O[k]===v?'sel':''}"
    onclick="optSet('${k}',${JSON.stringify(v).replace(/"/g,'&quot;')})">${n}</button>`).join('')}</div></div>`;
function optOpen(){const o=$('#opt');o.style.display='grid';o.onclick=e=>{if(e.target===o)optClose()};optRender()}
function optClose(){$('#opt').style.display='none'}
function optRender(){const g=H.style.display==='flex';$('#opt').innerHTML=`<div class="box" style="max-
    width:440px;width:100%;max-height:90vh;overflow:auto;margin:0"><h2 style="font-size:14px;margin-
    top:0">OPTIONS</h2>${seg('Sound
    fx','sound',[['On',1],['Off',0]])}${seg('Music','music',[['On',1],['Off',0]])}${seg('Text speed','ts',[['Slow',45]
    ,['Normal',28],['Fast',12],['Instant',0]])}${seg('Animations','anim',[['On',1],['Off',0]])}${seg('Theme','theme',[
    ['Dark','dark'],['Light','light']])}<div class="row"><button onclick="$('#mf').click()">Load credits
    song</button><button onclick="howTo()">How to play</button></div>${g?'<div class="row"><button
    onclick="quit()">Quit to main menu</button></div>':''}<div class="row"><button class="big"
    onclick="optClose()">CLOSE</button></div></div>`}
function optSet(k,v){O[k]=v;apply();save();optRender();blip(380);if(k==='music')mus(lastK||'menu')}
function quit(){if(confirm('Quit this run and go back to the main menu?')){optClose();menu()}}
function howTo(){const o=$('#opt');o.style.display='grid';o.onclick=e=>{if(e.target===o)optClose()};o.innerHTML=`<div
    class="box" style="max-width:440px;width:100%;max-height:90vh;overflow:auto;margin:0"><h2 style="font-
    size:14px;margin-top:0">HOW TO PLAY</h2><p>You have <b>$${START.toLocaleString()}</b> and need
    <b>$${GOAL.toLocaleString()}</b> for Mum's operation by Day ${DAYS}.</p><p>Each day: paint one piece. Five buyers
    rate it and bid. You choose who gets it. The state keeps 15% of every sale.</p><p>Food, bus fare and phone cost
    money every night.</p><p>Your painting is rated on colour harmony, shapes and lines, and gestalt design. Mix
    colours, fill shapes with the bucket, and read the notes in the saleroom.</p><p>Who you sell to most decides who
    stands beside you at the end, and how likely it is to end well.</p><p class="dim">Draw with mouse or finger. Click
     / Space / Enter advances the story.</p><div class="row"><button class="big" onclick="optClose()">GOT
    IT</button></div></div>`}
document.addEventListener('keydown',e=>{if(e.key==='Escape'&&optShown())optClose()});

const START=8500,GOAL=10000,DAYS=10,TAX=.15,TAXLO=.08,NECT=9,NEC=[['food',5],['bus to the
    hospital',2],['phone',2]];let S={};
const fresh=()=>S={name:S.name||O.name||'Rudra',g:S.g||O.g||'m',day:1,money:START,rep:{R:0,S:0,P:0},tools:[],trend:'bl
    ack',rat:[],net:[],lf:0,cur:0,panic:0,panicDay:0,prev:O.prev||null,same:false};fresh();
function blip(f=260){if(!O.sound)return;try{const a=blip.a||(blip.a=new AudioContext()),o=a.createOscillator(),g=a.cre
    ateGain();o.type='square';o.frequency.value=f+Math.random()*30;g.gain.value=.03;o.connect(g);g.connect(a.destinati
    on);o.start();o.stop(a.currentTime+.04)}catch(e){}}
const TOOLS={radar:['<U+1F4E1>',"Patron's Ledger",'Shows the colour the market wants each
    day.'],brush:['<U+1F58C><U+FE0F>','Gilded Brush','+10 rating on every
    piece.'],eye:['<U+1F441><U+FE0F>',"Connoisseur's Eye",'Live rating preview while you
    paint.'],charm:['<U+1F91D>','Silver Tongue','All offers +10%.'],coupon:['<U+1F9FE>','Tax Waiver','Tax on your
    sales drops from 15% to 8%.'],letter:['✉<U+FE0F>',"Patron's Letter",'+$80 right now.']};
const has=k=>S.tools.includes(k);
function
    hud(){H.style.display='flex';H.innerHTML=`<span>${S.g==='f'?'<U+1F469><U+200D><U+1F3A8>':'<U+1F468><U+200D><U+1F3A
    8>'} ${esc(S.name)}</span><span>Day <b>${S.day}</b>/${DAYS}</span><span title="The doctor's clock"><U+1F56F>
    <b>${Math.max(0,DAYS-S.day+1)}</b> days left</span><span>Need <b>$${Math.max(0,GOAL-S.money).toLocaleString()}</b>
     more</span><span title="Sold to: Collectors / Critics / Students"><U+1F4B0>${S.rep.R} <U+1F9D0>${S.rep.S}
    <U+1F393>${S.rep.P}</span><span>${S.tools.map(k=>`<span title="${TOOLS[k][1]}:
    ${TOOLS[k][2]}">${TOOLS[k][0]}</span>`).join(' ')||'<span class="dim">no tools</span>'}</span><button
    onclick="optOpen()">⚙ Options</button>`}
/* ===== scenes ===== */
const BGX=`<rect x="-600" width="1600" height="230" fill="var(--bg)"/><rect x="-600" y="190" width="1600" height="40"
    fill="var(--line)" opacity=".2"/>`;
const RN=`<g class="rain" stroke="#7fb2e5" stroke-width="1.5" opacity=".55"><path
    d="${Array.from({length:16},(_,i)=>`M${i*26+8} ${i*5%20}V230`).join('')}"/></g>`;
const HS=(lit,c)=>`<path d="M110 190V110L200 52L290 110V190Z" fill="#1b1f2c" stroke="#3a4054" stroke-width="3"/><path
```

```
    d="M96 112L200 40L304 112" stroke="#3a4054" stroke-width="5" fill="none"/><rect x="180" y="70" width="40"
    height="34" fill="${lit}" class="${c}"/><path d="M200 70v34M180 87h40" stroke="#1b1f2c" stroke-width="3"/><rect
    x="132" y="130" width="30" height="26" fill="#10131c"/><rect x="238" y="130" width="30" height="26"
    fill="#10131c"/><rect x="186" y="146" width="28" height="44" fill="#10131c"/>`;
const WIN=`<defs><clipPath id="wc"><rect x="30" y="30" width="100" height="92"/></clipPath></defs><rect x="30" y="30"
    width="100" height="92" fill="#16202c"/><g clip-path="url(#wc)"><g class="rain" stroke="#7fb2e5" stroke-width="2"
    opacity=".75"><path d="M40 30V122M56 30V122M72 30V122M88 30V122M104 30V122M120 30V122"/></g></g><rect x="28"
    y="28" width="104" height="96" fill="none" stroke="var(--line)" stroke-width="4"/><path d="M80 28v96M28 76h104"
    stroke="var(--line)" stroke-width="3"/><rect x="22" y="122" width="116" height="8" fill="var(--line)"/><g
    class="bob"><ellipse cx="104" cy="115" rx="11" ry="8" fill="#8d8a80" stroke="var(--line)" stroke-
    width="2"/><circle cx="114" cy="107" r="5" fill="#8d8a80" stroke="var(--line)" stroke-width="2"/><circle cx="116"
    cy="106" r="1.4" fill="var(--line)"/><path d="M119 108l5 2-5 1z" fill="var(--ac)"/></g>`;
const EASEL=`<g transform="translate(160 64)"><path d="M10 126L40 10M70 126L40 10M40 10V126" stroke="var(--line)"
    stroke-width="4"/><rect x="6" y="34" width="68" height="50" fill="#f2ecdd" stroke="var(--line)" stroke-
    width="3"/><path d="M14 72q12-28 24-8t26-18" stroke="var(--ac)" stroke-width="3" fill="none"/></g>`;
const DESK=`<rect x="250" y="150" width="150" height="10" fill="var(--line)"/><rect x="262" y="160" width="8"
    height="30" fill="var(--line)"/><rect x="380" y="160" width="8" height="30" fill="var(--line)"/><polygon
    class="flick" points="354,100 384,108 410,150 300,150" fill="var(--ac)" opacity=".22"/><g transform="translate(340
     150)"><path d="M0 0h26" stroke="var(--line)" stroke-width="5"/><path d="M12 0L6-38L30-60" stroke="var(--line)"
    stroke-width="3" fill="none"/><polygon points="20,-66 48,-50 38,-38 10,-54" fill="var(--ac)" stroke="var(--line)"
    stroke-width="3"/></g><g transform="translate(262 124)"><path d="M0 0h30l-5 26h-20z" fill="#9a4a38"
    stroke="var(--line)" stroke-width="3"/><path class="steam" d="M10-9q-6-10 0-18M21-9q6-10 0-18" stroke="var(--dim)"
     stroke-width="3" fill="none"/></g><g transform="translate(304 139)"><rect width="30" height="11" fill="#f4f0e4"
    stroke="var(--line)" stroke-width="2"/><path d="M0 0l15 7 15-7" fill="none" stroke="var(--line)" stroke-
    width="2"/><circle cx="15" cy="7" r="3" fill="var(--bad)"/></g>`;
const hair=()=>S.g==='f'?`<path d="M23 96Q21 75 40 75Q59 75 57 96L58 120H51V92Q40 84 29 92V120H22Z"
    fill="var(--fg)"/>`:`<path d="M24 93Q23 77 40 77Q57 77 56 93Q48 85 40 85Q32 85 24 93Z" fill="var(--fg)"/>`;
const FACE=`<circle cx="34" cy="97" r="2" fill="var(--fg)"/><circle cx="46" cy="97" r="2" fill="var(--fg)"/><path
    d="M35 105q5 -2 10 0" stroke="var(--fg)" stroke-width="2" fill="none"/>`;
const LB=(x,a)=>`<text x="${x}" y="62" text-anchor="middle" font-family="VT323" font-size="18"
    fill="var(--fg)">${esc(a)}</text>`;
const PL=(x,a)=>`<g transform="translate(${x} 40)"><circle cx="40" cy="95" r="16" fill="var(--card)"
    stroke="var(--line)" stroke-width="4"/>${hair()}${FACE}<rect x="22" y="113" width="36" height="37"
    fill="var(--line)"/>${LB(40,a)}</g>`;
const RB=(x,b)=>`<g transform="translate(${x} 40)"><circle cx="40" cy="95" r="16" fill="var(--card)"
    stroke="var(--line)" stroke-width="4"/><path d="M24 93Q26 76 42 78Q56 80 56 93Q50 84 40 85Q30 85 24 93Z"
    fill="var(--ac)"/>${FACE}<rect x="22" y="113" width="36" height="37" fill="#7a2e3a" stroke="var(--line)" stroke-
    width="3"/><rect x="28" y="111" width="24" height="6" fill="var(--ac)"/>${LB(40,b)}</g>`;
const SV=x=>`<svg viewBox="0 0 400 230" preserveAspectRatio="xMidYMax meet">${x}</svg>`;
const ICON={R:'<U+1F4B0>',S:'<U+1F4F0>',P:'<U+1F5BC><U+FE0F>'};let ET='R',EW=true;
const endSc=(t,w)=>SV(`<defs><linearGradient id="sk" x1="0" y1="0" x2="0" y2="1"><stop offset="0" stop-
    color="${w?'#3b4a7a':'#0b0e18'}"/><stop offset="1" stop-
    color="${w?'#f0b868':'#262c3a'}"/></linearGradient></defs><rect width="400" height="230"
    fill="url(#sk)"/>${w?'<circle class="sun" cx="200" cy="168" r="34" fill="#ffd98a"/>':'<circle cx="330" cy="46"
    r="16" fill="#aab" opacity=".45"/>'}<rect y="190" width="400" height="40" fill="${w?'#2a2230':'#07090f'}"/>${HS(w?
    '#ffd98a':'#10131c',w?'flick2':'')}${w?Array.from({length:9},(_,i)=>`<circle class="mote" cx="${60+i*35}"
    cy="${200+i%3*8}" r="2" fill="#ffe7a8" style="animation-delay:${i*.5}s"/>`).join(''):RN}<text class="bob" x="318"
    y="176" font-size="28">${ICON[t]}</text>`);
const SC={
studio:()=>SV(BGX+WIN+EASEL+DESK),
house:()=>SV(`<rect width="400" height="230" fill="#0b0e18"/><circle cx="330" cy="46" r="18" fill="#e9e4d6"
    opacity=".85"/><rect y="190" width="400" height="40" fill="#07090f"/>${HS('#f0c060','flick2')}${RN}`),
two:(a,b)=>SV(BGX+WIN+EASEL+PL(230,a)+RB(300,b)),
notice:()=>SV(BGX+`<polygon class="flick" points="120,0 280,0 400,230 0,230" fill="var(--ac)" opacity=".12"/><g
    transform="rotate(-3 200 115)"><rect x="95" y="14" width="210" height="190" fill="#e9e1c6" stroke="var(--line)"
    stroke-width="4"/><text x="200" y="44" text-anchor="middle" font-family="Press Start 2P,monospace" font-size="10"
    fill="#5e0f1a">ASHGROVE BANK</text><path d="M112 56h176M112 66h176M112 76h120" stroke="#241a04" stroke-width="2"
    opacity=".5"/><text x="200" y="118" text-anchor="middle" font-family="Press Start 2P,monospace" font-size="18"
    fill="#241a04">$10,000</text><text x="200" y="142" text-anchor="middle" font-family="Press Start 2P,monospace"
    font-size="10" fill="#241a04">WITHIN 10 DAYS</text><g transform="rotate(-8 200 175)"><rect x="124" y="158"
    width="152" height="30" fill="none" stroke="#b01e2f" stroke-width="3"/><text x="200" y="179" text-anchor="middle"
    font-family="Press Start 2P,monospace" font-size="11" fill="#b01e2f">FORECLOSURE</text></g></g>`),
hall:()=>SV(`<rect width="400" height="230" fill="#1a1520"/><rect width="60" height="230" fill="#5a1a26"/><rect
    x="340" width="60" height="230" fill="#5a1a26"/><rect y="190" width="400" height="40" fill="#2a2030"/><path
    d="M200 0V34" stroke="var(--ac)" stroke-width="3"/><path d="M160 62Q200 22 240 62Z" fill="var(--ac)"/><g
    fill="#ffe7a8">${[168,184,200,216,232].map((x,i)=>`<circle class="tw" cx="${x}" cy="${68+i%2*6}" r="3"
    style="animation-delay:${i*.4}s"/>`).join('')}</g><rect x="170" y="140" width="60" height="50" fill="#3a2a18"
    stroke="var(--ac)" stroke-width="3"/><rect x="204" y="128" width="26" height="8" fill="var(--ac)"
    transform="rotate(-20 204 128)"/>`)};
const SCN=k=>k==='two'?SC.two(S.name,'Robin'):k==='end'?endSc(ET,EW):SC[k]();

/* ===== v2 scenes: hospital, corridor, door, letter ===== */
const MONI=f=>`<rect x="332" y="20" width="60" height="42" fill="#0a1a14" stroke="var(--line)" stroke-width="3"/><path
     ${f?'':'class="ecg"'} d="${f?'M336 42H388':'M336 42h10l4-12 6 24 4-12h8l4-8 4 8h6'}"
    stroke="${f?'#d1566a':'#6fcf97'}" stroke-width="2" fill="none"/>`;
const HWIN=`<defs><clipPath id="hw"><rect x="90" y="24" width="80" height="64"/></clipPath></defs><rect x="90" y="24"
    width="80" height="64" fill="#16202c"/><g clip-path="url(#hw)"><g class="rain" stroke="#7fb2e5" stroke-width="2"
    opacity=".7"><path d="M100 24V88M114 24V88M128 24V88M142 24V88M156 24V88"/></g></g><rect x="88" y="22" width="84"
    height="68" fill="none" stroke="var(--line)" stroke-width="4"/><path d="M130 22v68" stroke="var(--line)" stroke-
    width="3"/>`;
const BEDF=`<rect x="14" y="150" width="176" height="10" fill="var(--line)"/><rect x="14" y="160" width="6"
    height="30" fill="var(--line)"/><rect x="184" y="160" width="6" height="30" fill="var(--line)"/><rect x="20"
    y="108" width="42" height="20" fill="#e9e4d6" stroke="var(--line)" stroke-width="3"/>`;
const MUMH=`<circle cx="42" cy="114" r="13" fill="var(--card)" stroke="var(--line)" stroke-width="3"/><path d="M29
    112Q30 98 44 99Q56 100 55 112Q49 105 42 105Q34 105 29 112Z" fill="#aaa"/><circle cx="37" cy="116" r="1.5"
    fill="var(--fg)"/><circle cx="47" cy="116" r="1.5" fill="var(--fg)"/><rect x="20" y="126" width="170" height="26"
    fill="#2f3a52" stroke="var(--line)" stroke-width="3"/>`;
const DOC=x=>`<g transform="translate(${x} 40)"><circle cx="40" cy="95" r="16" fill="var(--card)" stroke="var(--line)"
     stroke-width="4"/><path d="M25 92Q28 78 41 79Q54 80 56 92Q48 86 40 86Q32 86 25 92Z" fill="#6a6a70"/>${FACE}<rect
    x="22" y="113" width="36" height="37" fill="#e9e4d6" stroke="var(--line)" stroke-width="3"/><path d="M40 113v37"
    stroke="var(--line)" stroke-width="2"/>${LB(40,'Doctor')}</g>`;
```

```
SC.hosp=()=>SV(BGX+HWIN+BEDF+MUMH+PL(176,S.name)+RB(246,'Robin')+MONI(0));
SC.hospd=()=>SV(BGX+HWIN+BEDF+MUMH+PL(160,S.name)+RB(222,'Robin')+DOC(300)+MONI(0));
SC.hflat=()=>SV(BGX+HWIN+BEDF+`<rect x="20" y="106" width="170" height="46" fill="#cfd3dc" opacity=".9"
    stroke="var(--line)" stroke-width="3"/>`+MONI(1)+`<path d="M362 62V150Q362 176 340 180" stroke="var(--dim)"
    stroke-width="3" fill="none"/><rect x="330" y="178" width="14" height="7" fill="var(--dim)"/><rect width="400"
    height="230" fill="#000" opacity=".25"/>`);
SC.corr=()=>SV(`<rect width="400" height="230" fill="#0f151b"/><rect y="190" width="400" height="40"
    fill="#1a232c"/><rect width="400" height="14" fill="#1a232c"/><rect x="170" y="14" width="60" height="5"
    fill="#cfe" class="flick2"/><rect x="10" y="50" width="52" height="140" fill="#1b2430" stroke="var(--line)"
    stroke-width="3"/><rect x="296" y="56" width="64" height="134" fill="#1b2a3a" stroke="var(--line)" stroke-
    width="3"/><rect x="304" y="66" width="48" height="62" fill="#7fb2e5" class="flick2" opacity=".5"/><rect x="304"
    y="140" width="48" height="10" fill="#0a0e14"/>`+PL(100,S.name)+RB(184,'Robin'));
SC.door=()=>SV(`<rect width="400" height="230" fill="#0b0e18"/><rect y="190" width="400" height="40"
    fill="#07090f"/><polygon points="152,190 248,190 310,230 90,230" fill="#f0c060" opacity=".08"/><rect x="150"
    y="30" width="100" height="160" fill="#1b1f2c" stroke="#3a4054" stroke-width="4"/><circle cx="234" cy="116" r="4"
    fill="var(--ac)"/><rect x="152" y="187" width="96" height="3" fill="#f0c060" class="flick2"/>`);
SC.letter=()=>SV(BGX+`<g transform="rotate(-2 200 115)"><rect x="120" y="14" width="160" height="196" fill="#e9e1c6"
    stroke="var(--line)" stroke-width="4"/><text x="140" y="42" font-family="VT323" font-size="24"
    fill="#241a04">Robin -</text><g stroke="#241a04" stroke-width="2" opacity=".5"><path d="M140 62h110M140 80h120M140
     98h96M140 116h116M140 134h88M140 152h110M140 170h60"/></g></g>`);
/* ===== dialogue ===== */
function dialog(lines,scene,skip){return new Promise(res=>{let i=0,t,full,typing=false;
if(!scene&&lines[0][2])scene=SCN(lines[0][2]);
W.innerHTML=(scene?`<div class="scene${skip?' cine':''}">${scene}</div>`:'')+`<div class="dbox${skip?' hs':''}"
    id="db"><div class="sp"></div><p></p><i>click / space</i></div>`;
const db=$('#db'),sp=db.querySelector('.sp'),p=db.querySelector('p'),sc=W.querySelector('.scene');
function fin(){clearInterval(t);typing=false;document.removeEventListener('keydown',key)}
function show(){const[w,x,k,f]=lines[i];if(k&&sc)sc.innerHTML=SCN(k);full=fx(x);const
    wf=fx(w);sp.textContent=wf;db.classList.toggle('meta',w==='???');
p.style.fontStyle=(!w||w.includes('('))?'italic':'normal';p.textContent='';clearInterval(t);
if(f==='shake'&&sc){sc.classList.remove('shk2');void sc.offsetWidth;sc.classList.add('shk2')}
const pitch=w==='???'?80:w.startsWith('Robin')?170:w.startsWith('Mum')?230:w===='Doctor'?130:300;
if(!O.ts){p.textContent=full;typing=false;return}
let n=0;typing=true;t=setInterval(()=>{p.textContent=full.slice(0,++n);if(n%2)blip(pitch);if(n>=full.length){clearInte
    rval(t);typing=false}},O.ts)}
function
    next(){if(typing){clearInterval(t);p.textContent=full;typing=false;return}if(++i>=lines.length){fin();res()}else
    show()}
const key=e=>{if(optShown())return;if(e.code==='Space'||e.code==='Enter'){e.preventDefault();next()}};
document.addEventListener('keydown',key);db.onclick=next;
if(skip){const sk=document.createElement('button');sk.className='sk';sk.textContent='skip
    ▸▸';sk.onclick=e=>{e.stopPropagation();fin();res()};db.appendChild(sk)}
show()})}

/* ===== menu / dev ===== */
let TMR=0,FORCEPANIC=false;
function menu(){clearInterval(TMR);H.style.display='none';W.className='';const p=O.prev;
W.innerHTML=`<div class="box c"><h1>LAST BID</h1><p class="serif dim">Ten days. One bed in Room 214. Whatever you can
    paint.</p>${p?`<p class="dim" style="font-size:18px">${p.w?'Welcome back.':'...welcome back.'}</p>`:''}<p>Name the
     player:</p><input id="nm" maxlength="14" placeholder="Rudra" autocomplete="off"
    value="${esc(S.name==='Rudra'?'':S.name)}"><p class="dim" id="hn" style="min-height:24px;font-size:18px;margin:4px
     0"></p><p class="dim" style="margin-bottom:4px">Gender</p><div class="row" id="gd"><button data-g="m">♂
    Male</button><button data-g="f">♀ Female</button></div><p><button class="big" id="go">START</button></p><div
    class="row"><button onclick="optOpen()">⚙ Options</button><button onclick="howTo()">How to play</button><button
    onclick="dev()">Dev menu</button></div><p class="dim" style="font-size:18px;margin-bottom:0">Endings found: ${new
    Set(O.seen).size}/6</p><p class="note">Content notes: terminal illness, grief, death, and suicide (never shown on
    screen). If this is a heavy week, it is fine to play another day.</p></div>`;
const gd=()=>document.querySelectorAll('#gd button').forEach(b=>b.classList.toggle('sel',b.dataset.g===S.g));gd();
document.querySelectorAll('#gd button').forEach(b=>b.onclick=()=>{S.g=b.dataset.g;gd();blip(360)});
$('#nm').oninput=()=>{const
    v=$('#nm').value.trim().toLowerCase();$('#hn').textContent=(p&&v&&v===p.n.toLowerCase())?(p.w?'That name
    again.':'The same name. Are you sure?'):''};$('#nm').oninput();
$('#go').onclick=()=>{const n=$('#nm').value.trim()||'Rudra',g=S.g;fresh();S.name=n;S.g=g;S.same=!!(O.prev&&O.prev.n.t
    oLowerCase()===n.toLowerCase());O.name=n===='Rudra'?'':n;O.g=g;save();intro()};$('#nm').onkeydown=e=>{if(e.key==='E
    nter')$('#go').click()};mus('menu')}
let DEV=false,devRating=null;
function dev(){DEV=true;$('#dvb').style.display='block';$('#dvp').style.display='block';dvRender()}
function dvToggle(){const p=$('#dvp');p.style.display=p.style.display==='block'?'none':'block';dvRender()}
function dvRender(){const p=$('#dvp');if(p&&p.style.display==='block')p.innerHTML=devHtml()}
const dm=n=>{S.money+=n;hud();dvRender()},dt=a=>{S.tools=a?Object.keys(TOOLS):[];hud();dvRender()},dr=k=>{if(k)S.rep[k
    ]++;else S.rep={R:0,S:0,P:0};hud();dvRender()};
function drate(c){const v=$('#drv').value;devRating=(c||v==='')?null:Math.max(1,Math.min(100,+v));dvRender()}
function dgo(){S.day=+$('#dd').value;hud();startDay()}
function devHtml(){return `<div class="px" style="font-size:11px">DEV TOOLS</div><p class="dim">Endings (win / fail),
    current odds:</p>${[['R','<U+1F4B0> Golden Cage'],['S','<U+1F9D0> Long Review'],['P','<U+1F393> Street
    Gallery']].map(([t,n])=>`<div class="row"><b>${n} ${Math.round(odds(t))}%</b><button
    onclick="prev('${t}',true)">Win</button><button onclick="prev('${t}',false)">Fail</button></div>`).join('')}<hr><p
     class="dim">Money: $${S.money.toLocaleString()} (day ${S.day})</p><div class="row"><button
    onclick="dm(-100)">-100</button><button onclick="dm(100)">+100</button><button
    onclick="dm(500)">+500</button><button onclick="dm(1500)">+1500</button></div><div class="row"><button
    onclick="dt(true)">All tools</button><button onclick="dt(false)">Clear tools</button></div><p class="dim">Sales (R
     ${S.rep.R} / S ${S.rep.S} / P ${S.rep.P}):</p><div class="row"><button onclick="dr('R')">R+1</button><button
    onclick="dr('S')">S+1</button><button onclick="dr('P')">P+1</button><button onclick="dr()">Reset</button></div><p
    class="dim">Force art rating ${devRating!=null?'(ON: '+devRating+')':'(off)'}:</p><div class="row"><input id="drv"
     type="number" min="1" max="100" style="max-width:90px;font-size:18px"><button
    onclick="drate()">Set</button><button onclick="drate(1)">Off</button></div><div class="row"><select id="dd"
    style="font:inherit">${[1,2,3,4,5,6,7,8,9,10].map(n=>`<option
    ${n===S.day?'selected':''}>${n}</option>`).join('')}</select><button onclick="dgo()">Go to day</button></div><div
    class="row"><button onclick="FORCEPANIC=!FORCEPANIC;dvRender()">Panic: ${FORCEPANIC?'ON':'off'}</button><button
    onclick="self.FAST=!self.FAST;dvRender()">Fast: ${self.FAST?'ON':'off'}</button><button onclick="menu()">Main
    menu</button></div>`}
```

```
function prev(t,w){S.money=w?GOAL+30:GOAL-380;ending(t,w)}

/* ===== intro ===== */
async function titleCard(){W.innerHTML=`<div class="tc"><svg class="tcs" viewBox="0 0 400 230"
    preserveAspectRatio="xMidYMax slice"><rect y="190" width="400" height="40"
    fill="#07090f"/>${HS('#f0c060','flick2')}${RN}</svg><div class="tx"><div class="px t1">${[...'LAST
    BID'].map((c,i)=>`<span style="animation-delay:${.4+i*.2}s">${c===' '?' ':c}</span>`).join('')}</div><p
    class="t2">Room 214. Ten days. $${GOAL.toLocaleString()}.</p><p class="t3">${esc(S.name)},
    painter</p></div></div>`;await Promise.race([sleep(6000),new Promise(r=>W.firstChild.onclick=r)])}
const TN={R:'collectors',S:'critics',P:'students'};
function replayLines(){const p=O.prev,same=S.same,L=[],k=(O.plays||1)+1;
if(!p.w&&!same)L.push(['???','Oh. A new name.'],['???','The last one was {prev}. You remember what happened to {pHim}.
     Or you remember being there.'],['???','{prev} does not get to start over. That is the strange part. You
    do.'],['???','You pressed a button and a mother, a brother, a whole house came back, and none of them asked
    why.'],['???','I am not angry. I just thought someone should say {prev}\'s name before you begin.']);
else if(!p.w&&same)L.push(['???','{n}.'],['???','The same name. The same spelling.'],['???','People usually change it.
     It is easier the second time if the name is not the one on the letter.'],['???','You did not change
    it.'],['???','...Fine. Then you know where it went wrong. I will not tell Robin.']);
else if(p.w&&!same)L.push(['???','A new name. The last one, {prev}, got {pHis} mother through the
    surgery.'],['???','She did not live forever. That was never on the menu. She lived, and that was the
    point.'],['???','So why come back? To see if it was luck? Or to see if you can buy another week?']);
else L.push(['???','{n}. You already did this one right.'],['???','So this is not about winning. You want to see it
    again. I understand. I will not ask which part.']);
if(k>=4)L.push(['???',`This is attempt ${k}. I keep count so you do not have to.`]);
return L}
async function intro(){await titleCard();
if(O.prev)await dialog(replayLines());mus('mother');
const dj=O.prev?[['Robin','Sorry. Strange feeling. As if we have stood in this corridor
    before.'],['{n}','...Yeah.']]:[];
await dialog([
['','* St. Anselm\'s Hospital, Room 214. Rain on the window. A monitor counting in a patient green.','hospd'],
['Doctor','The scans are back. I would rather be direct than gentle, if that is all right.'],
['Robin','Please.'],
['Doctor','Without the operation, your mother has about ten days before her breathing begins to struggle. After that
    we can keep her comfortable. We cannot do more.'],
['{n}','And with it?'],
['Doctor',`With it, she has a real chance. The surgery and the first month of care come to $${GOAL.toLocaleString()},
    paid before we can schedule. I wish the number were smaller.`],
['','* He says something else, something kind. You nod in the right places. You do not hear any of it.'],
['','* The door clicks shut. For a while nobody uses the room\'s silence.','hosp'],
['Mum','Stop looking at the door, both of you. It will not open with better news.'],
['Mum','Come here. Sit. Robin, you too. You hover like a nurse and nobody is paying you.'],
['Robin','Mum.'],
['Mum','Your father asked me once to end it. Did I ever tell you? The last winter. The pain was so bad he could not
    hold a spoon.'],
['Mum','He begged me. Not once. I said no, and I have never known whether no was love or cowardice.'],
['Mum','I think about his hands. Thirty years in that attic, and brushes were the only thing they ever held gently.'],
['Mum','{n}, you hold a brush the way he did. Thumb tucked in. You do not know you do it. He would be so proud to see
    you up there, doing what he did.'],
['Mum','And Robin. You sit with strangers on the worst day of their lives and still come home and make tea. I do not
    know where you got that. Not from me.'],
['Mum','I am proud of you both. I should have said it on an ordinary day. This is a bad day for it. I am saying it
    anyway.'],
['Mum','Now I have to say an unfair thing. You will not like it.'],
['Mum','If it comes to it, if the ten days run out, I want it to be fast. I cannot do the slow part again. I watched
    it once. I cannot.'],
['{n}','No.'],
['Robin','Mum, no. If we find the money you live. It is a number. Numbers can be found.'],
['{n}','We will find it. Do not ask us that again.'],
['Mum','...Just in case.'],
['','* ......'],['','* ......'],['','* ......'],
['Mum (to herself)','(It will not happen. I know how numbers like that behave. Nobody finds ten thousand dollars in
    ten days.)'],
['Mum (to herself)','(But they will spend the days trying. Together. That is the only thing I still have to give
    them.)'],
['Mum (to herself)','(Look at them. They are both so much his. I am so proud of them it is hard to breathe around
    it.)'],
['','* The corridor. A vending machine hums. Neither of you buys anything.','corr'],
...dj,
['Robin',`I have been running numbers since the scan. I can put in $${START.toLocaleString()}. Savings, and I sold the
     car this morning.`],
['{n}','You sold the car.'],
['Robin','I will also cover your rent and the big bills, so all you carry is food, bus fare and your phone. That is
    not generosity. That is so you can work.'],
['Robin',`That leaves $${(GOAL-START).toLocaleString()}.`],
['{n}','I paint pictures nobody asked for. I have not sold a canvas in a year.'],
['Robin','There is an auction house across the river. Halden House. Anyone can bring work. Collectors, critics,
    students. They bid on the spot.'],
['Robin','I got you a seller pass. One piece a day, ten days. And before you ask, yes, there is a tax. Fifteen percent
     of every sale goes to the state. I checked. It is not negotiable.'],
['{n}','So I need more than fifteen hundred in sales.'],
['Robin','...Yes. A bit more.'],
['{n}','If I had taken commissions last year, we would not be standing here.'],
['Robin','Do not.'],
['{n}','Tell me it is not on me.'],
['','* He opens his mouth. He closes it. It is perhaps a second. You will remember the second.'],
['Robin','It is not on you. Go home. Sleep. Tomorrow you paint.'],
['','* The vending machine hums. In the room behind you, the monitor keeps
    counting.','studio']],SCN('hospd'),true);pamphlet()}
function pamphlet(){hud();W.innerHTML=`<div class="pam"><h2>HALDEN HOUSE</h2><p class="serif">Notes for first-time
    sellers.</p><p><b>Reach $${GOAL.toLocaleString()} by Day ${DAYS}.</b> You begin with the
```

```
    $${START.toLocaleString()} Robin put in. Paint one piece a day. Five buyers come to look. They rate it out of 100
    and make offers. <b>You choose who takes it.</b> The state keeps <b>15%</b> of every sale. Food, bus fare and
    phone cost $${NECT} a night.</p><hr><ul>
<li><U+1F642> <b>Locals</b> pay fair money for honest work. Safe.</li>
<li><U+1F4B0> <b>Collectors</b> pay the most. They buy the name more than the painting.</li>
<li><U+1F9D0> <b>Critics</b> pay little but hand you a <b>tool</b>. Harsh scorers.</li>
<li><U+1F393> <b>Students</b> can barely afford it. Tiny offers, real feeling.</li></ul><hr>
<p>Who you sell to most decides who stands beside you at the end. Choose with care.</p><p><i>The market favours one
    colour family a day (any shade counts). Try several until you find it. Buyers also judge harmony, shapes and
    design.</i></p><span class="stamp">NO REFUNDS</span></div><p class="c"><button class="big" id="s">BEGIN DAY
    ${S.day}</button></p>`;$('#s').onclick=startDay}

/* ===== days: the lore between painting days (4-5 lines each) ===== */
const DYN={R:['Text from Robin: "Collector money is clean money. The hospital does not ask whose wall it hangs
    on."','Text from Robin: "Your collector sent flowers to the front desk. Not to Mum. To the desk. I think it is how
     they say sorry."','Text from Robin: "A collector\'s assistant called. Asked if you do portraits. I said you do
    whatever the day needs."'],
S:['Text from Robin: "A critic asked about you at the saleroom. Called you someone to watch. I said you were someone
    to feed."','Text from Robin: "They read your piece aloud in the room. People applauded. I stood there not knowing
    where to put my hands."','Text from Robin: "Someone in print wrote five words about you. A hand worth following.
    It is on the fridge."'],
P:['Text from Robin: "A student left a note for you at the desk. It says: it helped my week."','Text from Robin: "The
    baker down the road asked after Mum. She sent bread. Mum ate a crumb and gave a thumbs up like a pilot."','Text
    from Robin: "Three students came to the hospital. They did not come in. They sat in the lobby an hour and left a
    jar of coins with the guard."']};
function dyn(d){if(!S.rep.R&&!S.rep.S&&!S.rep.P)return[];const i=d===3?0:d===6?1:2;return[['',DYN[topType()][i]]]}
function EVN(d){const D={
1:[['','* Day 1. Room 214, seven in the morning. Mum is sitting up, offended by the hospital
    porridge.','hosp'],['Mum','Robin says you start today. Do not paint it for me. Paint it the way your father did,
    like you owe nobody anything.'],['{n}','I will paint it like I owe everybody.'],['Mum','That is worse. But I will
    take it.'],['','* Her grip is still strong when she squeezes your hand. You will remember that it was strong.']],
2:[['','* Day 2. The attic smells of turpentine and cold tea. You paint with the door open so you can hear the
    phone.','studio'],['','Text from Robin: "She ate half the toast. Tell the toast it is a hero."'],['','* The phone
    sits on the desk, face up. You check it every time the lamp flickers, which is every time.'],['','* On the shelf,
    your father\'s last canvas. A field, one corner of sky left grey. He never came back to finish it.'],['','* You
    think: if I were faster. You do not think it as a sentence. It sits in the hand that holds the brush.']],
3:[['','* Day 3. Mum tells the story of Dad\'s first exhibition. She has told it before. You let
    her.','hosp'],['Mum','Eleven people came. Nine were relatives. One was lost, and one was there for the wine. He
    painted over every canvas the next morning.'],['Mum','I made him coffee and said, now we can start. He laughed so
    hard he got paint in his hair.'],['Mum','It is a good story. You do not need to tell it well. You just need to
    keep telling it. Promise me.'],['{n}','I promise.'],...dyn(3)],
4:[['','* Day 4. Two in the morning, the corridor. The vending machine is the only thing awake.','corr'],['Robin','I
    tell people for a living that it is not their fault. I have a whole voice for it. Soft. Slow.'],['Robin','I tried
    it on myself in the car. It sounded like a stranger.'],['{n}','Robin.'],['Robin','I am fine. I am saying it out
    loud so it is not in the car anymore.']],
5:[['','* Day 5. Halfway. A nurse has pinned a paper snowflake to Mum\'s curtain. It is not
    winter.','hosp'],['Mum','Open the window. I want to hear the rain being rude.'],['','* You open it. The room fills
     with buses and wet leaves.'],['Mum','Your father drew you once, you know. You were six. You had painted the dog
    blue and he was furious. Then he hung it in the hall.'],['Mum','I want you to remember that you were allowed to
    paint the dog blue.']],
6:[['','* Day 6. You catch yourself counting in paintings now. Four more. Each one a day. Each one a day she does not
    get back.','studio'],['','* Her breathing has a sound now. A small dry catch, like a page stuck in a
    book.'],['','Text from Robin: "Mum asked where you were. I said at work. She said: good. Let him."'],['','* Let
    him. As if painting were something you were doing instead of her.'],['','* You pick up the brush. Your hand is
    already a little unsteady. You tell yourself it is the cold.'],...dyn(6)],
7:[['','* Day 7. The doctor takes you aside in the corridor. "The timeline may shorten," he says. "I want you
    prepared."','corr'],['Doctor','Some patients hold on. Some do not. I do not want to promise you more days than
    there are.'],['Mum','Do not look at the clock. Look at me. I am bad at being looked at, so it will be a short
    exercise.','hosp'],['','* She tries to laugh. It catches. The oxygen line hisses and you stop breathing until she
    does again.'],['','* You will count that breath later. You will count all of them.']],
8:[['','* Day 8. An hour of the old Mum. Clear eyes, sharp tongue, hair she insisted on brushing
    herself.','hosp'],['Mum','If I go early, it is not your fault. Either of you. That is the one thing I will say
    twice.'],['Mum','If I go early, it is not your fault.'],['','* She says it twice, and you count the second time.
    You are good at counting now.'],['Robin','...Mum, please stop saying it like it is already settled.']],
9:[['','* Day 9. The corridor again. Robin hands you the awful hospital tea. You both hold it and neither
    drinks.','corr'],['Robin','Whatever happens tomorrow, we did the thing. We tried. Yes?'],['{n}','We have not done
    it yet.'],['Robin','...No. We have not.'],['','* He rubs his eyes with the heel of his hand, the way Dad did after
     a long night on a canvas. You have never told him that.'],...dyn(9)],
10:[['','* Day 10. Mum can barely speak. She presses something into your palm: the smallest of your father\'s
    brushes.','hosp'],['Mum','Go. Paint something that is yours.'],['{n}','I will be back before dark.'],['Mum','Of
    course you will.'],['','* She closes her eyes. Not sleep, not yet. You stand there a minute longer than you have
    time for.']]};return D[d]}
function metaDay(){const p=O.prev;if(!p)return null;const d=S.day;
if(d===3)return['???',`Last time, ${p.n} sold to the ${TN[p.t]} most. I wonder if you will.`];
if(d===5)return['???',`${p.n} finished with $${p.m.toLocaleString()}. You are at $${S.money.toLocaleString()}.`];
if(d===7&&p.pd)return['???',`${p.n}'s hands started shaking on day ${p.pd}. Yours are fine. For now.`];
if(d===9&&!p.w)return['???','{prev} is not in this corridor. You are. I wanted you to notice the difference.'];return
    null}
async function startDay(){hud();mus(S.day===DAYS?'d10':'mother');S.trend=pick(TRENDS);S.panic=panicNow();if(S.panic&&!
    S.panicDay)S.panicDay=S.day;
const L=[...EVN(S.day)],m=metaDay();if(m)L.push(m);await dialog(L);drawScreen()}
function panicNow(){if(FORCEPANIC)return 1;if(S.day<7)return 0;const
    avg=S.net.length?S.net.reduce((a,b)=>a+b,0)/S.net.length:140,left=DAYS-S.day+1,proj=S.money+left*(avg-
    NECT);return(proj<GOAL*.985||odds(topType())<34)?1:0}
/* ===== v3: colour palette (additive + subtractive), mixing, pixel analysis, rating ===== */
const PAL={
add:{tip:'Light adds up: red + green = yellow. Mix all three and you head for white.',groups:[
 ['Primary',[['#ff0000','Red'],['#00ff00','Green'],['#0000ff','Blue']]],
 ['Secondary',[['#ffff00','Yellow'],['#00ffff','Cyan'],['#ff00ff','Magenta']]],
 ['Tertiary',[['#ff8000','Orange'],['#80ff00','Chartreuse'],['#00ff80','Spring
    green'],['#0080ff','Azure'],['#8000ff','Violet'],['#ff0080','Rose']]]]},
```

```
sub:{tip:'Pigment soaks up light: blue + yellow = green. Mix too much and it turns muddy.',groups:[
 ['Primary',[['#e32636','Red'],['#ffd400','Yellow'],['#1f4fd8','Blue']]],
 ['Secondary',[['#ff7f11','Orange'],['#2fa84f','Green'],['#7a3db8','Purple']]],
 ['Tertiary',[['#f2501f','Red-orange'],['#ffaa00','Amber'],['#9acd32','Yellow-
    green'],['#0fa3a3','Teal'],['#4b3fc4','Indigo'],['#b32d7a','Red-violet']]],
 ['Printer inks (CMY)',[['#00aeef','Cyan ink'],['#ec008c','Magenta ink'],['#fff200','Yellow ink']]]]};
const NEU=[['#111111','Black'],['#555555','Dark grey'],['#999999','Grey'],['#d0d0d0','Light
    grey'],['#ffffff','White'],['#7b4a2a','Brown'],['#d4af37','Gold']];
const CN={};[...Object.values(PAL).flatMap(p=>p.groups.flatMap(g=>g[1])),...NEU].forEach(([h,n])=>{CN[h]=n});
let PM='add',MIXED=[],tool='brush',SD=[],FL=[],UN=[],fbDone=false;
const BGHEX='#fdfdf8',GOLD='#d4af37';
const FAMH={red:'#e63946',orange:'#ff8c1a',yellow:'#f5c518',green:'#2fb344',cyan:'#14b8c8',blue:'#2a6fdb',purple:'#8e4
    4ad',magenta:'#e0359e',black:'#111111',gold:'#d4af37'};
const TRENDS=Object.keys(FAMH);
const h2r=h=>[1,3,5].map(i=>parseInt(h.slice(i,i+2),16));
const r2h=(r,g,b)=>'#'+[r,g,b].map(v=>Math.max(0,Math.min(255,Math.round(v))).toString(16).padStart(2,'0')).join('');
/* additive mixing adds light; subtractive mixing multiplies pigment (weighted geometric mean) */
function mixCol(a,b,w,mode){const A=h2r(a),B=h2r(b);return mode==='add'?r2h(...A.map((v,i)=>v*2*w+B[i]*2*(1-
    w))):r2h(...A.map((v,i)=>Math.pow(Math.max(v,1)/255,w)*Math.pow(Math.max(B[i],1)/255,1-w)*255))}

/* ---- colour space: RGB -> HSL (hue 0..360) ---- */
const hd=(a,b)=>{const d=Math.abs(a-b)%360;return d>180?360-d:d};
const
    hn=h=>h<15||h>=345?'red':h<40?'orange':h<70?'yellow':h<165?'green':h<200?'cyan':h<255?'blue':h<290?'purple':'magen
    ta';
function rgbToHsl(r,g,b){r/=255;g/=255;b/=255;const mx=Math.max(r,g,b),mn=Math.min(r,g,b),l=(mx+mn)/2,d=mx-
    mn;if(!d)return{h:0,s:0,l};const s=d/(1-Math.abs(2*l-1));let
    h=mx===r?((g-b)/d)%6:mx===g?(b-r)/d+2:(r-g)/d+4;h*=60;if(h<0)h+=360;return{h,s,l}}
function famOf(r,g,b,c){c=c||rgbToHsl(r,g,b);if(Math.abs(r-212)+Math.abs(g-175)+Math.abs(b-
    55)<60)return'gold';if(c.l<.16)return'black';if(c.s<.14)return c.l>.85?'white':'gray';return hn(c.h)}

/* ---- 1. pixel data extraction ---- */
function extractPixels(cv){return cv.getContext('2d').getImageData(0,0,cv.width,cv.height)}

/* ---- 2. smart sampling + filtering: every 2nd pixel, skip paper, bucket hues, drop noise bins ---- */
function analyze(cv){
 const img=extractPixels(cv),d=img.data,W_=img.width,H_=img.height,ST=2,GX=24,GY=16,cw=W_/GX,chh=H_/GY,NB=36;
 const bc=new Float32Array(NB),bs=new Float32Array(NB),bo=new Float32Array(NB),lh=new Uint32Array(20),fm={},gl=new
    Float32Array(GX*GY),gn=new Uint16Array(GX*GY);
 let tot=0,pt=0,chroma=0,neutral=0,sx=0,sy=0,ls=0;
 for(let y=0;y<H_;y+=ST)for(let x=0;x<W_;x+=ST){const i=(y*W_+x)*4;if(d[i+3]<8)continue;tot++;
  const r=d[i],g=d[i+1],b=d[i+2];
  if(Math.abs(r-253)+Math.abs(g-253)+Math.abs(b-248)<24)continue;
  const c=rgbToHsl(r,g,b);pt++;sx+=x;sy+=y;ls+=c.l;lh[Math.min(19,c.l*20|0)]++;
  const f=famOf(r,g,b,c);fm[f]=(fm[f]||0)+1;
  const ci=(y/chh|0)*GX+(x/cw|0);gl[ci]+=c.l;gn[ci]++;
  if(c.s>=.2&&c.l>.1&&c.l<.93){const
    bi=(c.h/10|0)%NB,rad=c.h*Math.PI/180;bc[bi]++;bs[bi]+=Math.sin(rad);bo[bi]+=Math.cos(rad);chroma++}else neutral++}
 const thr=Math.max(2,chroma*.02),on=Array.from(bc,v=>v>=thr),runs=[];let cur=null;
 if(on.every(Boolean))runs.push([...Array(NB).keys()]);
 else{const s0=on.indexOf(false);for(let k=1;k<=NB;k++){const ix=(s0+k)%NB;if(on[ix])(cur=cur||[]).push(ix);else
    if(cur){runs.push(cur);cur=null}}if(cur)runs.push(cur)}
 let cl=runs.map(r=>{let w=0,sn=0,cs=0;r.forEach(i=>{w+=bc[i];sn+=bs[i];cs+=bo[i]});let h=Math.atan2(sn,cs)*180/Math.P
    I;if(h<0)h+=360;return{h,share:chroma?w/chroma:0}}).filter(c=>c.share>=.05).sort((a,b)=>b.share-a.share);
 const nCl=cl.length;cl=cl.slice(0,4);
 const pc=q=>{if(!pt)return 0;let a=0;const t=pt*q;for(let i=0;i<20;i++){a+=lh[i];if(a>=t)return(i+.5)/20}return 1};
 return{cov:tot?pt/tot:0,pt,tot,fm,clusters:cl,nCl,chroma,neutralShare:pt?neutral/pt:0,p10:pc(.1),p90:pc(.9),cx:pt?sx/
    pt:240,cy:pt?sy/pt:160,ls,gl,gn,GX,GY}}

/* ---- 3. colour harmony: complementary, split, triadic, tetradic, analogous, monochrome + value contrast ---- */
function harmony(A){if(A.pt<30)return{s:0,note:'Nothing to judge yet.'};
 const cl=A.clusters,N=cl.length,Hs=cl.map(c=>c.h),nm=Hs.map(hn),det=[],pairs=[];
 for(let i=0;i<N;i++)for(let j=i+1;j<N;j++)pairs.push([i,j,hd(Hs[i],Hs[j])]);
 for(const[i,j,dd]of pairs){if(dd>=155)det.push(`complementary ${nm[i]} and ${nm[j]}`);else
    if(dd<=45&&nm[i]!==nm[j])det.push(`analogous ${nm[i]} and ${nm[j]}`)}
 let sch='Achromatic',sc=.5;
 if(N===1){sch='Monochromatic ('+nm[0]+')';sc=.55}
 else if(N===2){const dd=pairs[0][2];if(dd>=155){sch='Complementary';sc=1}else if(dd>=125){sch='Split-
    complementary';sc=.8}else if(dd>=90){sch='Contrasting hues';sc=.6}else if(dd>45){sch='Near
    neighbours';sc=.5}else{sch='Analogous';sc=.85}}
 else if(N===3){const ds=pairs.map(p=>p[2]),span=Math.max(...ds);
  const split=[0,1,2].some(i=>{const o=[0,1,2].filter(k=>k!==i);return o.every(k=>{const v=hd(Hs[i],Hs[k]);return
    v>=125&&v<=175})&&hd(Hs[o[0]],Hs[o[1]])<=75});
  if(ds.every(v=>v>=100&&v<=140)){sch='Triadic';sc=1}else if(span<=70){sch='Analogous';sc=.85}else
    if(split){sch='Split-complementary';sc=.95}else if(ds.some(v=>v>=155)){sch='Complementary with an
    accent';sc=.7}else{sch='Mixed palette';sc=.45}}
 else if(N===4){const pr=(a,b,c,d)=>hd(Hs[a],Hs[b])>=150&&hd(Hs[c],Hs[d])>=150;if(pr(0,1,2,3)||pr(0,2,1,3)||pr(0,3,1,2
    )){sch='Tetradic';sc=.9}else{sch='Busy palette';sc=.35}}
 if(A.nCl>4){sch='Rainbow mix';sc=.3}
 const vc=Math.min(1,(A.p90-A.p10)/.55),anchor=N>=1&&A.neutralShare>=.08&&A.neutralShare<=.6;
 const s=Math.min(25,sc*15+vc*7+(anchor?3:0));
 const note=[sch,det.slice(0,2).join(', '),vc>.6?'strong light and dark contrast':vc<.25?'flat values: add lights or
    darks':'',anchor?'neutrals anchor the colours':''].filter(Boolean).join('; ');
 return{s,note}}

/* ---- 4. shapes and lines ---- */
const pd=(p,a,b)=>{const dx=b.x-a.x,dy=b.y-a.y,L=Math.hypot(dx,dy);if(!L)return Math.hypot(p.x-a.x,p.y-a.y);return
    Math.abs(dy*p.x-dx*p.y+b.x*a.y-b.y*a.x)/L};
function rdp(p,e){if(p.length<3)return p;let mx=0,ix=0;const a=p[0],b=p[p.length-1];for(let
    i=1;i<p.length-1;i++){const v=pd(p[i],a,b);if(v>mx){mx=v;ix=i}}return
    mx>e?rdp(p.slice(0,ix+1),e).slice(0,-1).concat(rdp(p.slice(ix),e)):[a,b]}
```

```
function strokeInfo(s){const p=s.p,n=p.length;let len=0,x0=1e9,x1=-1e9,y0=1e9,y1=-1e9;
 for(let i=0;i<n;i++){const q=p[i];if(q.x<x0)x0=q.x;if(q.x>x1)x1=q.x;if(q.y<y0)y0=q.y;if(q.y>y1)y1=q.y;if(i)len+=Math.
    hypot(q.x-p[i-1].x,q.y-p[i-1].y)}
 const a=p[0],b=p[n-1],chord=Math.hypot(b.x-a.x,b.y-a.y),dg=Math.hypot(x1-x0,y1-y0);
 const o={len,chord,cx:(x0+x1)/2,cy:(y0+y1)/2,dg,c:s.c,z:s.z,f:famOf(...h2r(s.c)),t:'dot',p};
 if(len<12)return o;
 const rs=[a];for(let i=1;i<n;i++){const l=rs[rs.length-1];if(Math.hypot(p[i].x-l.x,p[i].y-l.y)>=10)rs.push(p[i])}
 const hs=[];for(let i=1;i<rs.length;i++)hs.push(Math.atan2(rs[i].y-rs[i-1].y,rs[i].x-rs[i-1].x));
 const dth=[];for(let i=1;i<hs.length;i++){let
    v=hs[i]-hs[i-1];while(v>Math.PI)v-=2*Math.PI;while(v<-Math.PI)v+=2*Math.PI;dth.push(v)}
 const sm=dth.map((v,i)=>((dth[i-1]??v)+v+(dth[i+1]??v))/3),turn=dth.reduce((a,b)=>a+b,0);
 let corners=0,csg=[],pv=false;for(const v of dth){const
    hit=Math.abs(v)>1.15;if(hit&&!pv){corners++;csg.push(Math.sign(v))}pv=hit}
 let infl=0,lsg=0;for(const v of sm){if(Math.abs(v)>.12){const g=Math.sign(v);if(lsg&&g!==lsg)infl++;lsg=g}}
 const st=chord/len;
 if(len>60&&(chord<.14*len&&Math.abs(turn)>4.5||chord<.1*len&&corners>=3)){
  let ar=0;for(let i=0;i<n;i++){const q=p[i],r=p[(i+1)%n];ar+=q.x*r.y-r.x*q.y}ar=Math.abs(ar)/2;
  const circ=4*Math.PI*ar/(len*len),v=Math.max(2,rdp(p,Math.max(5,.06*dg)).length-1);
  o.closed=true;o.t=v===3?'triangle':v===4?'rect':circ>.8?'circle':circ>.5?'polygon':'blob';return o}
 if(len>60&&chord>=.14*len&&chord<=.38*len&&Math.abs(turn)>3.6){o.implied=true;o.t='curve';return o}
 if(st>.93){o.t='straight';const ang=Math.abs(Math.atan2(b.y-a.y,b.x-
    a.x)*180/Math.PI);o.dir=ang<15||ang>165?'horizontal':ang>75&&ang<105?'vertical':'diagonal';return o}
 if(corners===0)o.t=infl>=2?'wave':'curve';
 else if(corners<=2)o.t='angle';
 else{let alt=0;for(let
    i=1;i<csg.length;i++)if(csg[i]!==csg[i-1])alt++;o.t=alt>=.7*(csg.length-1)?'zigzag':'scribble'}
 return o}
const SHL={straight:['straight line','straight lines'],curve:['curve','curves'],wave:['wave','waves'],zigzag:['zigzag'
    ,'zigzags'],angle:['angle','angles'],circle:['circle','circles'],rect:['rectangle','rectangles'],triangle:['triang
    le','triangles'],polygon:['polygon','polygons'],blob:['blob','blobs'],scribble:['scribble','scribbles']};
function shapesEval(){const inf=SD.filter(s=>s.p.length).map(strokeInfo),longs=inf.filter(o=>o.len>=30),cnt={};
 inf.forEach(o=>{cnt[o.t]=(cnt[o.t]||0)+1});const dots=cnt.dot||0;
 const kinds=Object.keys(cnt).filter(k=>k!=='scribble'&&k!=='dot');if(dots>=6)kinds.push('dots');
 const good=longs.filter(o=>o.t!=='scribble').length,clar=longs.length?good/longs.length:(dots>=6?.6:0),zs=new
    Set(SD.map(s=>s.z)).size;
 if(!inf.length&&!FL.length)return{s:0,inf,note:'No lines yet.'};
 const s=Math.min(20,Math.min(1,kinds.length/4)*8+clar*7+(zs>=2?1:0)+(zs>=3?1:0)+Math.min(3,FL.length*1.5));
 const parts=Object.entries(cnt).filter(([k])=>k!=='dot').map(([k,v])=>v+'
    '+SHL[k][v>1?1:0]);if(dots)parts.push(dots+' dots');if(FL.length)parts.push(FL.length+(FL.length>1?' filled
    areas':' filled area'));
 const note=(parts.join(', ')||'just marks')+(longs.length>=3&&clar<.4?'; shaky hands turn lines into scribbles':'');
 return{s,inf,note}}

/* ---- 5. gestalt principles ---- */
function gestalt(A,inf){const cl=(v,a=0,b=1)=>Math.max(a,Math.min(b,v)),dist=(a,b)=>Math.hypot(a.x-b.x,a.y-b.y);
 const marks=[...inf.map(o=>({x:o.cx,y:o.cy})),...FL.map(f=>({x:f.x,y:f.y}))];
 let prox=0;if(marks.length>=4){const par=marks.map((_,i)=>i),fd=i=>par[i]===i?i:(par[i]=fd(par[i]));
  for(let i=0;i<marks.length;i++)for(let j=i+1;j<marks.length;j++)if(dist(marks[i],marks[j])<60)par[fd(i)]=fd(j);
  const g={};marks.forEach((_,i)=>(g[fd(i)]=g[fd(i)]||[]).push(i));const gs=Object.values(g);
  if(gs.length>=2&&gs.length<=6&&gs.some(a=>a.length>1)){let gap=1e9;for(let a=0;a<gs.length;a++)for(let
    b=a+1;b<gs.length;b++)gs[a].forEach(i=>gs[b].forEach(j=>{gap=Math.min(gap,dist(marks[i],marks[j]))}));
   let ws=0,wn=0;gs.forEach(a=>{if(a.length>1)a.forEach(i=>{let
    m=1e9;a.forEach(j=>{if(j!==i)m=Math.min(m,dist(marks[i],marks[j]))});ws+=m;wn++})});
   prox=cl((gap/Math.max(10,ws/wn)-1)/3)}}
 const key={};inf.forEach(o=>{const k=o.f+'|'+o.t+'|'+o.z;key[k]=(key[k]||0)+1});const
    vs=Object.values(key).sort((a,b)=>b-a);
 const sim=vs[0]>=3?cl((vs[0]-1)/4+(vs[1]>=2?.15:0)):vs[0]===2?.25:0;
 const imp=inf.filter(o=>o.implied).length,clo=inf.filter(o=>o.closed).length,closure=cl(imp*.6+clo*.25+(FL.length&&cl
    o?.2:0));
 const
    sl=inf.filter(o=>['straight','curve','wave'].includes(o.t)&&o.len>=30),lc=sl.filter(o=>o.len>=100).length,ends=[];
 sl.forEach((o,k)=>{const p=o.p,n=p.length,m=Math.min(5,n-1);ends.push({k,x:p[0].x,y:p[0].y,a:Math.atan2(p[0].y-
    p[m].y,p[0].x-p[m].x)*180/Math.PI},{k,x:p[n-1].x,y:p[n-1].y,a:Math.atan2(p[n-1].y-p[n-1-m].y,p[n-1].x-p[n-1-
    m].x)*180/Math.PI})});
 let cp=0;for(let i=0;i<ends.length;i++)for(let
    j=i+1;j<ends.length;j++)if(ends[i].k!==ends[j].k&&dist(ends[i],ends[j])<18&&hd(ends[i].a,ends[j].a+180)<35)cp++;
 const cont=cl(lc*.35+cp*.5);
 const cenS=cl(1-Math.hypot(A.cx-240,A.cy-160)/150),meanL=A.pt?A.ls/A.pt:1,con=cl(Math.abs(.99-
    meanL)/.4),covF=A.cov<.05?A.cov/.05:A.cov>.65?cl((.95-A.cov)/.3):1;
 const fg=A.pt<30?0:cl(cenS*1.2)*con*covF;
 let best=0;for(const mode of[0,1]){let cons=0,mt=0;for(let cy=0;cy<A.GY;cy++)for(let
    cx=0;cx<A.GX;cx++){if(mode?cy>=A.GY/2:cx>=A.GX/2)continue;
  const
    i=cy*A.GX+cx,j=mode?(A.GY-1-cy)*A.GX+cx:cy*A.GX+(A.GX-1-
    cx),pa=A.gn[i]>=8,pb=A.gn[j]>=8;if(!pa&&!pb)continue;cons++;
  if(pa&&pb&&Math.abs(A.gl[i]/A.gn[i]-A.gl[j]/A.gn[j])<.2)mt++}if(cons>=10)best=Math.max(best,mt/cons)}
 const sym=cl((best-.55)/.35);
 const list=[['Proximity',prox],['Similarity',sim],['Closure',closure],['Continuity',cont],['Figure and
    ground',fg],['Symmetry',sym]];
 const det=list.filter(l=>l[1]>=.5).map(l=>l[0]);
 return{s:Math.min(20,list.reduce((a,l)=>a+l[1]*4,0)),note:det.length?'Uses '+det.join(', ').toLowerCase():'No clear
    principle yet. Try grouping, repeating or closing shapes.'}}

/* ---- overall rating (the "quality" the rest of the game already uses) ---- */
function quality(cv){const A=analyze(cv),hm=harmony(A),sh=shapesEval(),ge=gestalt(A,sh.inf);
 const cov=A.pt?Math.min(10,A.cov*30)+Math.min(5,strokes*.5):0;
 const fams=Object.keys(A.fm).filter(f=>f!=='gray'&&A.fm[f]/A.pt>=.02).length,vr=Math.min(10,fams*2.5);
 const tr=A.pt&&(A.fm[S.trend]||0)/A.pt>=.03?12:0,br=has('brush')?10:0,raw=hm.s+sh.s+ge.s+cov+vr+tr+br;
 const rep=[{k:'Colour harmony',s:hm.s,max:25,note:hm.note},{k:'Shapes and
    lines',s:sh.s,max:20,note:sh.note},{k:'Gestalt',s:ge.s,max:20,note:ge.note},{k:'Coverage and
    effort',s:cov,max:15,note:Math.round(A.cov*100)+'% of the canvas painted'},{k:'Colour
```

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    variety',s:vr,max:10,note:fams+(fams===1?' colour family':' colour families')}];
 const bonus=[tr?`Market colour (${S.trend}) +12.`:'',br?'Gilded Brush +10.':''].filter(Boolean);
 return{q:(devRating!=null?devRating:Math.max(1,Math.min(100,Math.round(raw)))),cov:A.cov,gold:(A.fm.gold||0)/Math.max
    (1,A.pt)>=.01,rep,bonus}}
const repHtml=m=>`<div class="rep">${m.rep.map(r=>`<div
    class="rr"><span>${r.k}</span><b>${Math.round(r.s)}/${r.max}</b><div class="rb"><div style="width:${Math.round(r.s
    /r.max*100)}%"></div></div><small>${esc(r.note)}</small></div>`).join('')}${m.bonus.length?`<p class="dim"
    style="margin:6px 0 0;font-size:18px">${m.bonus.map(esc).join(' ')}</p>`:''}</div>`;

/* ---- fill bucket: scanline flood fill with tolerance + 1px halo so anti-aliased edges are covered ---- */
function bucket(ctx,sx,sy,hex){const W_=480,H_=320;sx|=0;sy|=0;if(sx<0||sy<0||sx>=W_||sy>=H_)return 0;
 const im=ctx.getImageData(0,0,W_,H_),d=im.data,si=(sy*W_+sx)*4,tr=d[si],tg=d[si+1],tb=d[si+2],[fr,fg,fb]=h2r(hex);
 if(Math.abs(tr-fr)+Math.abs(tg-fg)+Math.abs(tb-fb)<6)return 0;
 const df=i=>Math.abs(d[i]-tr)+Math.abs(d[i+1]-tg)+Math.abs(d[i+2]-tb),match=i=>df(i)<=60,seen=new
    Uint8Array(W_*H_),stack=[sx,sy];let cnt=0;
 const put=p=>{const i=p*4;d[i]=fr;d[i+1]=fg;d[i+2]=fb;d[i+3]=255};
 while(stack.length){const y=stack.pop();let x=stack.pop();while(x>=0&&!seen[y*W_+x]&&match((y*W_+x)*4))x--;x++;let
    up=false,dn=false;
  for(;x<W_&&!seen[y*W_+x]&&match((y*W_+x)*4);x++){seen[y*W_+x]=1;put(y*W_+x);cnt++;
   if(y>0){const m=!seen[(y-1)*W_+x]&&match(((y-1)*W_+x)*4);if(m&&!up){stack.push(x,y-1);up=true}else if(!m)up=false}
   if(y<H_-1){const m=!seen[(y+1)*W_+x]&&match(((y+1)*W_+x)*4);if(m&&!dn){stack.push(x,y+1);dn=true}else
    if(!m)dn=false}}}
 const grow=[];for(let y=0;y<H_;y++)for(let x=0;x<W_;x++){const p=y*W_+x;if(seen[p])continue;
  if((x>0&&seen[p-1]||x<W_-1&&seen[p+1]||y>0&&seen[p-W_]||y<H_-1&&seen[p+W_])&&df(p*4)<=130)grow.push(p)}
 grow.forEach(put);ctx.putImageData(im,0,0);return cnt}

let col,sz,strokes,used,drawing=false;
const PNC=['Your hands are not listening.','The line wanders. You did not move.','The room tilts, just slightly.','You
     taste copper.','Mum\'s breathing is in your wrists.','Straight. Just draw one straight line.','Your stomach turns
     over. You keep painting.'];
function drawScreen(){clearInterval(TMR);strokes=0;used=new
    Set();SD=[];FL=[];UN=[];tool='brush';fbDone=false;col='#111111';sz=6;const P=S.panic,J=P?2.2+(S.day-6)*1.1:0;
mus(S.day===DAYS?'d10':'draw');
const hint=has('radar')?`Today the market wants <b style="border-bottom:6px solid ${FAMH[S.trend]}">${S.trend}</b>.
    Any shade of it counts. Use it for a big bonus.`:`The market wants one colour today. You do not know which. <span
    class="dim">(The Patron's Ledger would tell you.)</span>`;
W.className=P?'pw':'';
W.innerHTML=`<div class="box"><div class="px" style="font-size:12px">DAY ${S.day}: PAINT</div><p>${hint} Gold is
    flashy; collectors like it.</p>${has('eye')?'<p>Live rating: <b id="lv">0</b>/100</p>':''}${P?'<div class="bar"
    style="max-width:100%"><div id="tmb" style="width:100%;background:var(--bad)"></div></div>':''}<div id="msg"
    class="${P?'pnc':''}" style="${P?'':'color:var(--bad)'}">${P?pick(PNC):''}</div></div><canvas id="cv"
    class="${P?'nau':''}" width="480" height="320"></canvas>
<div class="row"><button data-t="brush"><U+1F58C><U+FE0F> Brush</button><button data-t="bucket"><U+1FAA3>
    Fill</button><button data-t="erase">Eraser</button><button id="un">↶ Undo</button><button
    id="clr">Clear</button></div>
<div class="row"><button data-s="3">Thin</button><button data-s="7">Mid</button><button data-s="16">Fat</button><span
    class="curw"><span class="cur" id="cur"></span><span id="curn"></span></span></div>
<div class="row"><button class="ptab" data-pm="add">✳ Additive</button><button class="ptab" data-pm="sub"><U+1F3A8>
    Subtractive</button><button id="mixb"><U+1F9EA> Mix</button></div>
<p class="c dim" id="ptip" style="margin:2px 0;font-size:18px"></p><div id="pal"></div>
<div class="box mixp" id="mixp" style="display:none"><div class="row"><button class="slot" id="sa" aria-label="Colour
    A"></button><b>+</b><button class="slot" id="sb" aria-label="Colour B"></button><b>=</b><span class="slot res"
    id="sr"></span></div><div class="row dim" style="font-size:18px">more A <input type="range" id="mr" min="10"
    max="90" value="50"> more B</div><div class="row"><button id="madd">Use this mix</button></div><p class="dim c"
    style="font-size:17px;margin:0">Tap colours above to fill slot A, then slot B. The mix follows the tab you are
    on.</p></div>
<div class="row"><input id="ti" maxlength="30" placeholder="Title (blank = untitled)" style="max-width:380px;font-
    size:20px"></div><p class="c"><button class="big" id="sub">TO THE SALEROOM</button></p>`;
const cv=$('#cv'),x=cv.getContext('2d',{willReadFrequently:true});x.fillStyle=BGHEX;x.fillRect(0,0,480,320);x.lineCap=
    x.lineJoin='round';
let mxA='#e32636',mxB='#ffd400',mxN='A',mxR=50,mxRes='#ff7f11',mxOpen=false,cs=null;
const sync=()=>{document.querySelectorAll('[data-
    t]').forEach(b=>b.classList.toggle('sel',b.dataset.t===tool));document.querySelectorAll('[data-
    pm]').forEach(b=>b.classList.toggle('sel',b.dataset.pm===PM));
 $('#cur').style.background=tool==='erase'?BGHEX:col;$('#curn').textContent=tool==='erase'?'Eraser':(CN[col]||'Mix');c
    v.style.cursor=tool==='bucket'?'cell':'crosshair'};
const mxUpd=()=>{mxRes=mixCol(mxA,mxB,mxR/100,PM);$('#sa').style.background=mxA;$('#sb').style.background=mxB;$('#sr')
    .style.background=mxRes;$('#sa').classList.toggle('nx',mxN==='A');$('#sb').classList.toggle('nx',mxN==='B')};
const sel=c=>{col=c;if(tool==='erase')tool='brush';renderPal();sync()};
const swClick=h=>{if(mxOpen){if(mxN==='A'){mxA=h;mxN='B'}else{mxB=h;mxN='A'}mxUpd()}else sel(h)};
function renderPal(){const sw=([h,n])=>`<button class="sw${h===col&&tool!=='erase'?' on':''}" data-c="${h}"
    style="background:${h}" title="${n}" aria-label="${n}"></button>`;
 $('#ptip').textContent=PAL[PM].tip;
 $('#pal').innerHTML=PAL[PM].groups.map(([g,a])=>`<div class="pgl"><span class="dim">${g}</span><div
    class="pal">${a.map(sw).join('')}</div></div>`).join('')+`<div class="pgl"><span class="dim">Neutrals and
    gold</span><div class="pal">${NEU.map(sw).join('')}</div></div>`+(MIXED.length?`<div class="pgl"><span
    class="dim">Your mixes</span><div class="pal">${MIXED.map(h=>sw([h,'Mix'])).join('')}</div></div>`:'');
 document.querySelectorAll('.sw').forEach(b=>b.onclick=()=>swClick(b.dataset.c));
 document.querySelectorAll('[data-pm]').forEach(b=>b.classList.toggle('sel',b.dataset.pm===PM))}
const live=()=>{if(has('eye')&&strokes)$('#lv').textContent=quality(cv).q};
const wipe=()=>{x.fillStyle=BGHEX;x.fillRect(0,0,480,320);strokes=0;used=new
    Set();SD=[];FL=[];UN=[];if(has('eye'))$('#lv').textContent=0};
const
    snap=()=>{UN.push({im:x.getImageData(0,0,480,320),sd:SD.length,fl:FL.length,st:strokes});if(UN.length>12)UN.shift(
    )};
document.querySelectorAll('[data-s]').forEach(b=>b.onclick=()=>sz=+b.dataset.s);
document.querySelectorAll('[data-t]').forEach(b=>b.onclick=()=>{tool=b.dataset.t;renderPal();sync()});
document.querySelectorAll('[data-pm]').forEach(b=>b.onclick=()=>{PM=b.dataset.pm;renderPal();mxUpd()});
$('#clr').onclick=wipe;
$('#un').onclick=()=>{const
    u=UN.pop();if(!u)return;x.putImageData(u.im,0,0);SD.length=u.sd;FL.length=u.fl;strokes=u.st;used=new
```

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    Set([...SD.map(s=>s.c),...FL.map(f=>f.c)]);live();blip(200)};
$('#mixb').onclick=()=>{mxOpen=!mxOpen;$('#mixp').style.display=mxOpen?'block':'none';$('#mixb').classList.toggle('sel
    ',mxOpen);mxUpd()};
$('#sa').onclick=()=>{mxN='A';mxUpd()};$('#sb').onclick=()=>{mxN='B';mxUpd()};
$('#mr').oninput=e=>{mxR=+e.target.value;mxUpd()};
$('#madd').onclick=()=>{if(!MIXED.includes(mxRes))MIXED.push(mxRes);if(MIXED.length>10)MIXED.shift();mxOpen=false;$('#
    mixp').style.display='none';$('#mixb').classList.remove('sel');sel(mxRes);blip(420)};
renderPal();mxUpd();sync();
/* panic: the pointer drifts and trembles, so no line is straight */
const pos=e=>{const r=cv.getBoundingClientRect(),tt=performance.now();let a=(e.clientX-
    r.left)*480/r.width,b=(e.clientY-r.top)*320/r.height;if(J){a+=Math.sin(tt/260)*J*.8+(Math.random()-
    .5)*J;b+=Math.cos(tt/310)*J*.8+(Math.random()-.5)*J}return[a,b]};
cv.onpointerdown=e=>{cv.setPointerCapture(e.pointerId);const[a,b]=pos(e);
 if(tool==='bucket'){snap();const
    n=bucket(x,a,b,col);if(n){strokes++;used.add(col);FL.push({c:col,x:a,y:b,n});blip(300);live()}else
    UN.pop();return}
 drawing=true;snap();x.strokeStyle=x.fillStyle=tool==='erase'?BGHEX:col;x.lineWidth=sz;x.beginPath();x.arc(a,b,sz/2,0,
    7);x.fill();x.beginPath();x.moveTo(a,b);strokes++;
 if(tool==='erase')cs=null;else{used.add(col);cs={p:[{x:a,y:b}],c:col,z:sz};SD.push(cs)}};
cv.onpointermove=e=>{if(!drawing)return;const[a,b]=pos(e);x.lineTo(a,b);x.stroke();if(cs){const
    l=cs.p[cs.p.length-1];if(Math.hypot(a-l.x,b-l.y)>=2)cs.p.push({x:a,y:b})}};
cv.onpointerup=cv.onpointercancel=()=>{if(!drawing)return;drawing=false;cs=null;live();
 /* mid-drawing flashback (not on Day 10, which has its own track) */
 if(!fbDone&&S.day!==DAYS&&strokes>=6){fbDone=true;mus('flash',true)}};
const send=()=>{clearInterval(TMR);S.lf=0;submit(cv,$('#ti').value.trim())};
$('#sub').onclick=()=>{if(strokes<2||!used.size){$('#msg').textContent='The terminal needs a painting. Draw something
    first.';return}
/* registration glitch: ~15% of the time, never twice in a row, never while panicking */
if(!S.lf&&!P&&Math.random()<.15){S.lf=1;wipe();blip(110);$('#msg').textContent='Registration error: the terminal could
     not read your piece. Nothing was saved. Paint it again.';scrollTo(0,0);return}
send()};
if(P){const T=Math.max(30,50-(S.day-7)*5)*1000,t0=Date.now();let ln=0;
TMR=setInterval(()=>{if(!$('#tmb')){clearInterval(TMR);return}const
    f=1-(Date.now()-t0)/T;$('#tmb').style.width=Math.max(0,f*100)+'%';
if(Date.now()-ln>5000){ln=Date.now();$('#msg').textContent=pick(PNC)}
if(f<=0){clearInterval(TMR);if(strokes<2||!used.size){x.strokeStyle='#111';x.lineWidth=6;for(let i=0;i<3;i++){const
    pts=[{x:rnd(40,440),y:rnd(40,280)}];x.beginPath();x.moveTo(pts[0].x,pts[0].y);for(let j=0;j<5;j++){const q={x:rnd(
    40,440),y:rnd(40,280)};pts.push(q);x.lineTo(q.x,q.y)}x.stroke();SD.push({p:pts,c:'#111111',z:6})}strokes=Math.max(
    strokes,3);used.add('#111111')}
$('#msg').textContent='Time. Your hands stop.';setTimeout(send,700)}},100)}}
const ROAST=['Untitled','Study in Grey','Attic, After Rain','For Mum','Unpaid Bills (Oil)','Still Life With
    Envelope','Late Light','Room 214'];
/* ===== buyers ===== */
const T={N:{e:'<U+1F642>',c:'#5b8def',l:'local',w:'raises,
    calmly.'},R:{e:'<U+1F4B0>',c:'#c9a24b',l:'collector',w:'raises without looking
    up.'},S:{e:'<U+1F9D0>',c:'#9b7be0',l:'critic (respected)',w:'raises,
    reluctantly.'},P:{e:'<U+1F393>',c:'#4fbf94',l:'student (loved)',w:'adds a few more coins.'}};
const NM={N:['Gary','Brenda','Dave','Linda','Mark','Priya','Tom','Helen','Joe','Karen'],R:['Mr. Whitmore','Lady
    Aldridge','Julian Harcourt','Mrs. Vandermeer','Edmund Crane'],S:['Prof. Vance','Odile Marchetti','Dr. Hale','Elias
     Quinn'],P:['Mina','Sol','Tobias','Zara']};
function makeCast(m){const nm=[...NM.N].sort(()=>Math.random()-.5),nf=(S.panic?.82:1)*.95;
return ['R','S','P','N','N'].map(t=>{let s,o;
if(t==='N'){s=m.q+rnd(-12,12);o=s*1.1}
else if(t==='R'){s=rnd(55,95);o=(rnd(60,150)+m.cov*50)*(m.gold?1.3:1)}
else if(t==='S'){s=m.q-rnd(8,18)-(S.panic?4:0);o=s>=45?s*.8:0}
else{s=m.q+rnd(0,8);o=s*.6+rnd(0,10)}
s=Math.max(1,Math.min(100,Math.round(s)));o=Math.round(o*nf*(has('charm')?1.1:1));
return{t,s,o,name:t==='N'?nm.pop():pick(NM[t])})}
const bracket=s=>s<45?0:s<75?1:2,L3=(a,b,c)=>[a,b,c].map(x=>x.split('|'));
const QX={
N:L3(`Is it finished?|I am sorry. I do not know where to look.|It is a lot of paint.|My daughter draws like this. She
    is four.|I wanted to like it. I did.|Hm. I need a minute. Or a chair.`,`It is nice. Warm.|I could hang this in the
     hallway.|My wife would like the colours.|Not bad. Not bad at all.|It looks like a quiet afternoon.|I would take
    it home.`,`Oh. That is really something.|I did not expect to feel anything today.|I will take it. Please do not
    sell it to anyone else.|I keep looking at the corner. It is good.|Where did you learn to do this?|My mother would
    have loved this.`),
R:L3(`Interesting. I am told that means something.|Modern, yes? I will take it regardless.|I do not understand it. I
    will pay for it anyway.|It will go in the east hallway. Nobody walks there.|It is an investment, I am sure.|My
    assistant says it is fashionable.`,`A fine piece. It will suit the study.|I do not know much about art. I know
    what I like to own.|Splendid. Wrap it.|I will tell guests it is from an emerging name.|Quite tasteful. The frame
    is another matter.|I will have it appraised later. Or never.`,`Remarkable. Name a figure.|I rarely say this: I
    would like to meet the artist.|It will be the centrepiece of the room.|I have a wall that has been waiting for
    this.|Put it on my account. Quickly.|Yes. This one. Whatever the cost.`),
S:L3(`The composition has no centre.|I see effort. I do not see intent.|There is an idea here. It has not
    arrived.|Colour without reason is just noise.|Technically complete. Emotionally absent.|I will not pretend. This
    needs work.`,`There is something honest here.|The edges are rough. The middle holds.|I see where you are headed.
    Keep going.|A careful piece, if a cautious one.|You have looked at real paintings. It shows.|Promising. I do not
    say that often.`,`This has a voice. That is rare.|I came to be unimpressed. I was not.|Every mark seems chosen.|I
    would write about this, if anyone read me.|Restraint. That is the hard part. You have it.|I have not felt this in
    years.`),
P:L3(`I cannot afford it. I am sorry.|It looks how I feel on a Sunday night.|I like that you tried. That counts.|I
    only have a little. It is yours if you want it.|It is rough. So is everything lately.|I will hang it somewhere
    dark. For luck.`,`This is kind of lovely.|It made my bus ride shorter.|I only have some coins, but I would like
    it.|It reminds me why I studied art.|I would frame it if I could afford a frame.|You made something warm. Thank
    you.`,`I am not crying. It is the lights.|This is why I chose art. Thank you.|Please take what I have. It is not
    much.|I am going to tell everyone I know.|It feels like someone finally said something kind.|I would skip meals
    for this. I already do.`)};
const RS=[`Not your best hour.|The room is quiet, and not in a good way.|There is paint. That is most of what can be
    said.|Even the easel looks away.|Start again tomorrow with a plan.`,`A rough draft with some nerve.|Something is
    trying to happen here.|Hesitant, but there is a pulse.|Not a disaster. That is progress.|You can see where it
    might go.`,`A solid piece. Honest.|It holds the eye a little while.|Careful hands. A steady mind.|People will
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    remember this until dinner.|Decent work. Your father would nod.`,`Strong. The room leaned in.|It has a voice
    now.|Confident, and a little dangerous.|Someone will ask who painted this.|Your hands remember what you
    forgot.`,`The room went still.|This one will outlive the winter.|Somebody sniffled. Nobody mentioned it.|A
    masterpiece, quietly.|You did not paint this. You let it out.`].map(x=>x.split('|'));
const BAG={},uniq=(k,a)=>{let b=BAG[k];if(!b||!b.length)b=BAG[k]=a.map((_,i)=>i).sort(()=>Math.random()-.5);return
    a[b.pop()]};
const rateLine=q=>{const i=q<20?0:q<40?1:q<60?2:q<80?3:4;return uniq('r'+i,RS[i])};

async function submit(cv,title){mus('bid');const
    m=quality(cv),img=cv.toDataURL();title=title||pick(ROAST);S.cur=m.q;const cast=makeCast(m);
W.innerHTML=`<div class="box c"><div class="px" style="font-size:12px">THE SALEROOM</div><img class="art" src="${img}"
     alt="your painting"><p><b>"${esc(title)}"</b></p><p>Your rating: <b style="color:var(--ac);font-
    size:30px">${m.q}/100</b></p><div class="bar"><div style="width:${m.q}%"></div></div>${repHtml(m)}<p
    class="dim"><i>* ${rateLine(m.q)}</i></p></div><div id="rm"></div>`;
const rm=$('#rm');await sleep(900);const els=[];
for(const b of cast){await sleep(850);const
    q=uniq('q'+b.t+bracket(b.s),QX[b.t][bracket(b.s)]);rm.insertAdjacentHTML('beforeend',`<div class="npc"
    style="border-color:${T[b.t].c}"><em>${T[b.t].e}</em><div><b>${b.name}</b><span class="tag"
    style="color:${T[b.t].c}">${T[b.t].l}</span><br>"${q}"<br><span class="dim">Rates it ${b.s}/100.</span> <b
    class="off">${b.o?'Offer
    $'+b.o:'Passes.'}</b></div></div>`);els.push(rm.lastElementChild.querySelector('.off'));blip(420)}
await sleep(900);rm.insertAdjacentHTML('beforeend','<p class="c px" style="font-size:11px">THE BIDDING</p>');
const top=Math.max(...cast.map(b=>b.o)),order=cast.map((b,i)=>i).sort(()=>Math.random()-.5);
for(const i of order){const b=cast[i];if(b.o&&b.o>=top*.6&&Math.random()<.75){await
    sleep(750);b.o=Math.round(b.o*rnd(1.04,1.15));els[i].textContent='Offer
    $'+b.o;els[i].classList.add('flash');rm.insertAdjacentHTML('beforeend',`<div class="dim">${T[b.t].e} ${b.name}
    ${T[b.t].w} <b>$${b.o}</b></div>`);blip(520)}}
await sleep(700);
rm.insertAdjacentHTML('beforeend',`<div class="box" style="margin-top:12px"><p class="c px" style="font-size:11px">WHO
     TAKES IT?</p><p class="dim c">Collectors pay the most. Critics give you a tool. Students and locals pay what they
     can.</p>${cast.map((b,i)=>`<div class="row"><button data-i="${i}" ${b.o?'':'disabled'}>${T[b.t].e} ${b.name}:
    ${b.o?'$'+b.o:'passes'}</button></div>`).join('')}</div>`);
rm.querySelectorAll('button[data-i]').forEach(bt=>bt.onclick=()=>sell(cast[+bt.dataset.i],rm))}
function sell(b,rm){const t=b.t,rate=has('coupon')?TAXLO:TAX,tax=Math.round(b.o*rate),net=b.o-tax;let
    x='';S.money+=net;S.net.push(net);S.rat.push(S.cur);
if(t==='R'){S.rep.R++;x='The collector leaves with the painting. Nobody speaks.'}
else if(t==='S'){S.rep.S++;const free=Object.keys(TOOLS).filter(k=>!has(k));if(free.length){const
    k=pick(free);S.tools.push(k);if(k==='letter')S.money+=80;x=`${b.name} gives you a tool: <b>${TOOLS[k][0]}
    ${TOOLS[k][1]}</b>. ${TOOLS[k][2]}`}else{S.money+=40;x='You own every tool. They tip you $40 anyway.'}}
else{S.rep.P++;x=t==='P'?'They hold it with both hands, like it might spill.':'A fair, quiet sale.'}
S.money-=NECT;hud();
rm.innerHTML=`<div class="box c"><p>SOLD to ${T[t].e} <b>${b.name}</b> for <b>$${b.o}</b>.</p><p
    class="dim">${x}</p><p>State tax (${Math.round(rate*100)}%): -$${tax}. You keep <b>$${net}</b>.<br>Tonight:
    ${NEC.map(([n,v])=>n+' $'+v).join(', ')} (-$${NECT}).<br>Still need
    <b>$${Math.max(0,GOAL-S.money).toLocaleString()}</b>.</p><button class="big" id="nx">${S.day>=DAYS?'GO TO THE
    HOSPITAL':'NEXT DAY'}</button></div>`;
$('#nx').onclick=()=>{if(S.day>=DAYS)finale();else{S.day++;startDay()}}}

/* ===== endings ===== */
const topType=()=>['P','S','R'].sort((a,b)=>S.rep[b]-S.rep[a])[0];
/* ODDS (tuned so an average run lands near 40%): R 40-52 (money helps), S 32-48 (better paintings help), P 34-44.
    Falling short of the money costs 12 points. */
function odds(t){const
    mf=Math.min(1,S.money/GOAL),av=S.rat.length?S.rat.reduce((a,b)=>a+b,0)/S.rat.length/100:(devRating||50)/100;
let c=t==='R'?40+12*mf:t==='S'?32+16*av:34+10*(mf+av)/2;if(S.money<GOAL)c=Math.max(4,c-12);return c}
const COMIC_SRC='';/* <<< PUT YOUR COMIC IMAGE PATH/URL HERE (shown at the end of every bad ending) */
const EN={R:{W:'BORROWED SPRING',L:'THE CLEARING DAY'},S:{W:'IN PRINT',L:'A REVIEW, TOO LATE'},P:{W:'THE FULL
    ROOM',L:'THE EMPTY FRAME'}};
const VOUCH={R:'* Behind you, a collector in a long coat, a pen already in hand. He is not here for her. He was told
    it would matter.',S:'* Behind you, a critic with a folded newspaper under her arm. She has read the whole file.
    She has not said how she feels about it.',P:'* Behind you, the lobby fills. A teacher. A baker. A student holding
    a bag of coins out in front of her, not knowing where to put it.'};
function finale(){const t=topType(),win=Math.random()*100<odds(t);ending(t,win)}
function clerkLines(t,win,enough){const m=`$${S.money.toLocaleString()}`;
const L=[['',`* Day ${DAYS}, evening. The hospital finance office. ${m} on the desk, counted twice. ${enough?'It is
    enough.':'It is $'+(GOAL-S.money).toLocaleString()+' short.'}`,'corr'],['',VOUCH[t]]];
if(win&&enough)L.push(['Clerk','The deposit is sufficient. If you will sign here, we can schedule the theatre for the
    morning.']);
else if(win)L.push(['Clerk','I am sorry. We cannot schedule without the full deposit.'],...({R:[['','* The collector
    steps forward and signs a guarantee for the difference. He does not look at you while he does
    it.'],['Collector','An investment. Do not make it sound like kindness.']],S:[['','* The critic puts a cheque on
    the desk. "A review fee," she says. "An advance on the one I am going to write."']],P:[['','* They count it
    themselves in the lobby. Coins, folded notes, a child\'s money box. It is exactly the difference. Nobody remarks
    on how.']]}[t]),['Clerk','Then... yes. We can schedule for the morning.']);
else if(enough)L.push(['Clerk','The funds clear at nine tomorrow. The theatre is booked until Thursday. I am sorry. I
    know how that sounds.']);
else L.push(['Clerk','I am sorry. We cannot schedule without the full deposit. I wish it were different.']);
return L}
const GOODL={
R:['* The collector leaves before the news, a signed receipt on the chair. The painting you sold him hangs in a house
    you will never see. It bought your mother a spring.'],
S:['* Two days later the critic\'s piece runs. Mum reads it aloud in bed in a terrible theatre voice, skipping the
    adjectives she disagrees with.'],
P:['* The students come on visiting day. Six of them sit on the end of her bed. She asks each what they study, then
    tells them what they are doing wrong.']};
const GOODF={
R:'* A wreath with no card. You know who it is from. You hang it in the hallway anyway.',
S:'* The critic sends a card. Three lines. Not a review: just, I am sorry. She was clearly your first reader.',
P:'* The students fill the back row. One of them does not know where to stand and ends up holding the umbrella for the
     whole family.'};
const BADA={
```

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R:'* Behind you, the guarantee is being typed into a system that will accept it at nine. Tomorrow.',
S:'* Behind you, the critic folds her newspaper along its crease. She says nothing, and you are grateful, and ashamed
    of being grateful.',
P:'* Behind you, the students stand very still. One is still holding out her bag of coins, as if the gesture could be
    finished.'};
const BADR={
R:'You sold her to the people with chandeliers. They paid for a wall. Not one of them ever came to see her.',
S:'You wanted to be seen by people who write things down. She wanted to be seen. You were at the saleroom.',
P:'You took pennies from students and told me it was a plan. It was a kind plan. She is dead.'};
const BADL={R:'the money that pays for walls',S:'the people who write things down',P:'the people with the least to
    give'};
async function ending(t,win){hud();ET=t;EW=win;mus(win?'win':'mother');const enough=S.money>=GOAL,pv=0.prev;
await dialog(clerkLines(t,win,enough),SCN('corr'));
if(win){
await dialog([
['','* The surgery takes nine hours. You and Robin do not talk. At hour six he takes your hand without looking, the
    way he did when you were children.','corr'],
['','* At hour nine the door opens. The surgeon takes off his cap, which is how you know before he says.'],
['Doctor','She is through. It was difficult. She is asking for tea, and she is being very rude about it.','hosp'],
['',GOODL[t][0]],
['','* She lives. It is not forever. The doctor said so quietly in the corridor, and you both heard it and chose not
    to carry it.'],
['Mum','Take me home. I want to see the attic before I look any worse.'],
['','* You take her up the stairs one at a time. On the landing she stops and looks at the wall: your father\'s
    brushes, and yours beside them.','studio'],
['Mum','There. Two sets. That is the house.'],
['','* Eleven weeks. You will count them later. You will tell yourself it was more than the doctors gave. You will
    wonder if you could have bought twelve.'],
['','* She dies in October, in her own bed, with the window open. The rain is rude. She would have said so.','house'],
['',GODF(t)],
['','* Afterward you and Robin climb to the attic. Neither of you has cried. Not at the scan. Not at the bed. You are
    not sure either of you remembers how.','two'],
['Robin','I keep waiting to feel something big. I just feel tired. I tell clients that is allowed.'],
['{n}','They were good weeks. They were ours.'],
['Robin','They were. They were so good, {n}.'],
['','* His voice cracks on the second good. He puts a hand over his mouth, as if he could take it back.'],
['','* You break first. It comes up from somewhere under the ribs, from a place you did not know was load-bearing.'],
['','* Then he is crying too. Robin, who has a voice for everyone else\'s grief. He holds on to you like a man on a
    boat.'],
['','* Tears, the first for both of you. They are salt and warm, and as ordinary as anything.'],
['Robin','She would have hated this. We are making a mess.'],
['{n}','She would have said: finally.'],
['','* You laugh, wetly, in the one room where your father used to. Rain on the roof. The lamp does not flicker.'],
['','* You are still guilty. You will always be a little guilty. But you are guilty together, and that is a different
    weight.'],
...(pv&&!pv.w?[['???','{prev} never got this far. You did. I hope that feels like something.']]:[]),
...(pv&&pv.w?[['???','Mum lived again. A little longer, a little less. I am only telling you what I saw.']]:[])]);
}else{
mus('lose');
await dialog([
['Robin','Then tomorrow. Tomorrow we sign, we schedule, we...','corr'],
['Doctor','Tomorrow may be too late. I am sorry. Tonight I would go in and sit with her.'],
['',BADA[t]],
['Mum','Well? Do not make the face. I can read the face.','hosp'],
['{n}','Tomorrow. They said tomorrow.'],
['Mum','...Ah.'],
['Mum','I asked you. Just in case.'],
['Robin','Mum, we have until tomorrow. Please.'],
['Mum','Go and get tea. Both of you. The good tea, from the machine on the second floor. It takes ten minutes to walk
    there if you go slowly.'],
['{n}','Mum.'],
['Mum','Go. Let an old woman have one thing she asked for.'],
['','* You go. You both know. Neither of you says it. You walk toward the worst thing of your life slowly, like she
    said.'],
['','* The machine on the second floor is broken. You stand in front of it for a long time anyway.','corr'],
['','* When you come back, the monitor is quiet. It has never been quiet before.','hflat'],
['','* The plug is out of the wall. She was always strong in the hands. She reached the one thing in the room she
    still could.'],
['Robin','No. No, no, no. Mum.'],
['','* A nurse. Another. Someone says a time out loud. It does not sound like a time. It sounds like a word.'],
...(pv&&!pv.w?[['???','You know this part. You know what he says next.']]:[]),
['Robin','You said ten days. You said you could do it.','corr','shake'],
['Robin',BADR[t],'corr','shake'],
['{n}','You think I do not know what I did?'],
['Robin',`Eight thousand five hundred. I gave you everything I had, and you gave me a receipt and a flat
    line.`,'corr','shake'],
['','* He shoves you. You shove back. A chair goes over. A nurse in the doorway, then not in the
    doorway.','corr','shake'],
['','* Someone\'s fist finds someone\'s shoulder. You are not sure whose. It does not matter.','corr','shake'],
['','* Then he stops, mid-motion. His hands open. He sits down on the hospital floor like someone has switched him
    off.'],
['Robin','...Get out. Please. I cannot look at your hands.'],
['','* Three days. He does not answer the phone. You stand outside his flat once and walk away without
    knocking.','door'],
['','* The funeral is eight people and a priest who gets her name right on the second try. Robin stands at the far end
     of the row.','house'],
['','* The attic is quiet. The lamp is off. Nobody moves the brushes.','studio'],
['','* On the fifth day, someone climbs the attic stairs. You know the weight of the step.','two'],
['Robin','Hi.'],
['{n}','Hi.'],
['Robin','I came to say I am sorry. I said things in that corridor I do not think were true. Some were. Just not the
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    way I said them.'],
['Robin','Mum and Dad would have wanted us in the same room. Carrying this together. Not like...'],
['','* His phone rings. A number you both recognise: the hospital billing office.'],
['Robin','I have to take this. It is paperwork. I will come back tonight. Eat something. Please.'],
['','* He goes down the stairs. You hear his steps, then the front door, then the long quiet of a street.'],
['','* You sit where Mum stopped on the landing. The brushes are in their two rows. You do the sum again. It does not
    change.'],
['{n} (letter)','Robin. You will find this before the bank does. I am sorry for that too.','letter'],
['{n} (letter)','I know you came up the stairs to say sorry. I heard you. I believe you. It does not change the
    arithmetic in my head.'],
['{n} (letter)','It is my fault. I picked who to sell to. I painted while she slept. I counted days like a sum, and I
    got the sum wrong.'],
['{n} (letter)',`You said it in the corridor, and you were right about one thing. I chose ${BADL[t]}. I chose the wall
     over the clock.`],
['{n} (letter)','You gave me everything you had. I gave it back as a receipt. You should be angry. I would like you to
     be angry at me and not at yourself.'],
['{n} (letter)','This is the wrong answer to a hard question. I know that. I am giving it anyway, and it is the worst
    thing I could do to you.'],
['{n} (letter)','Please do not carry this the way I could not. Say it to yourself once, out loud, the way you say it
    to strangers. Not in the car. Look after the brushes. {n}.'],
['','* Robin climbs the attic stairs at eleven. The lamp is on. He calls {n}\'s name. He calls it again, in the voice
    he used when you were small and he knew where the keys were.','door'],
['','* He opens the door.'],
['','* He will not describe what he sees. He will not describe it to anyone, ever. He will only say the lamp was on
    and the room was quiet, and that he was late by the length of one phone call.'],
['','* {n} is dead. Robin is the one left standing in the attic, holding two names now, his own and the one he cannot
    say.'],
['','* The letter is on the desk with his name on it. He reads it once. He reads it again. The paper is soft where it
    was held a long time.'],
['Robin','It was not your fault. I said it was, in a corridor, and I did not mean it. I will say that for the rest of
    my life.','studio'],
['','* He sits on the attic floor. A bus passes outside, and the world does not stop for it.'],
['','* Mum. Dad. {n}. Three rooms with the lights off, and he is the only one still counting.'],
['','* It was not your fault, someone says at the funeral. Whoever it is, they mean it. It does not go
    in.','house']]);
await blame();
await dialog([
['','* Robin keeps the brushes. He keeps the field that was never finished, one corner still grey.','studio'],
['','* He goes back to work. A stranger sits across from him and says it is all their fault, and he opens his mouth to
     say the soft, slow thing.'],
['Robin','...It is not your fault. And I am going to say it again, because I have just found out I have to.'],
['','* He says it in the attic, out loud, to the empty room. It sounds like his own voice. It is too late to be a
    comfort, and he says it anyway, and that is all of it.'],
...(pv&&!pv.w&&S.same?[['???','Same name. Same ending. I did not think you would do it
    again.']]:pv&&!pv.w?[['???','Again. {prev}, and now you. Same ending, different hands. I cannot tell whether that
    is a pattern or a habit.']]:pv&&pv.w?[['???','You had it once. Now you do not. I am only the one who
    counts.']]:[])]);
}
H.style.display='none';
O.seen=[...new Set([...O.seen,t+(win?'W':'L')])];O.prev={n:S.name,g:S.g,w:win,t,m:S.money,pd:S.panicDay};O.plays=(O.pl
    ays||0)+1;save();
const e=EN[t][win?'W':'L'];
W.innerHTML=`<div class="scene fadein">${endSc(t,win)}</div><div class="box c fadein"><p class="dim">${win?'She had
    eleven weeks. They were enough to be sad about, which is the best kind.':'The house is quiet.'}</p><h1 class="rv"
    style="color:${win?'var(--good)':'var(--bad)'}">${e}</h1><p class="serif">${win?'A good ending':'A hard ending'} ·
     Collectors ${S.rep.R} · Critics ${S.rep.S} · Students ${S.rep.P}</p><p class="dim">Endings found: ${new
    Set(O.seen).size}/6</p>${win?'':'<p class="note">Robin and the guilt in this story are fictional. If any part of
    it felt close to home, you do not have to carry it alone. In India you can call Tele-MANAS at 14416. In the US you
     can call or text 988. Elsewhere, findahelpline.com lists lines by country.</p>'}<p><button class="big"
    onclick="credits()">CREDITS ▸</button></p></div>`}
const GODF=t=>GOODF[t];
function blame(){return new Promise(async res=>{W.innerHTML=`<div class="box c fadein"><div class="comic"
    id="cm">${COMIC_SRC?`<img src="${COMIC_SRC}" alt="">`:'[ COMIC PLACEHOLDER: set COMIC_SRC near the top of the
    endings code ]'}</div><p class="blame dim" id="b1" style="opacity:0">It was not Robin's fault.</p><p class="blame"
     id="b2" style="opacity:0">It was <mark>${esc(S.name)}</mark>'s.</p><p class="c" id="b3" style="opacity:0"><button
     class="big" id="bn">...</button></p></div>`;
await sleep(1500);$('#b1').style.opacity=1;await sleep(2600);$('#b2').style.opacity=1;await
    sleep(2400);$('#b3').style.opacity=1;$('#bn').onclick=()=>res()})}

/* ===== credits ===== */
let CRT=0;
function credits(){clearTimeout(CRT);H.style.display='none';W.className='';mus('credits');
 const dv=`<svg class="dv" viewBox="0 0 300 14" width="230" height="14" aria-hidden="true"><path d="M4 8Q40 0 76 8T150
     8T224 8T296 7" fill="none" stroke="#8a2b1a" stroke-width="3" stroke-linecap="round" stroke-
    dasharray="320"/></svg>`;
 const hum=[['Rudra Parashar','Ideation, Music Design'],['Prince Raj Rawat','Main Ideation, Music Finishing
    Touches'],['Rudrapratap Singh Chauhan','Testing, Ideation, Video']];
 const ai=[['Main AI assistance','Claude'],['Side AI assistance','Gemini']];
 const sky=`<svg class="bgs" viewBox="0 0 400 230" preserveAspectRatio="xMidYMid slice" aria-
    hidden="true"><defs><linearGradient id="cg" x1="0" y1="0" x2="0" y2="1"><stop offset="0" stop-
    color="#0a0c18"/><stop offset="1" stop-color="#2a1a14"/></linearGradient></defs><rect width="400" height="230"
    fill="url(#cg)"/>${Array.from({length:26},(_,i)=>`<circle class="tw" cx="${(i*53)%400}" cy="${(i*37)%110}"
    r="${i%4?.8:1.4}" fill="#ffe7a8" style="animation-delay:${(i%7)*.3}s"/>`).join('')}<polygon class="flick"
    points="330,0 400,0 400,230 190,230" fill="#f0c060"/><rect y="196" width="400" height="34" fill="#07090f"/><g
    transform="translate(24 120)" stroke="#3a2a1c" stroke-width="3" fill="none"><path d="M10 76L40 0M70 76L40 0M40
    0V76"/><rect x="8" y="24" width="64" height="38" fill="#120d0a"/></g></svg>`;
 const ST=[
  `<p class="ct">Last Bid</p><p class="cs">was made by</p>${hum.map(([n,r],i)=>`<div class="ce" style="animation-
    delay:${.8+i*1.2}s;--d:${.8+i*1.2}s"><div class="cn">${esc(n)}</div><div
    class="cro">${esc(r)}</div>${i<2?dv:''}</div>`).join('')}`,
  `<p class="ct">Shaped with help from</p>${ai.map(([r,n],i)=>`<div class="ce" style="animation-
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    delay:${.8+i*1.4}s;--d:${.8+i*1.4}s"><div class="cro">${esc(r)}</div><div class="cn
    big">${esc(n)}</div>${i<1?dv:''}</div>`).join('')}`,
  `<p class="ct">Thank you for playing</p><div class="ce" style="animation-delay:.9s;--d:.9s"><p
    class="cs">${esc(S.name)} painted ten days.<br>Endings found: ${new Set(O.seen).size} of 6.</p>${dv}<p class="cs"
    style="margin-top:14px"><button class="big" id="cpa">PLAY AGAIN</button> <button id="crp">Replay
    credits</button></p></div>`];
 let i=-1;const DUR=[11000,8500];
 const key=e=>{if(e.target&&e.target.tagName==='BUTTON')return;if(e.code==='Space'||e.code==='Enter'){e.preventDefault
    ();go()}};
 const leave=()=>{clearTimeout(CRT);document.removeEventListener('keydown',key)};
 function go(n){clearTimeout(CRT);i=n!==undefined?n:Math.min(i+1,ST.length-1);
  W.innerHTML=`<div class="cr fadein" id="cr">${sky}<div class="scrw"><div class="scr">${ST[i]}</div></div><div
    class="cdots">${ST.map((_,k)=>`<span class="${k===i?'on':''}"></span>`).join('')}</div></div>${i<ST.length-1?'<p
    class="c"><button id="cnx">next ▸</button> <button id="csk">skip ▸▸</button></p>':''}`;
  $('#cr').onclick=e=>{if(e.target.tagName!=='BUTTON'&&i<ST.length-1)go()};
  if($('#cnx')){$('#cnx').onclick=()=>go();$('#csk').onclick=()=>go(ST.length-1)}
  if($('#cpa')){$('#cpa').onclick=()=>{leave();menu()};$('#crp').onclick=()=>go(0)}
  if(i<ST.length-1)CRT=setTimeout(()=>go(),DUR[i]);scrollTo(0,0)}
 document.addEventListener('keydown',key);go(0)}

/* ===== replay: dev quick-jump ===== */
menu();
</script></body></html>
```